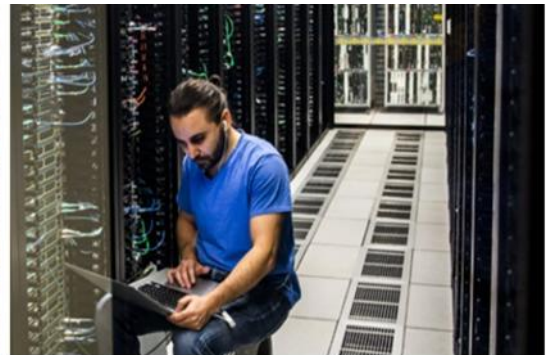




Cisco Intersight Managed Mode- Nutanix AHV or ESX



12 – March- 2026
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Disclaimer

Test results are suitable to the specific design, configuration, test procedure, and types of switches deployed in the environment described in this document.

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About this CPOC

Note: All printed copies and duplicate soft copies of this document are considered uncontrolled. Please refer to the original, online document link for the current version.

Cisco Intersight Standalone Mode is an **on-premises deployment** of Cisco's infrastructure management platform that operates **without requiring a connection to the Intersight cloud**. When integrated with **Nutanix hyperconverged infrastructure (HCI)**, it allows organizations to manage and monitor their Nutanix environments securely and locally.

In this mode, Intersight provides:

- **Visibility into Nutanix clusters** (via Prism API)
- **Monitoring of VMs, hosts, and cluster health**
- **Management of Cisco hardware** (e.g., UCS servers running Nutanix)
- **Single pane of glass** for hybrid environments
- **Air-gapped and compliance-friendly** operations

It's ideal for **secure, regulated environments** that require local control, while still benefiting from unified infrastructure management across Cisco and Nutanix systems.

Scope

Goals

After completing these scenarios, you will know how to claim, deploy and configure the Nutanix cluster with Cisco Intersight

IP Table

Host Name	IP Address	Credentials	Model
rtp-cpoc-dng-b16-c240m7-1	198.18.143.1	admin/C1sco12345!	HCIAF240C-M7SN
rtp-cpoc-dng-b16-c240m7-2	198.18.143.2	admin/C1sco12345!	HCIAF240C-M7SN
rtp-cpoc-dng-b16-c240m7-3	198.18.143.3	admin/C1sco12345!	HCIAF240C-M7SN
rtp-cpoc-dng-b16-c240m7-4	198.18.143.4	admin/C1sco12345!	HCIAF240C-M7SN
rtp-cpoc-dng-b16-c240m4-1	198.18.143.6	admin/C1sco12345!	UCSC-C240-M4SX
rtp-cpoc-dng-b16-n9336-1	198.18.143.7	admin/C1sco12345!	N9K-C9336C-FX2
rtp-cpoc-dng-b16-n9336-2	198.18.143.8	admin/C1sco12345!	N9K-C9336C-FX2
rtp-cpoc-dng-b16-fi6536-1	198.18.143.9	admin/C1sco12345!	HCIX-FI-6536
rtp-cpoc-dng-b16-fi6536-2	198.18.143.10	admin/C1sco12345!	HCIX-FI-6536
cpoc-win10-1	198.18.133.36	cpoc/cpoc123	Windows 10
cpoc-win10-2	198.18.133.37	cpoc/cpoc123	Windows 10
Ad1	198.18.133.1	dcloud/administrator /C1sco12345	Windows server
Vcenter 7	198.18.133.30	administrator@vsphere.local / C1sco12345!	VMWare Vsphere
Vcenter 8.0.3	198.18.133.31	administrator@vsphere.local / C1sco12345!	VMWare Vsphere
Cisco IMM	198.18.128.100	admin / C1sco12345	Cisco IMM Transition
Nutanix-pc.2022.6.0.12	198.18.128.150	admin / C1sco12345!	Nutanix-pc.2022.6.0.12
Intersight Assist	198.18.128.254	admin / C1sco12345	Intersight
Intersight pva	198.18.128.253	admin / C1sco12345	Intersight
Intersight cva	198.18.128.252	admin / C1sco12345	Intersight
VMware-ESXI	198.18.128.250	root / C1sco12345!	ESX 7.0

Host Name	IP Address	Credentials	Model
IMM Prism Element	198.18.136.2	admin / C1sco12345	
IMM Prism Element virtual IP	198.18.136.3	admin / C1sco12345	
IMM Nutanix PC-2024	198.18.136.4	admin / C1sco12345	
IMM Nutanix PC-2024 virtual IP	198.18.136.5	admin / C1sco12345	
IMM IP Pool for cluster	198.18.136.10-100		
Reserved IP Pool Intersight	198.18.136.220-255		

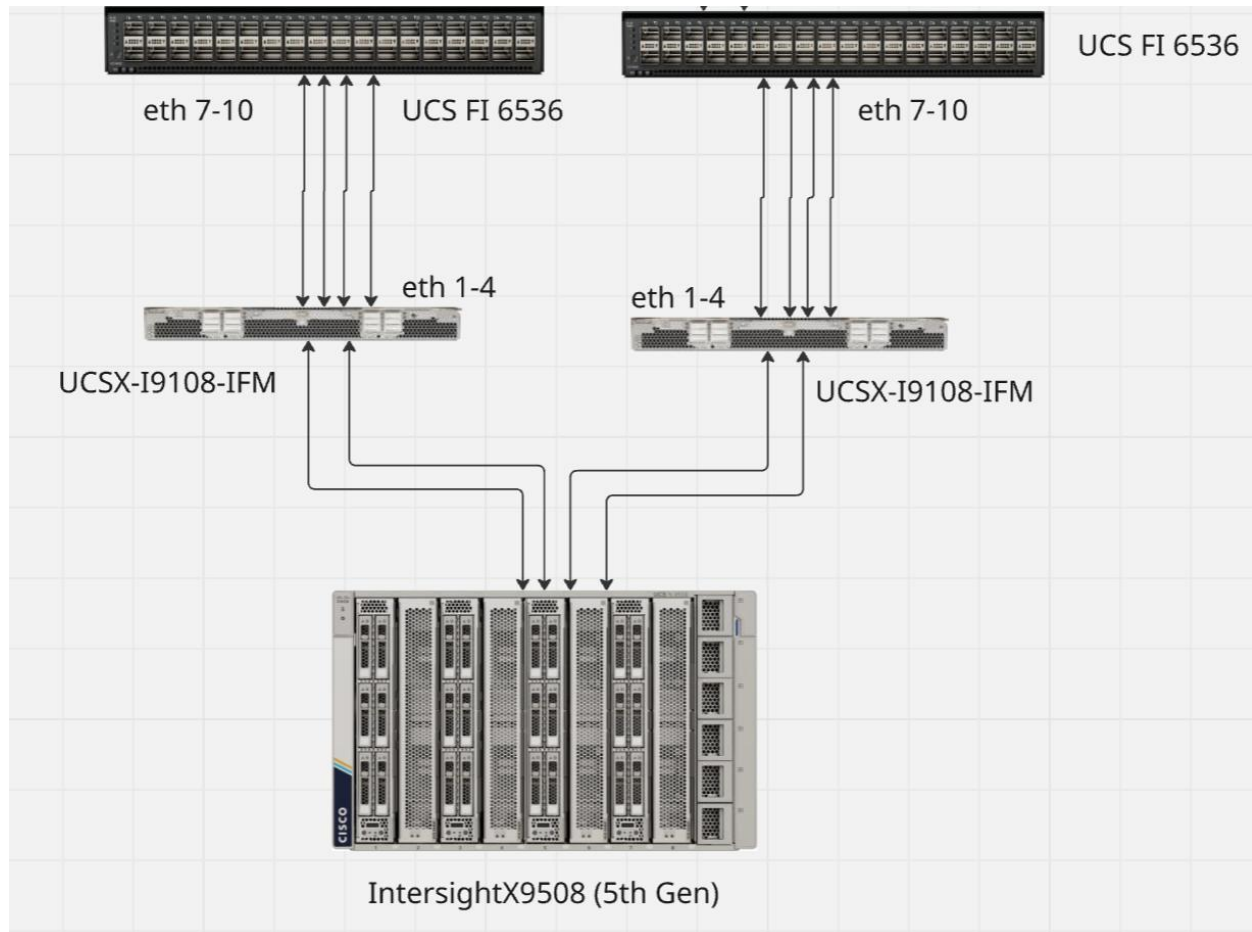
Equipment – CPOC Lab

Note: Be sure to include the Software Version in this table. Thank you!

Quantity Required	Model	Software Version
1	UCSX-9508	
2	UCS-FI-6536	4.3(5.250004)
2	UCSX-I-9108-100G	4.3(5c)
3	UCSX-210C-M7	5.3(0.240016)

Design and Topology Diagram

Introduction if helpful to user... We' ll replace the below with a generic Topology with no customer-facing data.



Get Started

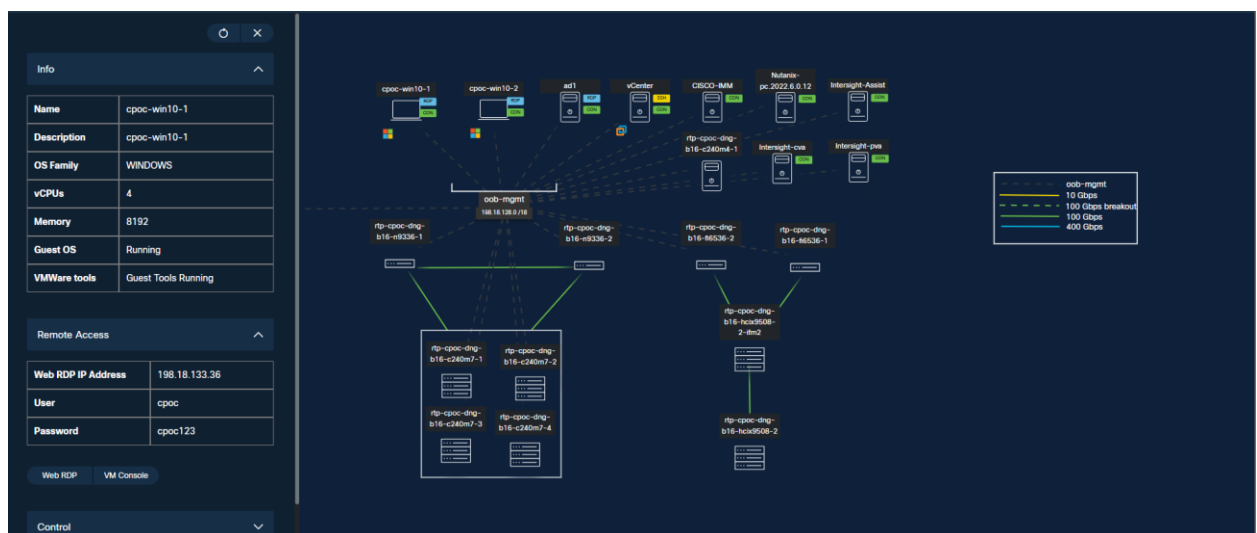
1. Initiate your dCloud session. [\[Show Me How\]](#)
2. For best performance, connect to the workstation with **Cisco Secure Client VPN** [\[Show Me How\]](#) and the **local RDP client** on your laptop. [\[Show Me How\]](#)
 - Workstation 1: **198.18.133.36**, Username: **cpoc**, Password: **cpoc123**

Scenario 1. Claim Servers and profile creation on Cisco Intersight

THIS SCENARIO IS JUST READ ONLY, ALL POLICIES, PROFILE, POOL HAS BEEN COMPLETED

Cisco Intersight Claim Servers is a process used to register or "claim" physical infrastructure devices (like Cisco UCS servers or Nexus switches) to the Cisco Intersight platform. Intersight is a cloud-based infrastructure management tool that provides centralized monitoring, configuration, and lifecycle management for Cisco data center hardware.

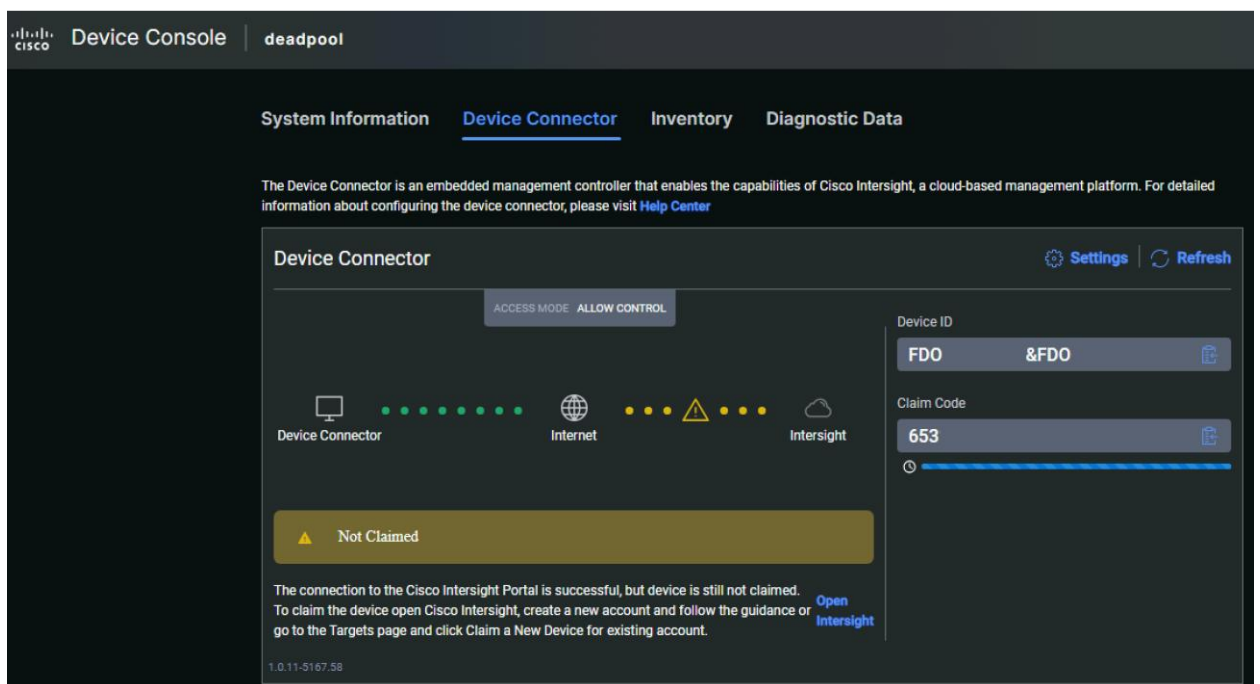
1. Click on cpoc-win10-1 on the main topology, on the left side select remote access and then click on web RDP to log in



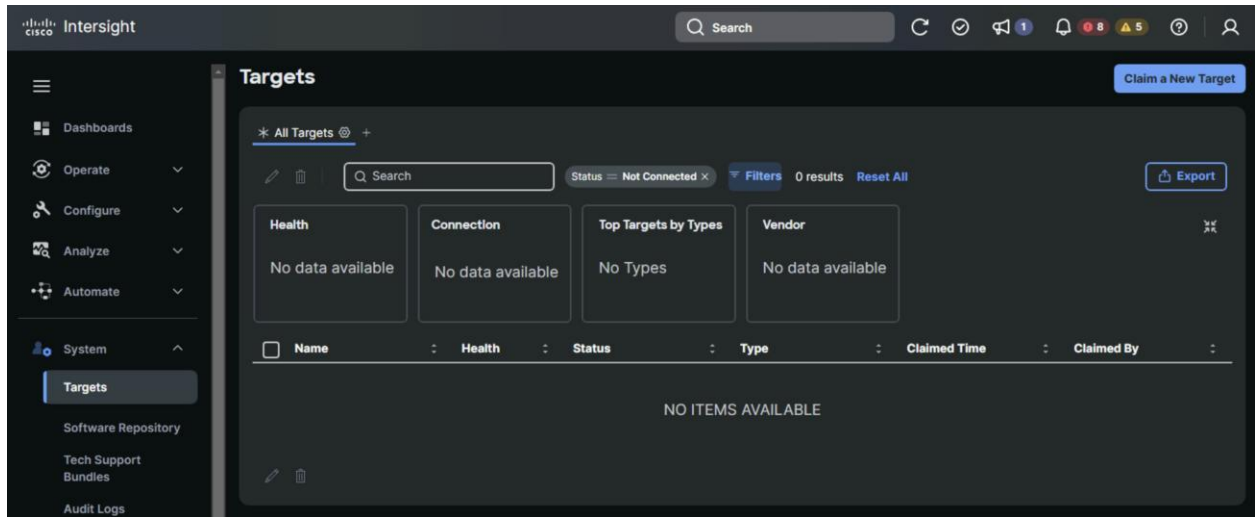
2. Once in the workstation open firefox browser



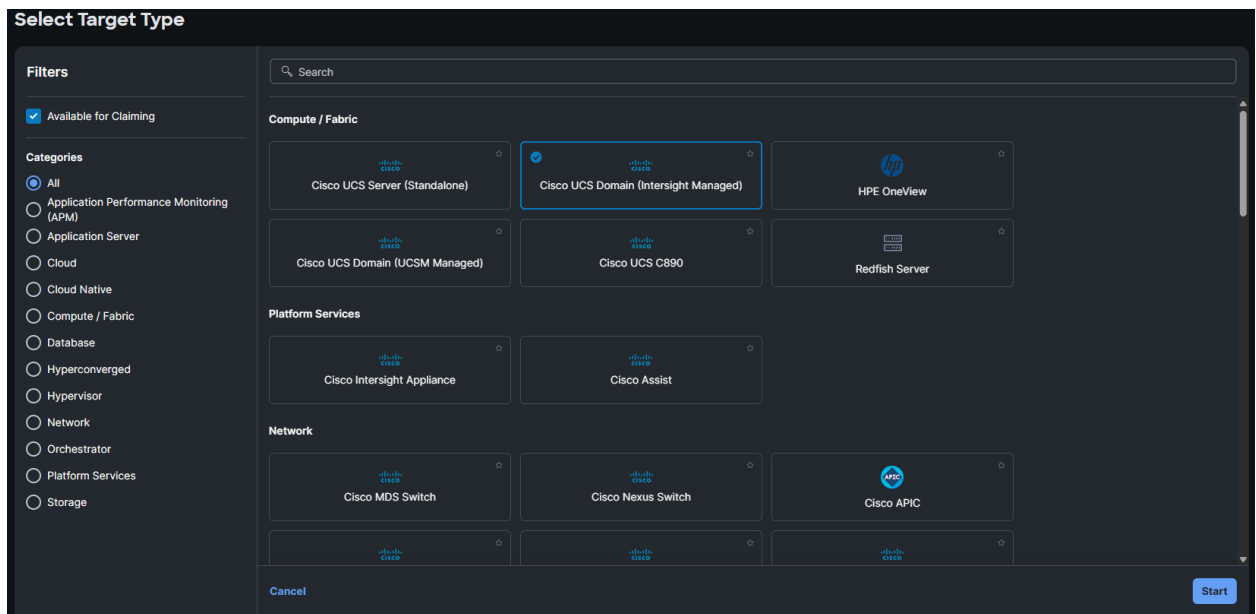
3. Log in to one Fabric Interconnect's web console via HTTPS with a web browser using the IP address **198.18.135.9 admin / C1sco12345!** - Retrieve the Device ID and the Claim Code from the console page by clicking on Device Connector at the top.



4. In Cisco Intersight, go to the System area, click on Targets, then Claim a New Target



5. Select Cisco UCS Domain (Intersight Managed) then enter the Device ID and Claim Code, then click Claim. Repeat for all the servers to be used in your new cluster



Claim Cisco UCS Server (Standalone) Target

To claim your target, provide the Device ID, Claim Code and select the appropriate Resource Groups.

General

Device ID * ⓘ Claim Code * ⓘ

Resource Groups

Select resource groups, if required. This is not mandatory, since by default, the claimed target will be added to "All" type resource groups.

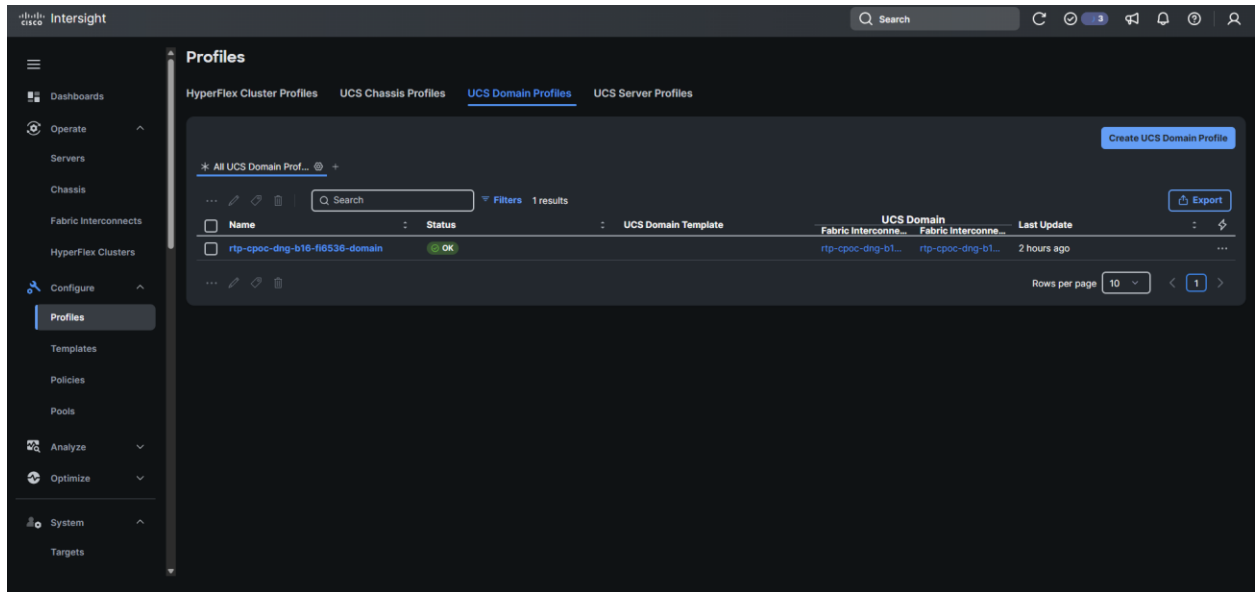
0 items found per page << < 0 of 0 > >> ⚙️

☐	Name	Usage	Description
NO ITEMS AVAILABLE			

<< < 0 of 0 > >>

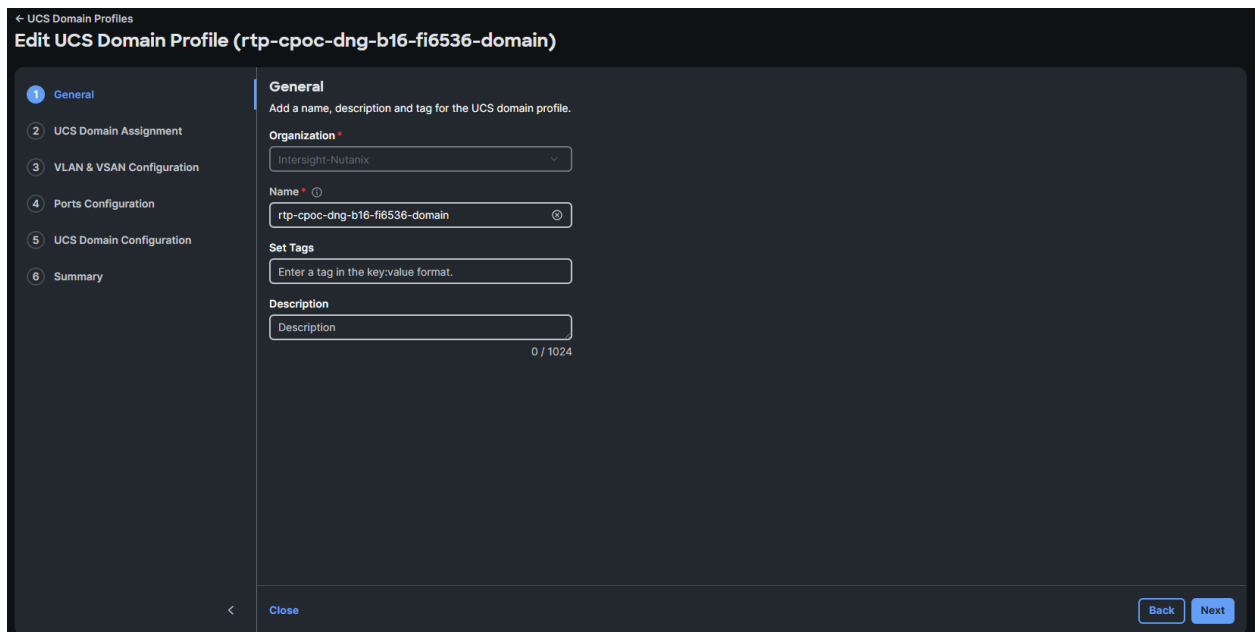
[Back](#) [Cancel](#) [Claim](#)

A Domain Profile must be created and deployed to the Fabric Interconnects. The Domain Profile defines the roles of the ports on the Fabric Interconnects, the VLANs used on the network and several other domain-wide policy settings such as QoS. After the Domain Profile is deployed the rackmount servers and/or modular blades will discover and can then be onboarded in Foundation Central and targeted for a Nutanix cluster deployment. Ensure that the profile and all the associated policies are created in the Organization that also contains the Resource Group for the Fabric Interconnects and servers.



Select the appropriate Organization in the Intersight account and give the policy a name.

Select the Fabric Interconnect pair the policy will be applied to.



← UCS Domain Profiles

Edit UCS Domain Profile (rtp-cpoc-dng-b16-fi6536-domain)

- General
- UCS Domain Assignment**
- VLAN & VSAN Configuration
- Ports Configuration
- UCS Domain Configuration
- Summary

UCS Domain Assignment

Choose to assign a fabric interconnect pair to the profile now or later.

[Assign Now](#) [Assign Later](#)

i Choose to assign a Fabric Interconnect pair now or later. If you choose Assign Now, select a pair that you want to assign and click Next . If you choose Assign Later, click Next to proceed to policy selection.

☒ Show Assigned

Filters 1 results

Domain N...	Model	Fabric Interconnect A			Fabric Interconnect B		
		Serial	Bundle Version	Model	Serial	Bundle Version	
☒ rtp-cpoc-dng-...	UCS-FI-6536	FDO28440769	4.3(5.250004)	UCS-FI-6536	FDO2844077P	4.3(5.250004)	

Selected 1 of 1 [Show All](#) [Unselect All](#)

Rows per page < 1 >

[Close](#) [Back](#) [Next](#)

Select the VLAN policy for Fabric Interconnect A, then select to create a new VLAN policy.

← UCS Domain Profiles

Create UCS Domain Profile

- General
- UCS Domain Assignment
- VLAN & VSAN Configuration**
- Ports Configuration
- UCS Domain Configuration
- Summary

VLAN & VSAN Configuration

Create or select a policy for the Fabric Interconnect pair.

^ Fabric Interconnect A 0 of 2 Policies Configured

VLAN Configuration [Select Policy](#)

VSAN Configuration [Select Policy](#)

^ Fabric Interconnect B 0 of 2 Policies Configured

VLAN Configuration [Select Policy](#)

VSAN Configuration [Select Policy](#)

[Close](#) [Back](#) [Next](#)

Select the appropriate organization, enter a name for the VLAN policy for Fabric Interconnects, click Next, then select to add a VLAN

Enter a prefix and the VLAN ID for the new VLAN, then click to select the Multicast policy. Select to create a new Multicast policy. Select the appropriate organization, enter a name for the new Multicast policy and click Next.

UCS Domain Profiles > Create UCS Domain Profile

Create VLAN

Add VLANs

Add VLANs to the policy

VLANs should have one Multicast policy associated to it

Configuration

Prefix *

mgmt

VLAN IDs *

10

Auto Allow On Uplinks

Enable VLAN Sharing

Multicast Policy *

Select Policy

Cancel

Add

UCS Domain Profiles > Create UCS Domain Profile > Create VLAN

Create Multicast Policy

1 General

2 Policy Details

General

Add a name, description, and tag for the policy.

Organization *

default

Name *

multicast-policy

Set Tags

Enter a tag in the key:value format.

Description

Description

0 / 1024

Cancel

Next

UCS Domain Profiles > Create UCS Domain Profile > Create VLAN

Create Multicast Policy

✓ General

2 Policy Details

Policy Details

Add policy details.

Multicast Policy

Shooping State

Querier State

Source IP Proxy State

< Cancel

Back Create

Leave the default settings for the Multicast policy and click Create. Click Add to create the new VLAN in the VLAN policy.

UCS Domain Profiles > Create UCS Domain Profile

Create VLAN

Add VLANs

Add VLANs to the policy

⚠ VLANs should have one Multicast policy associated to it

Configuration

Prefix *

VLAN IDs *

vlan-mgmt

10

Auto Allow On Uplinks

Enable VLAN Sharing

Multicast Policy *

Selected Policy: multicast-policy | Edit Selection |

Cancel

Add

UCS Domain Profiles > Create UCS Domain Profile

Create VLAN

General

Policy Details

Policy Details

Add policy details.

This policy is applicable only for UCS Domains

VLANs

Add VLANs

☐ Show VLAN ID Ranges

Search Filters 2 results

☐	VLAN ID	Name	Sharing Type	Primary VLAN ID	Multicast Policy	Auto Allow On Uplink	
☐	1	default	None			Yes	...
☐	10	vlan-mgmt_10	None		multicast-policy	Yes	...

☐ Set Native VLAN ID

Rows per page: 10 < 1 >

If any additional VLANs need to communicate via the uplinks from this Fabric Interconnect pair, click Add VLANs and create them as was done for the first VLAN. Click Create to finally finish creating the VLAN policy. Click to select the VLAN policy to apply for Fabric Interconnect B.

← UCS Domain Profiles

Create UCS Domain Profile

General

UCS Domain Assignment

VLAN & VSAN Configuration

Ports Configuration

UCS Domain Configuration

Summary

VLAN & VSAN Configuration

Create or select a policy for the Fabric Interconnect pair.

^ Fabric Interconnect A 1 of 2 Policies Configured

VLAN Configuration

VSAN Configuration

^ Fabric Interconnect B 0 of 2 Policies Configured

VLAN Configuration

VSAN Configuration

Select the VLAN policy that was just created so that both Fabric Interconnects have the same VLAN policy configured, then click Next.

← UCS Domain Profiles

Create UCS Domain Profile

- General
- UCS Domain Assignment
- VLAN & VSAN Configuration**
- Ports Configuration
- UCS Domain Configuration
- Summary

VLAN & VSAN Configuration

Create or select a policy for the Fabric Interconnect pair.

^ Fabric Interconnect A 1 of 2 Policies Configured

VLAN Configuration 🗑️ ✎ 👁️ • vlan-policy

VSAN Configuration Select Policy

^ Fabric Interconnect B 1 of 2 Policies Configured

VLAN Configuration 🗑️ ✎ 👁️ • vlan-policy

VSAN Configuration Select Policy

Close Back Next

Click to select the Port policy for Fabric Interconnect A then click Create New.

← UCS Domain Profiles

Create UCS Domain Profile

- General
- UCS Domain Assignment
- VLAN & VSAN Configuration
- Ports Configuration**
- UCS Domain Configuration
- Summary

Ports Configuration

Create or select a port policy for the Fabric Interconnect pair.

i Configure ports by creating or selecting a policy.

^ Fabric Interconnect A Not Configured

Ports Configuration Select Policy

^ Fabric Interconnect B Not Configured

Ports Configuration Select Policy

Close Back Next

Select the appropriate organization, enter a name for the Port policy and select the model of Fabric Interconnect matching the hardware in use, then click next. If Fibre Channel ports are required for use by other non-Nutanix modular servers then configure the unified ports, otherwise click Next to continue.

UCS Domain Profiles > Create UCS Domain Profile

Create Port

1 General

2 Unified Port

3 Breakout Options

4 Port Roles

General

Add a name, description, and tag for the policy.

Organization *
default

Name *
port-policy

Fabric Interconnect Model *
UCS-FI-6536

Set Tags
Enter a tag in the key-value format.

Description
Description
0 / 1024

Cancel Next

UCS Domain Profiles > Create UCS Domain Profile

Create Port

1 General

2 Unified Port

3 Breakout Options

4 Port Roles

Unified Port

Configure the port modes to carry FC or Ethernet traffic.

Move slider to configure unified ports.

Configuring unified port will result in the deletion of previously configured breakout ports, port roles and port channels roles.

Fibre Channel Ports
No Fibre Channel Ports

FC Ethernet Ports 1-36

Back Next

If Breakout ports are required for other non-Nutanix servers then select them and click the Configure button, otherwise click Next to continue.

UCS Domain Profiles > Create UCS Domain Profile

Create Port

- General
- Unified Port
- Breakout Options**
- Port Roles

Breakout Options
Configure breakout ports on FC or Ethernet.

Ethernet Fibre Channel

Configure

FC Ethernet Port Modes

☐	Port	Type	Speed	Breakout Ports	
☐	Port 1	Ethernet	-	-	
☐	Port 2	Ethernet	-	-	
☐	Port 3	Ethernet	-	-	
☐	Port 4	Ethernet	-	-	
☐	Port 5	Ethernet	-	-	
☐	Port 6	Ethernet	-	-	

Cancel Back Next

Select the ports which will be server ports, i.e. ports connected to the blade chassis, then click the Configure button.

UCS Domain Profiles > Create UCS Domain Profile

Create Port

- General
- Unified Port
- Breakout Options
- Port Roles**

Port Roles Port Channels Port Groups

Configure Selected Ports Port 3, Port 4, Port 5, Port 6 Clear Selection

Unconfigured

Export

☐	Name	Type	Role	Connected De...	Device Number	Port Channel	Mode	Auto Negotiation	
☐	port 1	Ethernet	Unconfigured			-			
☐	port 2	Ethernet	Unconfigured			-			
☒	port 3	Ethernet	Unconfigured			-			
☒	port 4	Ethernet	Unconfigured			-			
☒	port 5	Ethernet	Unconfigured			-			
☒	port 6	Ethernet	Unconfigured			-			
☐	port 7	Ethernet	Unconfigured			-			

Cancel Back Save

Select Server as the role for the ports connecting to the blade chassis or the rackmount servers then click Save. For rackmount servers it is recommended to enable "Manual Server Numbering" and setting the server number to match their physical order in the rack.

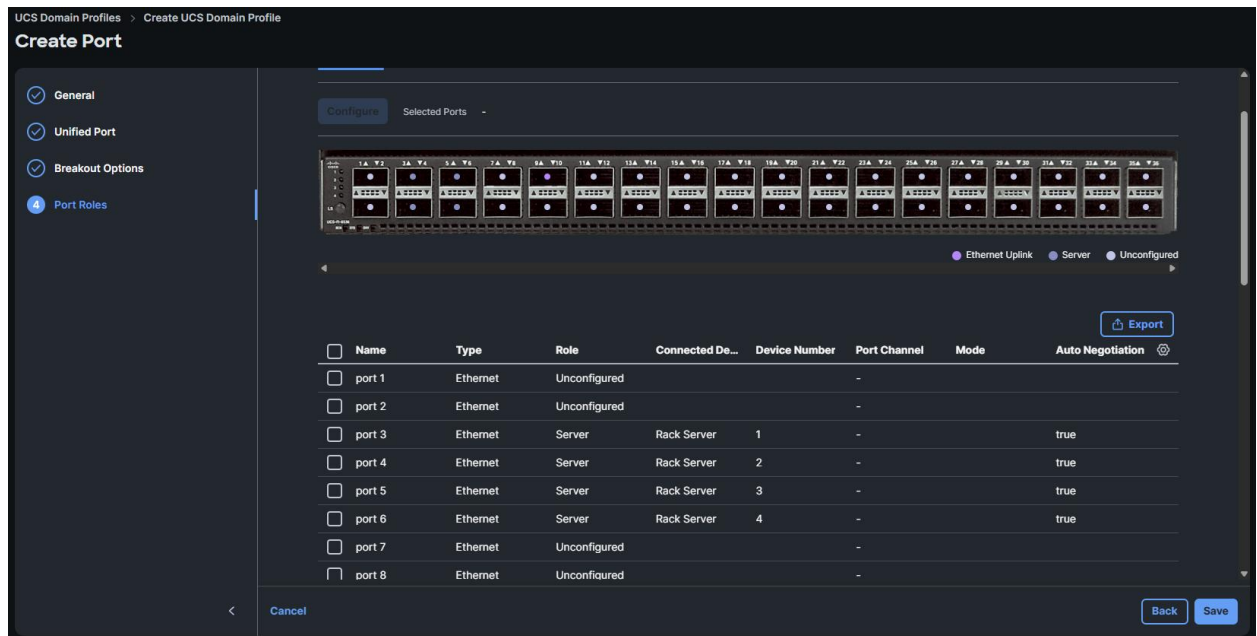
The screenshot shows the 'Create Port' configuration window. At the top, the breadcrumb is 'UCS Domain Profiles > Create UCS Domain Profile'. The title is 'Create Port'. A 'Role' dropdown is set to 'Server'. Below it, there are three toggle switches: 'Auto Negotiation' (checked), 'Manual Chassis/Server Numbering' (checked), and 'Auto-Fill Numbering' (unchecked). Two informational messages are displayed: one about Auto Negotiation not being supported on N9K-C93180YC-FX3 for 100G speed ports, and another about device type and numbering conflicts. Below these, there is a table with three rows for 'port 3', 'port 4', and 'port 5'. Each row has a 'Connected Device Type' dropdown set to 'Rack Server' and a 'Device Number' input field with values 1, 2, and 3 respectively. At the bottom, there are 'Cancel' and 'Save' buttons.

Port	Connected Device Type *	Device Number *
port 3	Rack Server	1
port 4	Rack Server	2
port 5	Rack Server	3

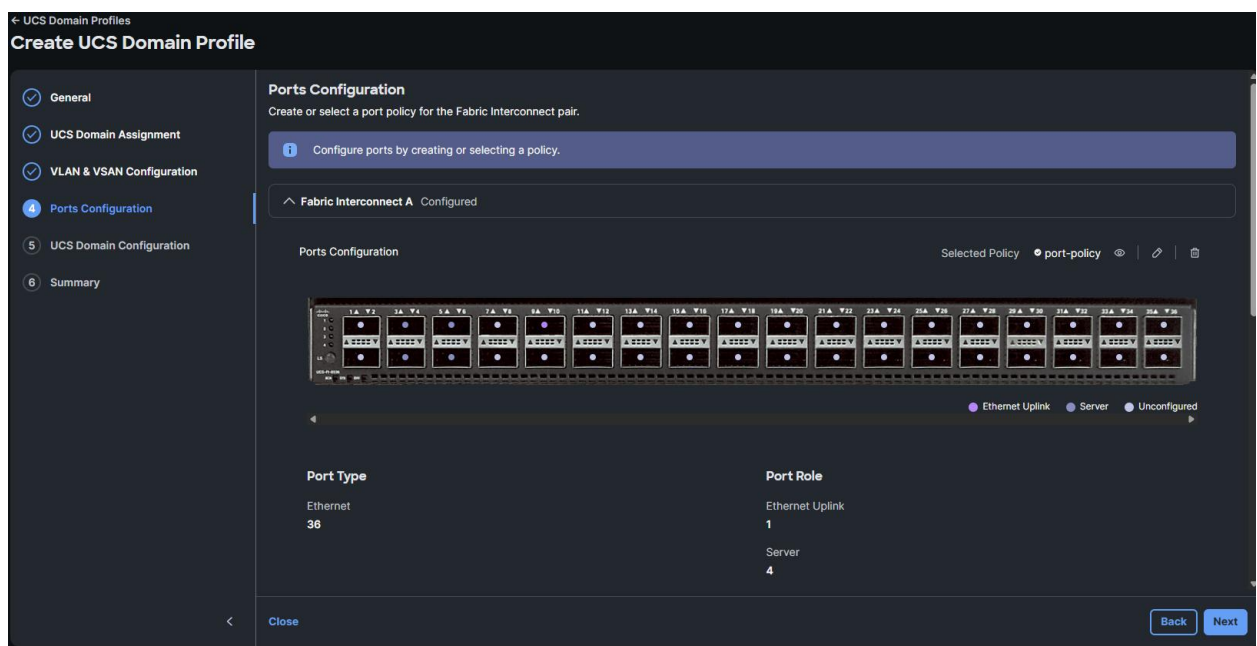
After returning to the previous screen select the ports which will be the Ethernet uplinks from the Fabric Interconnects and click the Configure button. Select Ethernet Uplink as the role then click Save.

The screenshot shows the 'Configure Port' configuration window. The breadcrumb is 'UCS Domain Profiles > Create UCS Domain Profile'. The title is 'Configure Port'. Under the 'Configuration' section, 'Selected Port' is 'Port 9'. The 'Role' dropdown is set to 'Ethernet Uplink'. Below it are 'Admin Speed' and 'FEC' dropdowns, both set to 'Auto'. There is an 'Ethernet Network Group' section with a 'Select Policies' link. Below that are 'Flow Control', 'Link Control', and 'MACsec Policy' sections, each with a 'Select Policy' link. At the bottom, there are 'Cancel' and 'Save' buttons.

If necessary, click on Port Channels and add the multiple Ethernet uplink ports to a port channel. After all the server and Ethernet uplink ports and their optional port channels are configured, click Save.



Click to select a port policy for Fabric Interconnect B, then select the policy just created so both FIs are configured with the same port policies.



Select to create an NTP policy. Select the appropriate organization, enter a name for the policy, then click Next.

Create UCS Domain Profile

← UCS Domain Profiles

General
UCS Domain Assignment
VLAN & VSAN Configuration
Ports Configuration
UCS Domain Configuration
Summary

UCS Domain Configuration
Select the compute and management policies to be associated with the Fabric Interconnect.

Show Attached Policies (0)

Management 0 of 6 Policies Configured

NTP	Select Policy
Syslog	Select Policy
Network Connectivity	Select Policy
SNMP	Select Policy
LDAP	Select Policy
Certificate Management	Select Policy

Network 0 of 2 Policies Configured

System QoS *	Select Policy
--------------	---------------

Close Back Next

UCS Domain Profiles > Create UCS Domain Profile

Create NTP

1 General
2 Policy Details

General
Add a name, description, and tag for the policy.

Organization *

default

Name *

Ntp-policy

Set Tags

Enter a tag in the key:value format.

Description

Description

0 / 1024

Cancel Next

Enable NTP, enter at least one NTP server and select the appropriate timezone then click Create

UCS Domain Profiles > Create UCS Domain Profile

Create NTP

1 General

2 Policy Details

Policy Details
Add policy details.

☒ Enable NTP ⓘ

NTP Servers * ⓘ
 +

Timezone ⓘ

[Cancel](#)
[Back](#)
[Create](#)

After returning to the previous screen click to select a System QoS Policy. Select the appropriate organization, enter a name for the policy, then click Next.

UCS Domain Profiles > Create UCS Domain Profile

Create System QoS

1 General

2 Policy Details

General
Add a name, description, and tag for the policy.

Organization *

Name *
 ⓘ

Set Tags

Description

0 / 1024

[Cancel](#)
[Next](#)

Set the Best Effort QoS class to MTU 9216 then click Create

UCS Domain Profiles > Create UCS Domain Profile

Create System QoS

General

2 Policy Details

Policy Details
Add policy details.

Configure Priorities

☐ Platinum

☐ Gold

☐ Silver

☐ Bronze

☐ Best Effort

☐ Fibre Channel

CoS Weight Bandwidth Percent ☒ Allow Packet Drops MTU

CoS Weight Bandwidth Percent ☐ Allow Packet Drops MTU

After returning to the previous screen click to select a Network Connectivity Policy. Select the appropriate organization, enter a name for the policy then click Next.

UCS Domain Profiles > Create UCS Domain Profile

Create Network Connectivity

1 General

2 Policy Details

General
Add a name, description, and tag for the policy.

Organization

Name

Set Tags

Description

Set the preferred primary DNS server address then click Create. Once an NTP policy, Network Connectivity policy and System QoS policy are configured, click Next to move to the domain summary

UCS Domain Profiles > Create UCS Domain Profile

Create Network Connectivity

General | **Policy Details**

Add policy details.

All Platforms | UCS Server (Standalone) | **UCS Domain**

Common Properties

IPv4 Properties

Preferred IPv4 DNS Server ⓘ 198.18.133.1 ⓘ

Alternate IPv4 DNS Server ⓘ 0.0.0.0 ⓘ

☐ Enable IPv6 ⓘ

[Cancel](#) [Back](#) [Create](#)

← UCS Domain Profiles

Create UCS Domain Profile

☐ Show Attached Policies (3)

General | **UCS Domain Assignment** | **VLAN & VSAN Configuration** | **Ports Configuration** | **UCS Domain Configuration** | **Summary**

Management 2 of 6 Policies Configured

NTP	Ntp-policy
Syslog	Select Policy
Network Connectivity	Network-Policy
SNMP	Select Policy
LDAP	Select Policy
Certificate Management	Select Policy

Network 1 of 2 Policies Configured

System QoS *	QoS-Policy
Switch Control	Select Policy

[Close](#) [Back](#) [Next](#)

Click Deploy and watch the domain profile progress through Validation and Configuration until the status is OK.

The image shows two screenshots from the Cisco UCS management interface. The top screenshot is the 'Create UCS Domain Profile' wizard, and the bottom screenshot is the 'Profiles' list.

Create UCS Domain Profile

Summary

Review the UCS domain profile details, resolve configuration errors and deploy the profile.

General

Name: Test Status: Not Assigned

Organization: default

Ports Configuration | VLAN & VSAN Configuration | UCS Domain Configuration | Errors/Warnings (0)

✓ Fabric Interconnect A

✓ Fabric Interconnect B

Close Back Deploy

Profiles

HyperFlex Cluster Profiles | UCS Chassis Profiles | **UCS Domain Profiles** | UCS Server Profiles

Create UCS Domain Profile

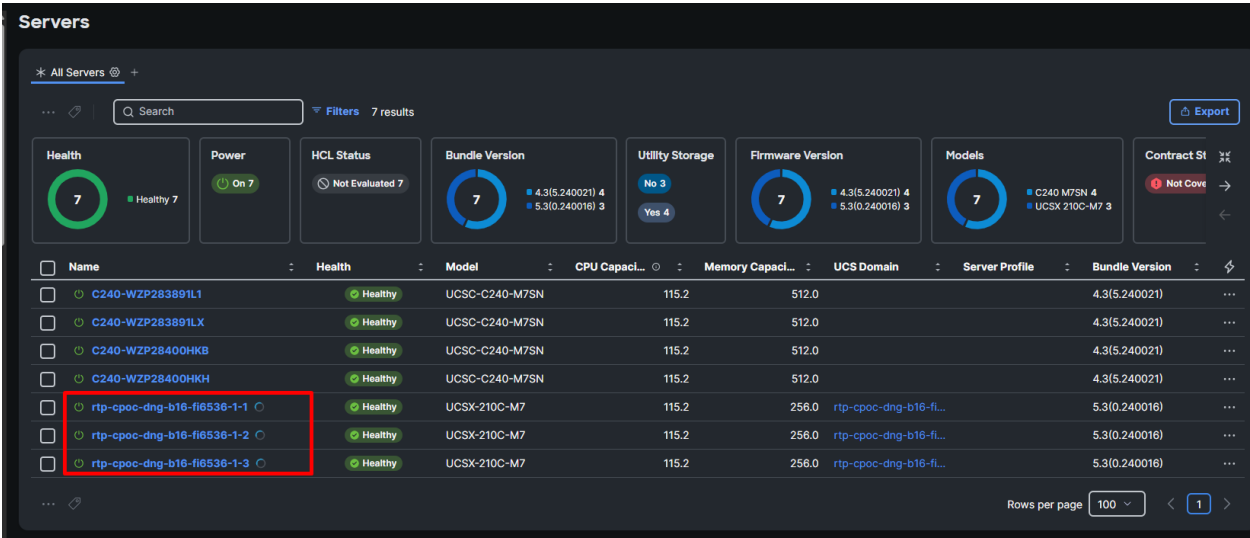
* All UCS Domain Prof... +

Search Filters 1 results Export

Name	Status	UCS Domain Template	UCS Domain	Last Update
rtp-cpoc-dng-b16-fi6536-domain	OK		Fabric Interconne... rtp-cpoc-dng-b1... rtp-cpoc-dng-b1...	2 hours ago

Rows per page 10 < 1 >

After the Domain profile is deployed, all modular chassis, the blades in the chassis and the rackmount servers will be discovered. Once all the chassis, blades and rackmounts have finished discovery the next steps can be completed.



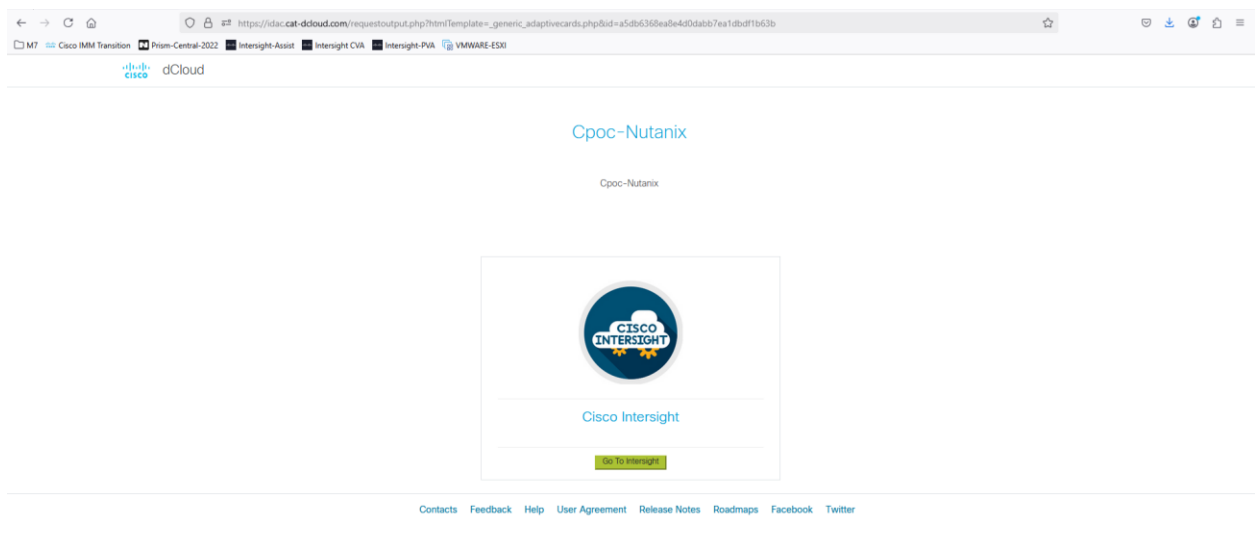
Scenario 2. Cisco Intersight API key generation

- On the Desktop click the Active.html icon to access the Intersight login, on the next page click on the Hyperlink "Click here to login"

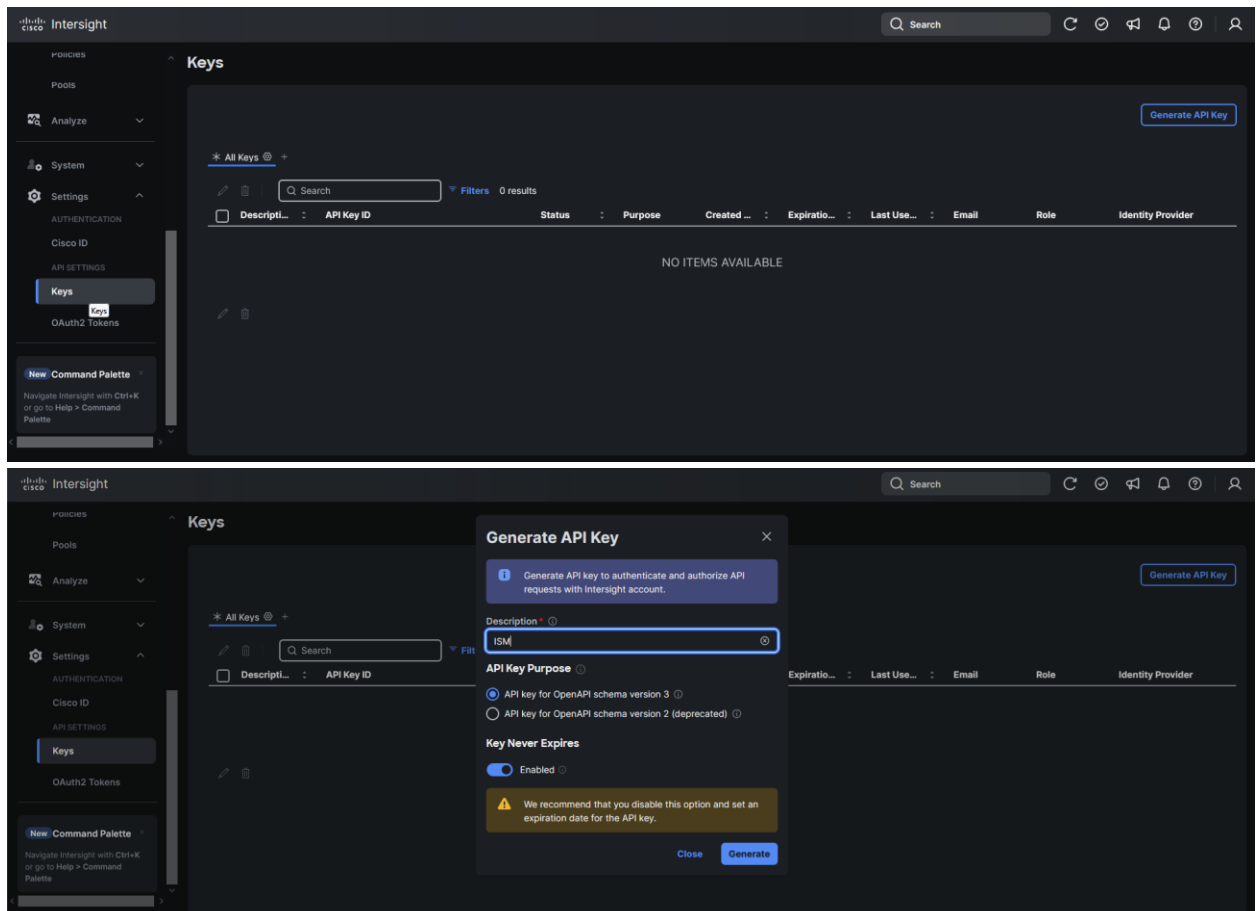


- In the resulting window click Go to intersight, this will do the auto-login

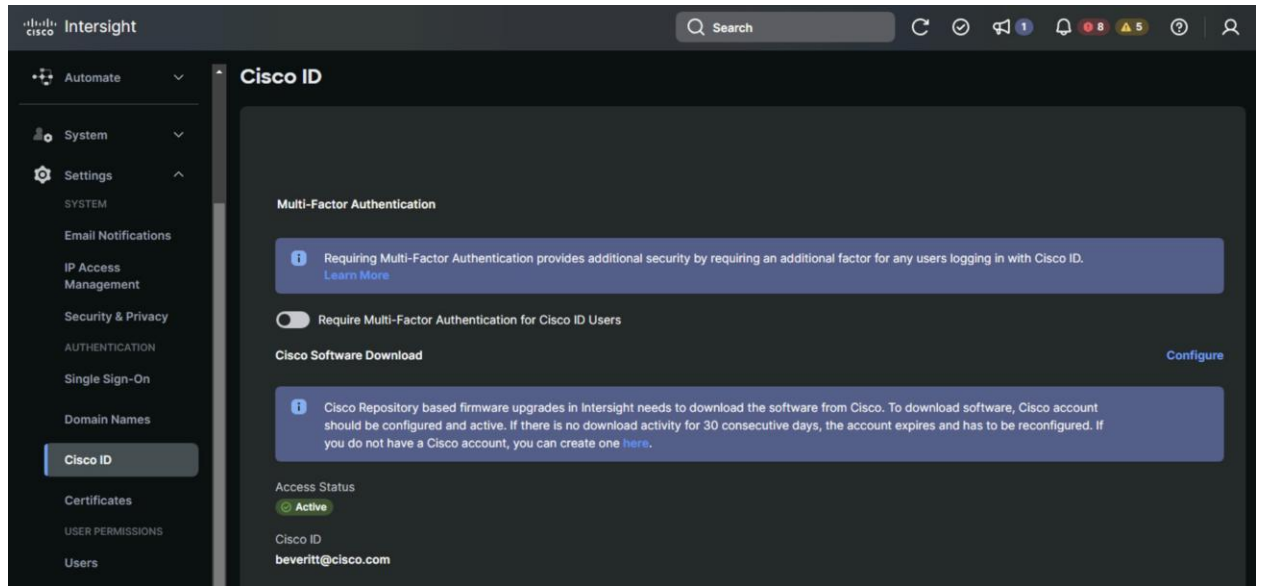
If require accessing Intersight from the cpoc-win10-2, copy the URL and paste on the second windows virtual machine, then click Go to Intersight



8. On the left page of Intersight, under Settings, click Keys
9. Click “Generate API key” using schema version 3 for use by Nutanix Foundation Central. Be sure to save the Secret Key to a file. It will only be shown once

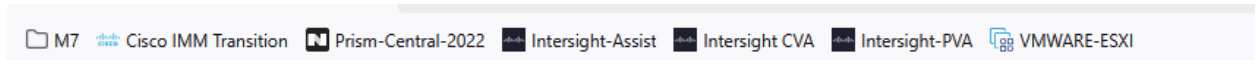


10. Save the API Key ID and Secret Key in the Desktop
11. Ensure your Cisco ID is granted access to download software from CCO. If not, click the Activate link and enter your CCO login credentials. This step is not required for an air-gapped Cisco Intersight private Virtual Appliance (PVA).

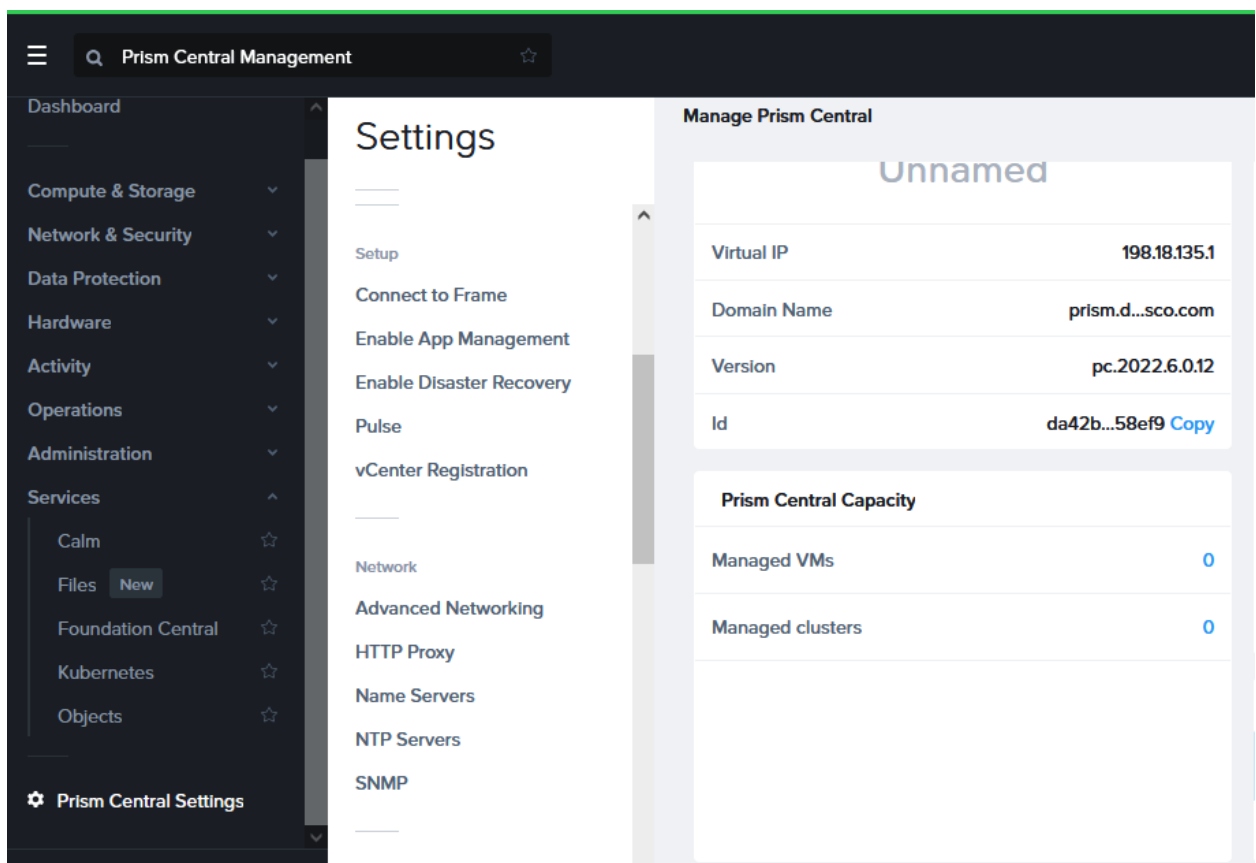


Scenario 3. Onboard nodes in Foundation Central

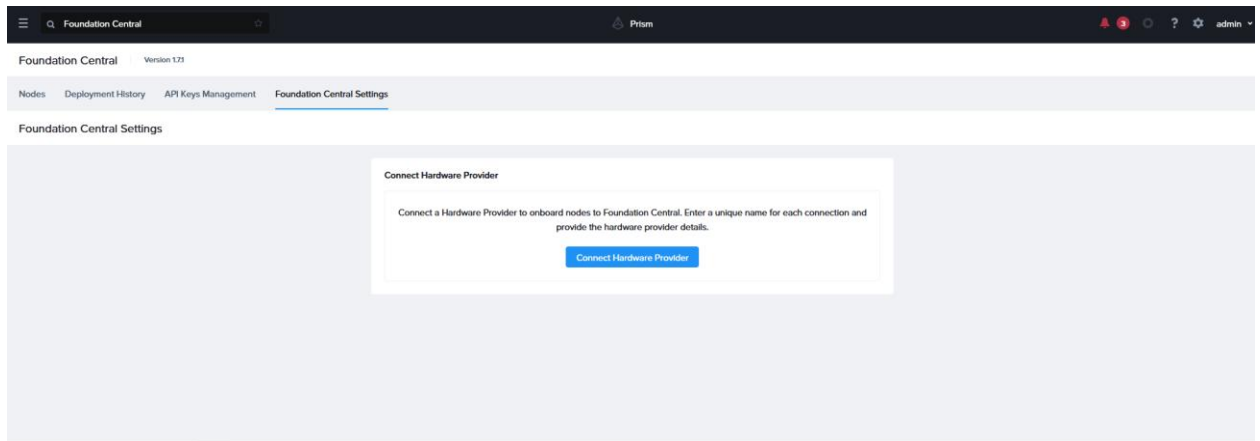
1. On Firefox browser bookmarks click over Prism-Central-2022, the credentials are admin / C1sco12345!



2. On this lab we' re using the Nutanix Prism 2022.6.0.12 with Foundation Central 1.7.1
3. Make sure the DNS, NTP, Virtual IP and Domain Name are set.

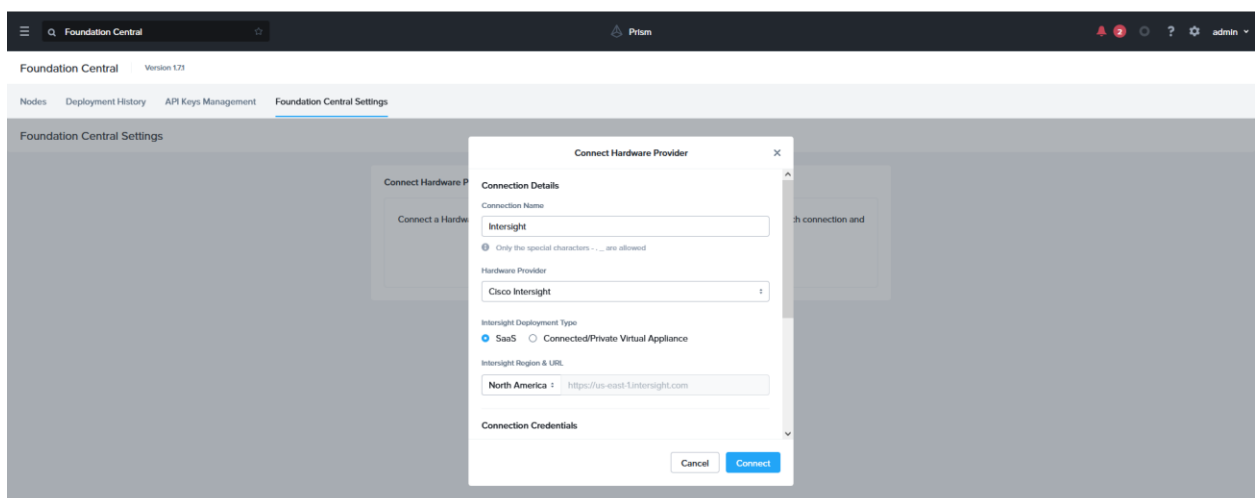


- Once all the parameters are confirmed, go to Foundation Central Settings to connect Prism Central with Intersight account, click “Connect Hardware Provider”

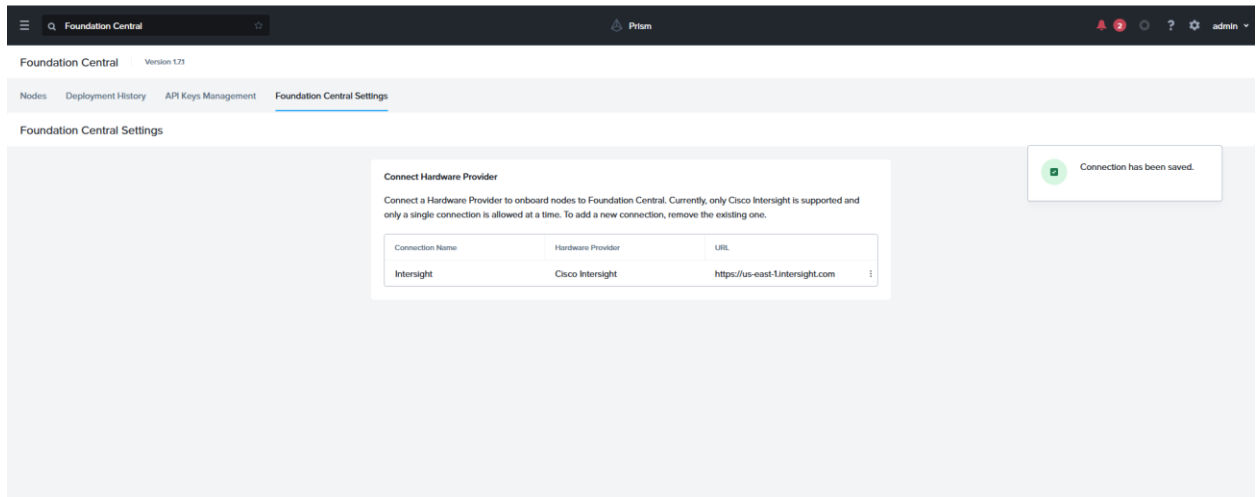


- Fill in the required information

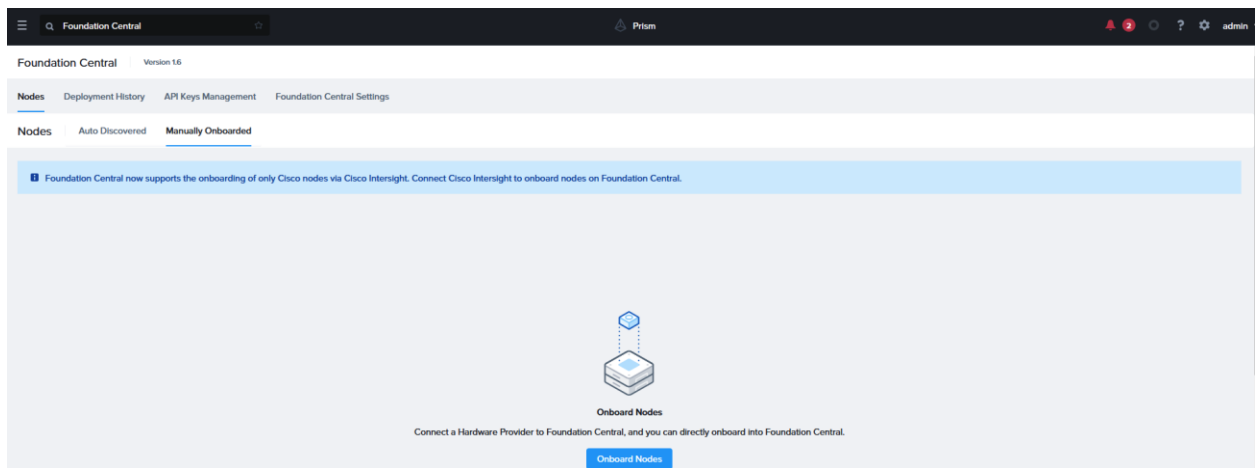
- Connection name= Intersight
- Hardware Provider = Cisco Intersight
- Intersight Deployment Type = SaaS
- Intersight Region & URL = North America
- Intersight API Key = <copy the one generated previously>
- Secret Key= <copy the one generated previously and saved to the desktop>



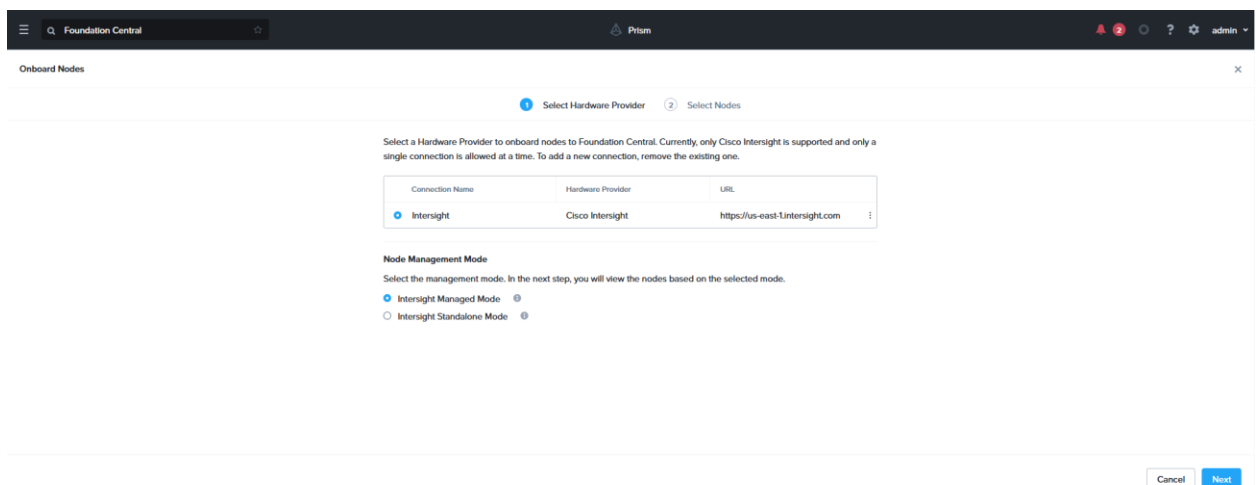
6. Prism Central will display a pop-up of "Connection has been saved"



7. Then click on Nodes > Manually Onboarded and click Onboard Nodes



8. For node management mode, choose **Intersight Managed Mode** then click Next



9. Select all Nodes you can to onboard and then click Onboard

Foundation Central

Onboard Nodes

Select Hardware Provider Select Nodes

Connection Name: **Intersight** Management Mode: **Intersight Managed Mode**

You are viewing Cisco Intersight nodes without any server profile attached. To update the nodes list, click [Refresh](#).

Type in a query

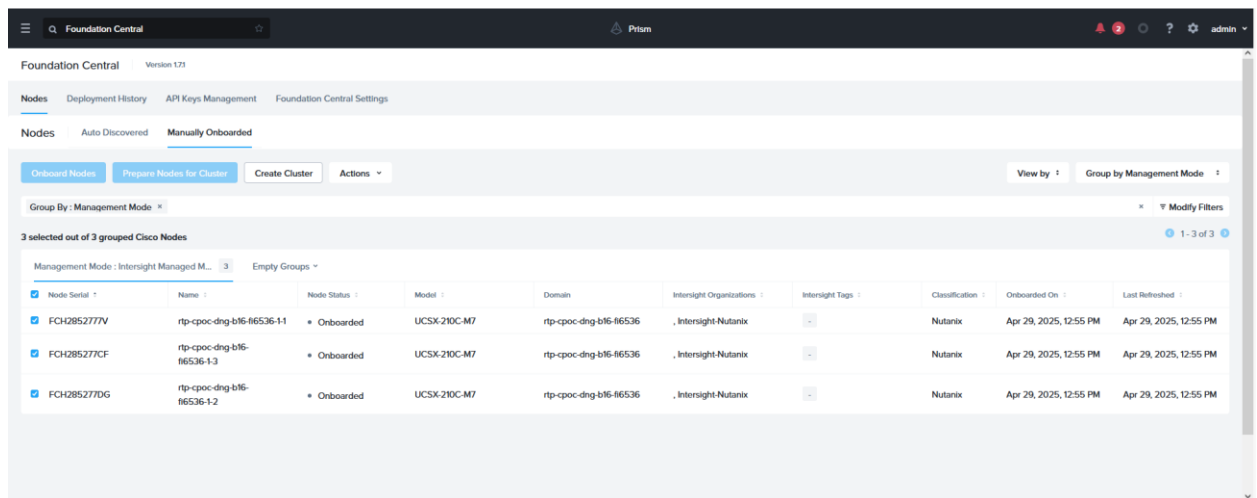
3 selected out of 3 Cisco Nodes

☒ Node Serial	Name	Model	Block Serial	Domain	Classification	Insight Organizations	Insight Tags	Memory (GB)	CPU (Cores)
☒ FCH285277TV	rtp-cpoc-dng-b16-86536-11	UCSX-210C-M7	FOX2907P2ZC	rtp-cpoc-dng-b16-86536	Nutanix	, Intersight-Nutanix	-	256	32
☒ FCH285277CF	rtp-cpoc-dng-b16-86536-13	UCSX-210C-M7	FOX2907P2ZC	rtp-cpoc-dng-b16-86536	Nutanix	, Intersight-Nutanix	-	256	32
☒ FCH285277DG	rtp-cpoc-dng-b16-86536-12	UCSX-210C-M7	FOX2907P2ZC	rtp-cpoc-dng-b16-86536	Nutanix	, Intersight-Nutanix	-	256	32

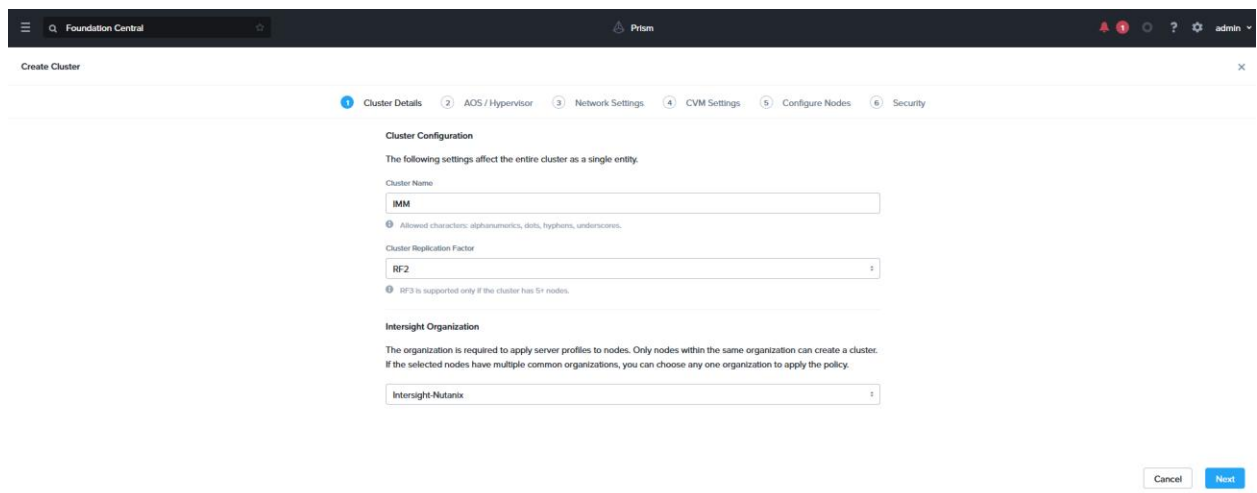
Back Cancel Onboard

Scenario 4. Nutanix Cluster Deployment with AHV or ESX Hypervisor

1. In Foundation Central under Nodes > Manually Onboarded, select the 3 nodes would like to use for the cluster deployment and click on Create Cluster

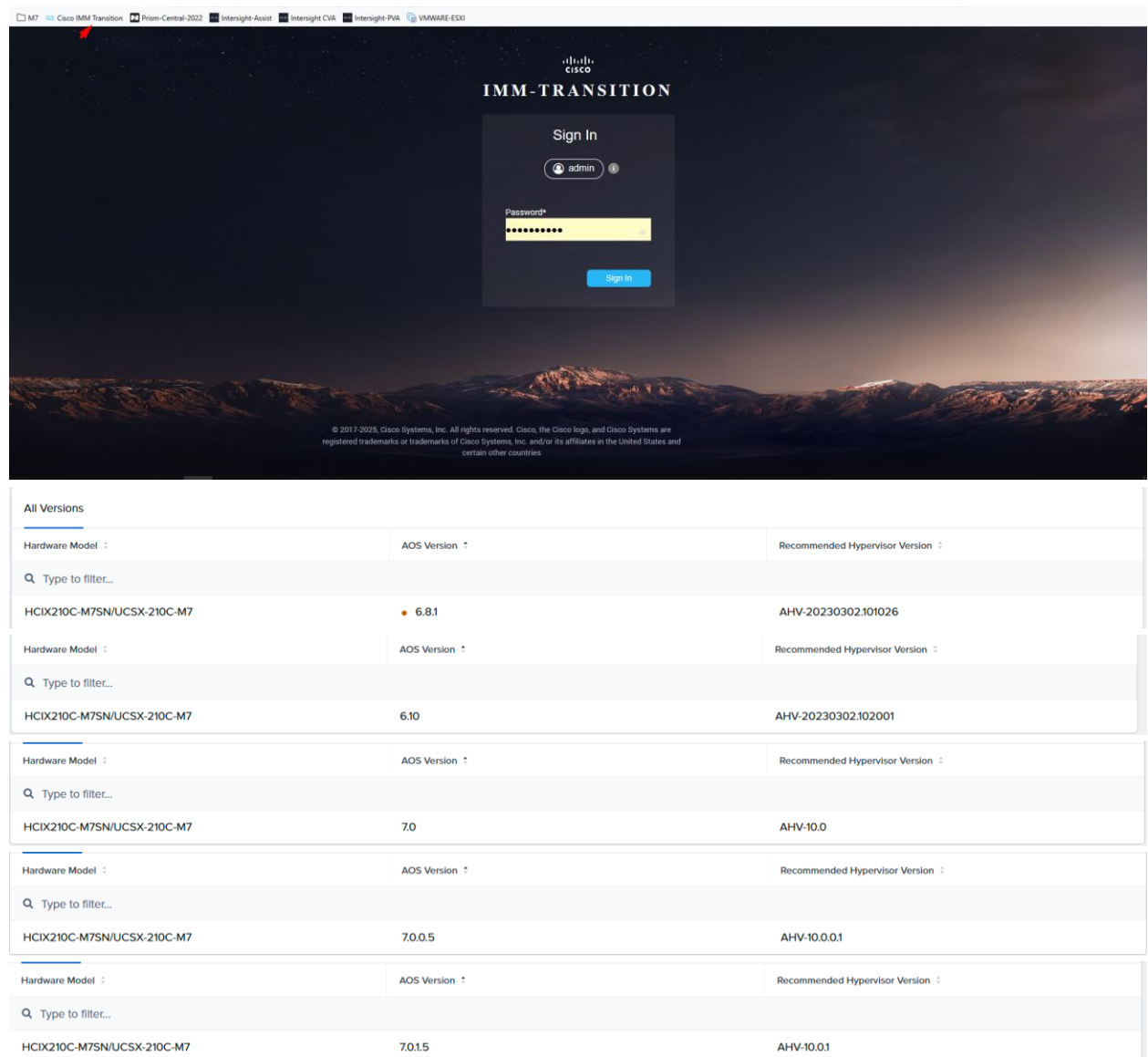


2. Enter the required information and click Next
 - Name = IMM
 - Intersight Organization= Intersight-Nutanix



3. Login into Cisco IMM Transition tool, the access is located on the bookmark, In the Application click in Software Repository, you will see some folders for different versions, for this deployment, we offer the option to select one of the Nutanix AOS and the hypervisor which is supported by the servers, check the versions and select the one you like to deploy using AOS with AHV hypervisor or check below the options of VMware ESXi

Check the versions here <https://portal.nutanix.com/page/compatibility-interoperability-matrix>



IMM-TRANSITION

Sign In

admin

Password*

Sign In

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Hardware Model	AOS Version	Recommended Hypervisor Version
HCIX210C-M7SN/UCSX-210C-M7	6.8.1	AHV-20230302.101026
HCIX210C-M7SN/UCSX-210C-M7	6.10	AHV-20230302.102001
HCIX210C-M7SN/UCSX-210C-M7	7.0	AHV-10.0
HCIX210C-M7SN/UCSX-210C-M7	7.0.0.5	AHV-10.0.0.1
HCIX210C-M7SN/UCSX-210C-M7	7.0.1.5	AHV-10.0.1

All Versions		
Hardware Model	AOS Version	Recommended Hypervisor Version
Q Type to filter...		
HCIX210C-M7SN/UCSX-210C-M7	6.8.1	ESXi 8.0 U2 Vendor
HCIX210C-M7SN/UCSX-210C-M7	6.8.1	ESXi 7.0 U3o Vendor
HCIX210C-M7SN/UCSX-210C-M7	6.8.1	ESXi 7.0 U3s
HCIX210C-M7SN/UCSX-210C-M7	6.8.1	ESXi 8.0 U2d

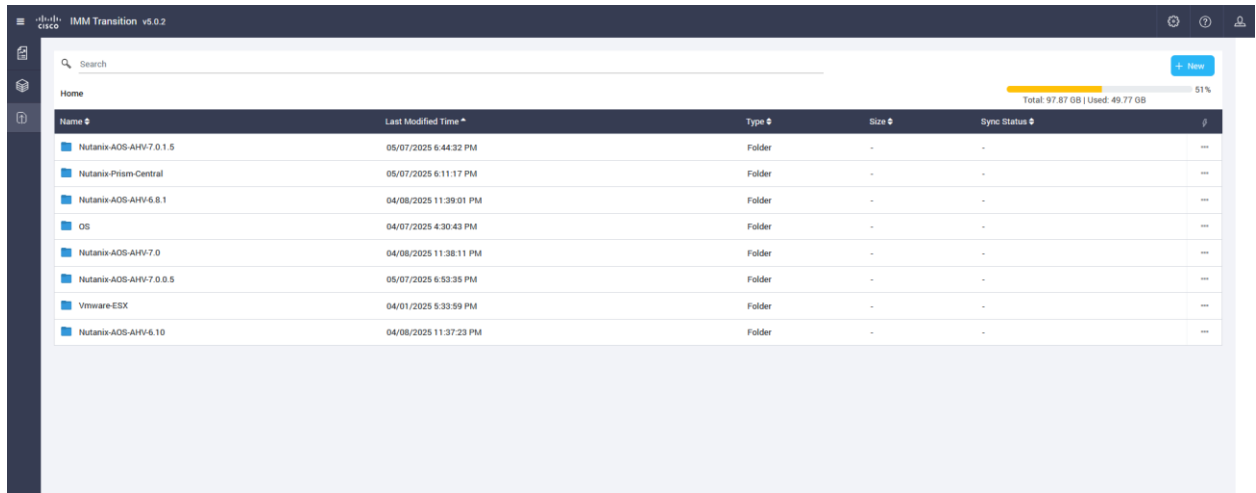
Hardware Model	AOS Version	Recommended Hypervisor Version
Q Type to filter...		
HCIX210C-M7SN/UCSX-210C-M7	6.10	ESXi 8.0 U2 Vendor
HCIX210C-M7SN/UCSX-210C-M7	6.10	ESXi 7.0 U3o Vendor
HCIX210C-M7SN/UCSX-210C-M7	6.10	ESXi 8.0 U3 Vendor
HCIX210C-M7SN/UCSX-210C-M7	6.10	ESXi 8.0 U3d
HCIX210C-M7SN/UCSX-210C-M7	6.10	ESXi 8.0 U2d
HCIX210C-M7SN/UCSX-210C-M7	6.10	ESXi 7.0 U3s

Hardware Model	AOS Version	Recommended Hypervisor Version
Q Type to filter...		
HCIX210C-M7SN/UCSX-210C-M7	7.0	ESXi 8.0 U2 Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0	ESXi 7.0 U3o Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0	ESXi 8.0 U3 Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0	ESXi 8.0 U3d
HCIX210C-M7SN/UCSX-210C-M7	7.0	ESXi 8.0 U2d
HCIX210C-M7SN/UCSX-210C-M7	7.0	ESXi 7.0 U3s

Hardware Model	AOS Version	Recommended Hypervisor Version
Q Type to filter...		
HCIX210C-M7SN/UCSX-210C-M7	7.0.0.5	ESXi 8.0 U2 Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0.0.5	ESXi 7.0 U3o Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0.0.5	ESXi 8.0 U3 Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0.0.5	ESXi 8.0 U3d
HCIX210C-M7SN/UCSX-210C-M7	7.0.0.5	ESXi 8.0 U2d
HCIX210C-M7SN/UCSX-210C-M7	7.0.0.5	ESXi 7.0 U3s

Hardware Model	AOS Version	Recommended Hypervisor Version
Q Type to filter...		
HCIX210C-M7SN/UCSX-210C-M7	7.0.1.5	ESXi 8.0 U3 Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0.1.5	ESXi 8.0 U2 Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0.1.5	ESXi 7.0 U3o Vendor
HCIX210C-M7SN/UCSX-210C-M7	7.0.1.5	ESXi 8.0 U3d
HCIX210C-M7SN/UCSX-210C-M7	7.0.1.5	ESXi 8.0 U2d
HCIX210C-M7SN/UCSX-210C-M7	7.0.1.5	ESXi 7.0 U3s

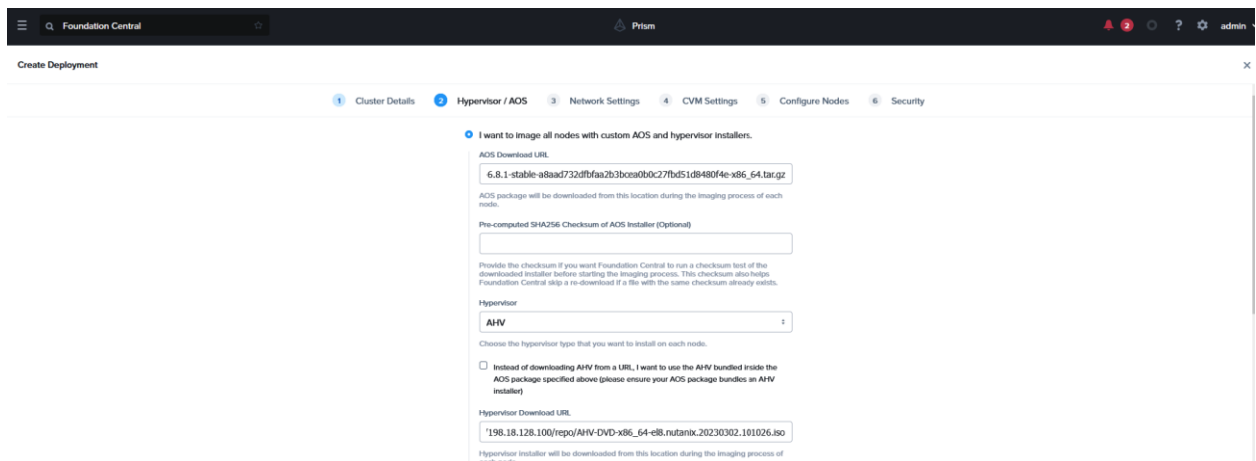
- Choose one of the options to deploy the new cluster, access the desired folder and copy the AOS link, click the Ellipsis at the right side, then select share link



The screenshot shows the Cisco IMM Transition v5.0.2 interface. At the top, there is a search bar and a '+ New' button. Below the search bar, a progress bar indicates 'Total: 97.87 GB | Used: 49.77 GB' with a 51% completion status. The main area displays a table of files and folders.

Name	Last Modified Time	Type	Size	Sync Status	
Nutanix-AOS-AHV-7.0.1.5	05/07/2025 6:44:32 PM	Folder	-	-	---
Nutanix-Prism-Central	05/07/2025 6:11:17 PM	Folder	-	-	---
Nutanix-AOS-AHV-6.8.1	04/08/2025 11:39:01 PM	Folder	-	-	---
OS	04/07/2025 4:30:43 PM	Folder	-	-	---
Nutanix-AOS-AHV-7.0	04/08/2025 11:38:11 PM	Folder	-	-	---
Nutanix-AOS-AHV-7.0.0.5	05/07/2025 6:53:35 PM	Folder	-	-	---
VMware-ESX	04/01/2025 5:33:59 PM	Folder	-	-	---
Nutanix-AOS-AHV-6.10	04/08/2025 11:37:23 PM	Folder	-	-	---

5. Return to Prism Central and paste the copied link into the AOS URL, then from the hypervisor option click on it and select AHV, then return to Cisco IMM to copy the link of AHV, select the ISO and click Next



The screenshot shows the 'Create Deployment' wizard in Foundation Central, specifically the 'Hypervisor / AOS' step. The wizard has six steps: 1. Cluster Details, 2. Hypervisor / AOS (current), 3. Network Settings, 4. CVM Settings, 5. Configure Nodes, and 6. Security.

The main content area shows the following configuration options:

- I want to image all nodes with custom AOS and hypervisor installers.**
 - AOS Download URL:** `6.8.1-stable-a8aad732df8aa2b3b0ca080c27bd51d8480f4e-x86_64.tar.gz`
 - AOS package will be downloaded from this location during the imaging process of each node.**
 - Pre-computed SHA256 Checksum of AOS Installer (Optional):** (Empty field)
 - Provide the checksum if you want Foundation Central to run a checksum test of the downloaded installer before starting the imaging process. This checksum also helps Foundation Central skip a re-download if a file with the same checksum already exists.**
 - Hypervisor:** `AHV`
 - Choose the hypervisor type that you want to install on each node.**
 - ☐ **Instead of downloading AHV from a URL, I want to use the AHV bundled inside the AOS package specified above (please ensure your AOS package bundles an AHV installer)**
 - Hypervisor Download URL:** `198.18.128.100/repos/AHV-DVD-x86_64-el8.nutanix.20230302.101026.iso`
 - Hypervisor installer will be downloaded from this location during the imaging process of each node.**

6. Fill in the networking information
 - a. Gateway = 198.18.128.1
 - b. Netmask = 255.255.192.0 / 18
 - c. Cluster Virtual IP = 198.18.136.2
 - d. Host CVM VLAN = 10

Foundation Central | Prism

Create Cluster

Cluster Details | AOS / Hypervisor | **Network Settings** | CVM Settings | Configure Nodes | Security

Nutanix requires all hosts and CVMs of a cluster to have static IP addresses in the same subnet.

Gateway: 198.18.128.1

Network: 255.255.192.0 (/18)

Cluster Virtual IP (Optional)
This IP will always point to an online CVM, even in case of a node failure.
198.18.136.2

Host and CVM VLAN
If your host CVM subnet has a VLAN configuration, enter the tag below. All packets leaving the hosts and the CVMs will be wrapped with this VLAN tag. This input is required for Interight Managed Mode.
10
Must be an integer between 1 and 4093 (inclusive)

Back Cancel Next

- e. Mac Pool = click the bar and select the existing
- f. IMC Access Configuration
 - IMC Access Type = Out-of-Band
 - Out-of-Band IP Pool = choose the one from the dropdown menu
- g. Click next to continue

Foundation Central | Prism

Create Cluster

Cluster Details | AOS / Hypervisor | Network Settings | **CVM Settings** | Configure Nodes | Security

MAC Pool
It is a collection of MAC addresses that can be allocated to vNICs of a server profile. Select a MAC pool from the MAC pools created in Cisco Interight.
A node requires 2 unique MAC addresses for each adapter.
ntp-cpoc-dng-b16-f6536-mac-pool (64 available)

IMC Access Configuration
To enable IMC access, you should configure In-Band, Out-of-Band, or both.
IMC Access Type: Out-of-Band
Out-of-Band IP Pool: ntp-cpoc-dng-b16-f6536-IP-Pool (25 IPv4 available)
Out-of-Band only supports IPv4, not IPv6.

Back Cancel Next

7. CVM Settings use the following information:

- Timezone= America/New_York

- NTP=198.18.128.1
- DNS= 198.18.133.1
- Click Next

Time Configuration

Time settings apply to every CVM, and also apply to hosts depending on the hypervisor you chose.

Timezone
America/New_York

Only available when you choose to form a cluster, because of technical reasons, Nutanix concluded AHV and ESX don't properly support host timezone.

NTP Servers (Optional)

198.18.128.1

Enter a comma-separated list of IPs or domain names. Applies to the host too if the host is AHV. For ESX, Nutanix concluded it is best to configure host NTP servers in vCenter.

DNS Settings

DNS settings apply to every CVM, and also apply to hosts depending on the hypervisor you chose.

DNS Servers (Optional)

198.18.133.1

Enter a comma-separated list of IPs. Required if any NTP server above is a domain name, or if AOS/Hypervisor package download URLs are using domain names. For ESX, Nutanix concluded it is best to configure host DNS settings in vCenter.

8. To complete the cluster Host IP, CVM IP and Host name use the image below as an example, you can use any IP address from 198.18.136.10 up to 198.18.136.100

UID Pool (Optional)

It is a collection of UID items that can be allocated to server profiles. Select a pool from the UID pools created in Cisco Intersight.

UID Pool (Optional)
Select UID Pool

3 Nodes

Node Serial	Name	Hypervisor IP Set Range	CVM IP Set Range	Host Name Set Range
FCH285277CF	rtp-cpoc-dng-b16-#6536-1-3	198.18.136.10	198.18.136.13	210c-node-1
FCH285277V	rtp-cpoc-dng-b16-#6536-1-1	198.18.136.11	198.18.136.14	210c-node-2
FCH285277DG	rtp-cpoc-dng-b16-#6536-1-2	198.18.136.12	198.18.136.15	210c-node-3

Set up an incremental Hypervisor Hostname for the entire column in the input below.
210c-node-1

Back Cancel Next

9. For the Foundation Central API Key, click the “+ Generate API key”, add a name, select the new API from the drop-down menu and click Submit

1 Cluster Details

2 Hypervisor / AOS

3 Network Settings

4 CVM Settings

5 Configure Nodes

6 Security

Foundation Central API Key

Foundation Central provides an API key to authenticate the remote nodes. It is recommended that a distinct API key be created for each remote site. You can create a new key or select from the existing ones.

Foundation Central API Key [+ Generate API Key](#)

Intersight

[Back](#) [Submit](#)

10. The cluster deployment begins and takes approximately 1h15m to complete

Foundation Central

Prism

admin

It may take up to 15 minutes for a deployment progress to start being reported. [Why?](#)

Viewing 2 deployments Show Only Non-archived

IMM

Deployment in progress

Start Date and Time: 5/7/2025, 04:45 PM

Abort

Phase 1A: Node Preparation

Phase 1B: Node Imaging

Phase 2: Cluster Formation

9% 3 nodes in progress

0% Not started

0% Waiting for Phase 1 to finish

Cluster Details

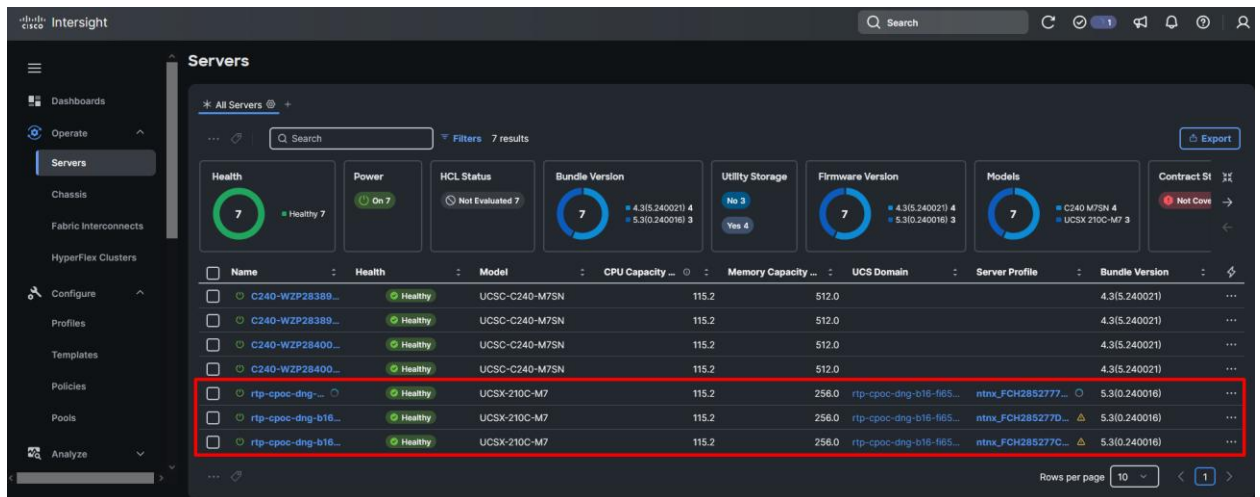
Redundancy Factor	Host CVM Subnet	CVM NTP Servers	AOS Installer URL
2	198.18.128.1 / 255.255.192.0	198.18.128.1	https://198.18.128.100/repo/Nutanix-AOS-AHV-7.0/nutanix_installer_package-release-ganges-7.0-stable-2b790c6331a28149892d8d46287b782761f4906-x86_64.tar.gz
Cluster External IP	Intersight Organization	CVM DNS Servers	Hypervisor Installer URL
198.18.136.2	Intersight-Nutanix	198.18.133.1	https://198.18.128.100/repo/Nutanix-AOS-AHV-7.0/AHV-DVD-x86_64-10.0-808.iso
Host/CVM VLAN ID	Deployment UUID	LACP	SAC Out of Band IP Pool
10	754e08fa-76e1-48c4-46be-b7da8abc4d47	No	rtp-cpoc-dng-b16-f6536-IP-Pool
MAC Pool	VLAN ID for vNICs	VLAN Pool	
rtp-cpoc-dng-b16-f6536-mac-pool	10	Not provided	

3 Nodes in This Deployment

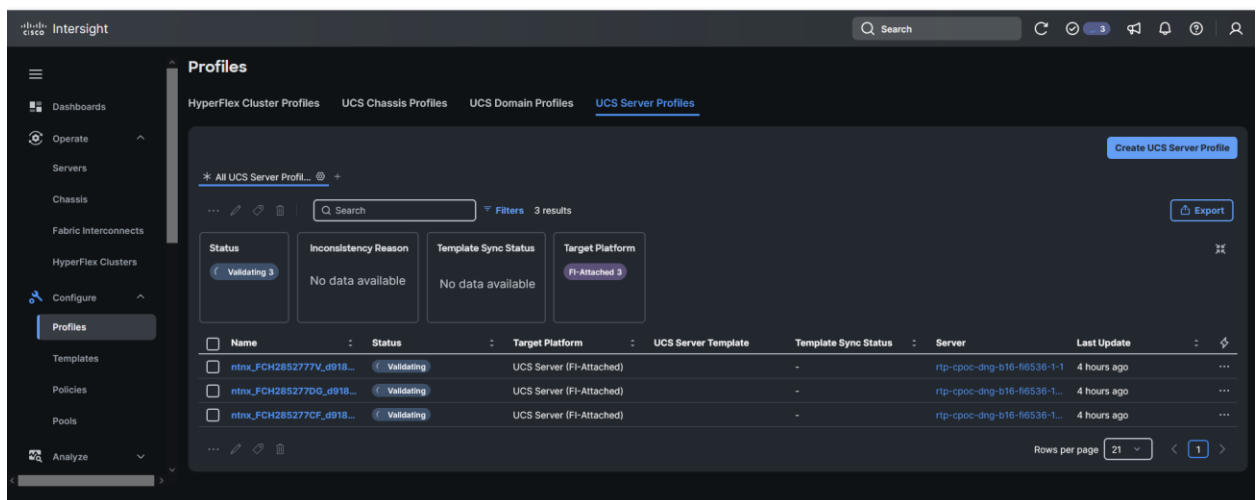
Node Serial	Block Serial *	Position	Progress of Phase 1	Status	Host IP	CVM IP	Host Type	Host Name
FCH285277V	FOX2907P2ZC	A	4%	[NodeConfiguration] Deploying and activating the profile of the Node	198.18.136.11	198.18.136.14	AHV	210c-node-2
FCH285277DG	FOX2907P2ZC	B	4%	[NodeConfiguration] Deploying and activating the profile of the Node	198.18.136.12	198.18.136.15	AHV	210c-node-3
FCH285277CF	FOX2907P2ZC	C	2%	[NodeConfiguration] Turning on the node	198.18.136.10	198.18.136.13	AHV	210c-node-1

[View Original Configuration](#)

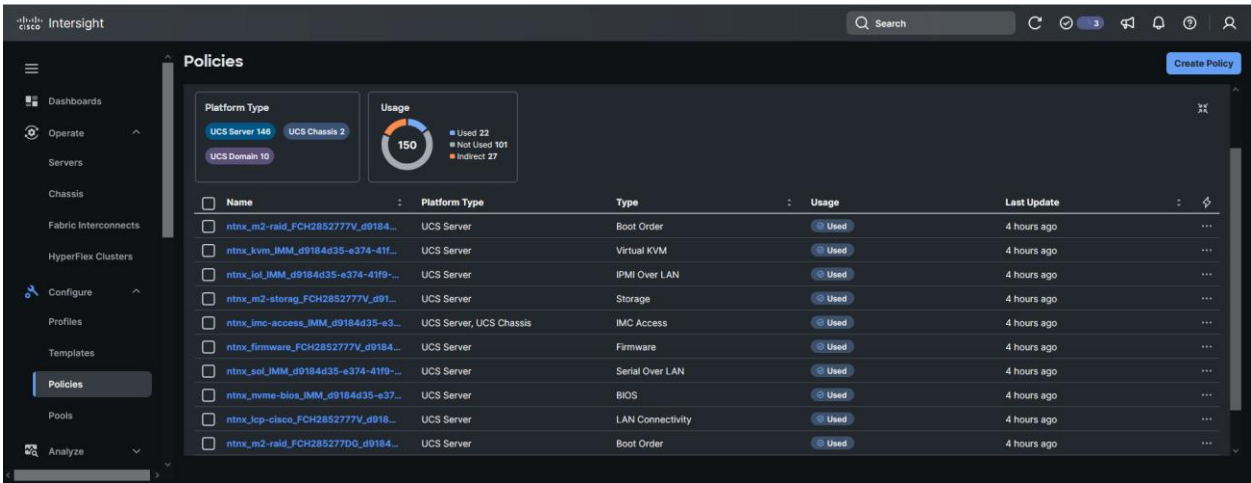
11. Return to Intersight, under Operate select Servers to check the Server Profile provisioning progress.



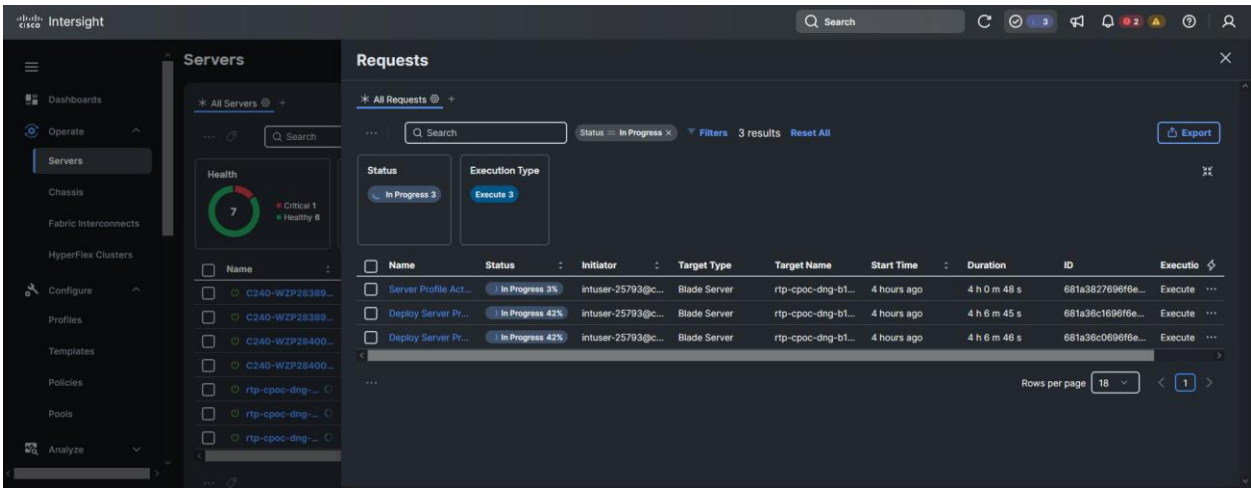
12. Let's explore Intersight, under Configure select Profiles > UCS Server Profiles. Note the profiles are being created/configured under Activating status.



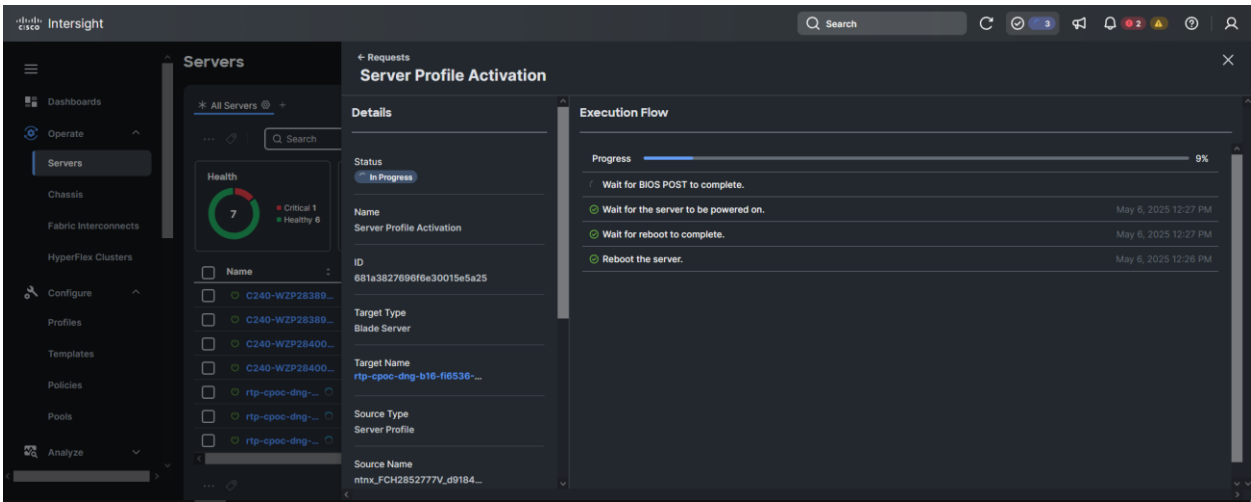
13. In Configure > Policies, note all the UCS Server policies created as part of the automated process between Foundation Central and Intersight.



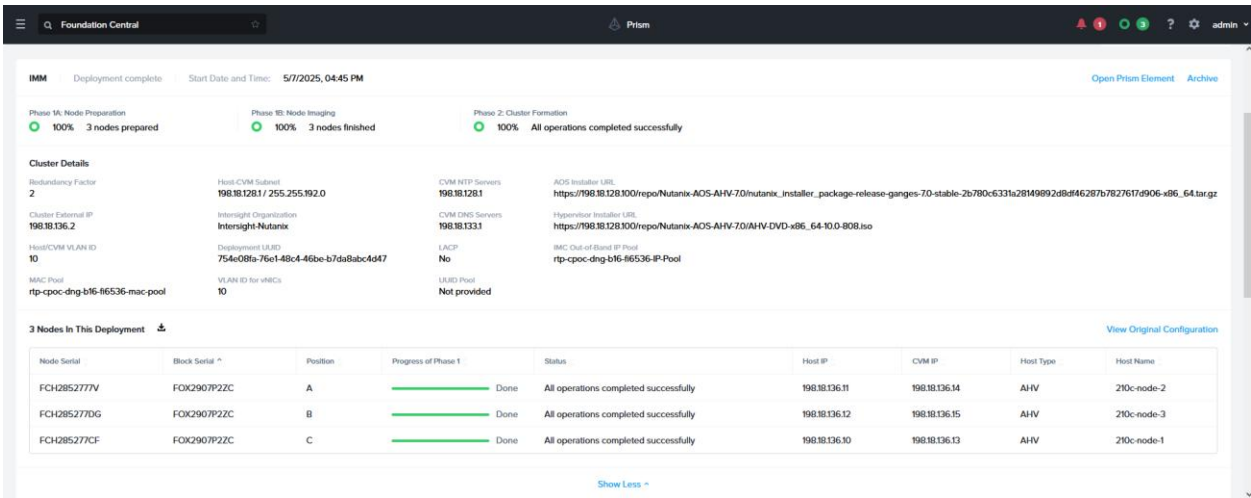
14. In the top right of the screen, click the Requests icon to monitor the undergoing tasks.



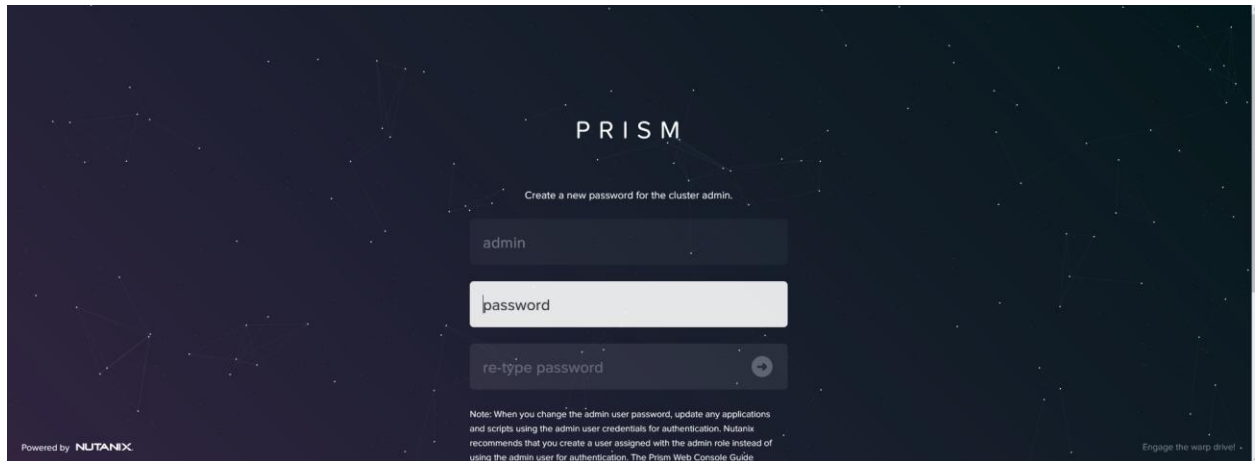
15. Click on any of the tasks to check the progress of the server configuration



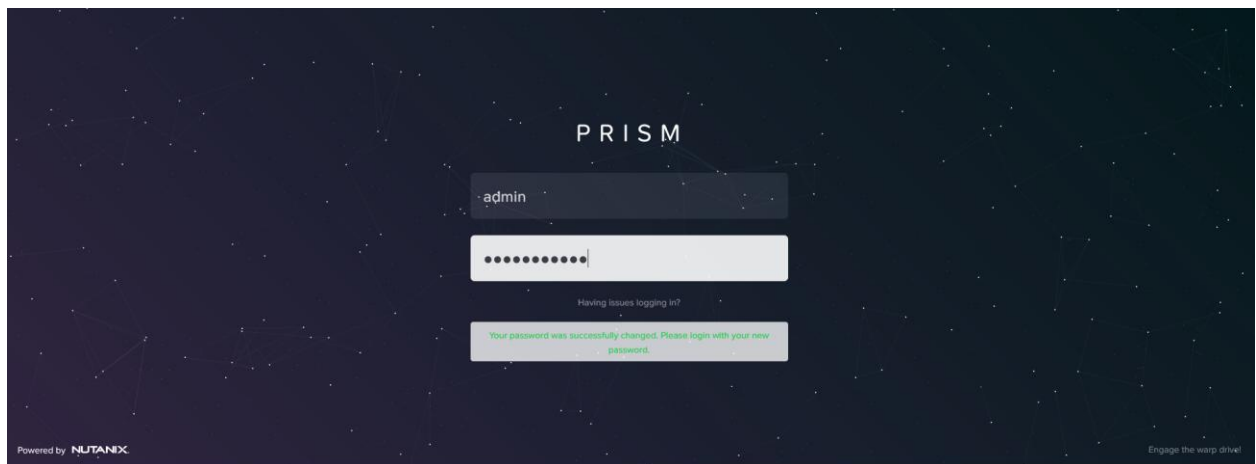
16. Return to Prism Central to monitor the cluster deployment progress.
17. Once the cluster deployment is completed, click **“Open Prism Element”**



18. On Prism Element UI, login with user **“admin”** (password: **Nutanix/4u**) and change the password to **C1sco12345!**



19. After the password has been successfully changed, login with user **admin** and password **C1sco12345!**



20. Complete the Nutanix EULA,

- Name = Cpoc
- Company = Cisco
- Job title = IMM
- Check the box and click accept and continue

Nutanix End User License Agreement (EULA) and Terms of Use

Nutanix is referred to in this Agreement as "Us", "We" or "Our" and "Your organization" is referred to as "You" or "Your". This Nutanix License and Services Agreement ("Agreement") governs Your use of any Products (as defined below) and becomes effective when You (a) issue a purchase order for the Products; (b) execute or accept this Agreement; or (c) access, download, install, or use the Software or Cloud Services. This Agreement is entered into between You and Nutanix Inc., located at 1740 Technology Dr. Ste. 150, San Jose, CA 95110, United States, if You are contracting in the Americas, or Nutanix Netherlands B.V., located at Mercuriusplein 1, 2132 HA Hoofddorp, The Netherlands, if You are contracting anywhere else in the world or both if applicable. You represent and warrant that You have the authority to enter into this Agreement on behalf of Your organization.

1. Definitions.

1.1. "Affiliates" means any corporation or other business entity which controls, is controlled by or is under common control through the ownership of more than fifty percent (50%) of the

☒ I have read and agree to the terms and conditions.

Decline
Accept

21. Keep the Pulse option by default, click Continue

Pulse will be enabled

Pulse continuously monitors cluster health and periodically sends usage data to Nutanix's cloud based analytics engine. It automatically takes action when a problem has occurred, or is about to occur. [Learn More](#)

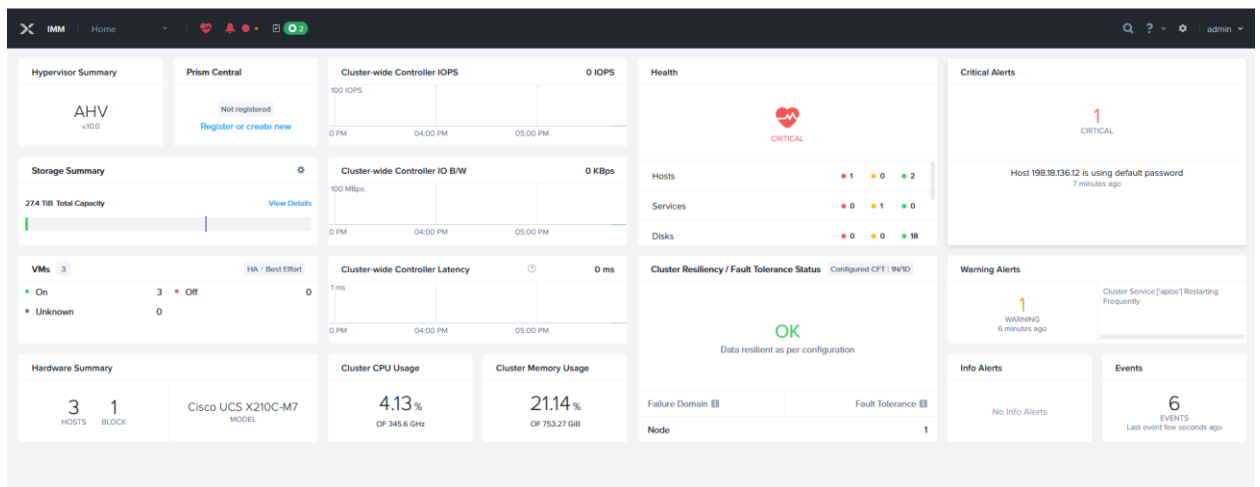
Nutanix strongly recommends not disabling this feature to improve your support experience in the event of a failure or critical system issue.

If you disable Pulse, Nutanix will not be able to proactively reach out to you in the event of failures, and your issue resolution time may increase significantly.

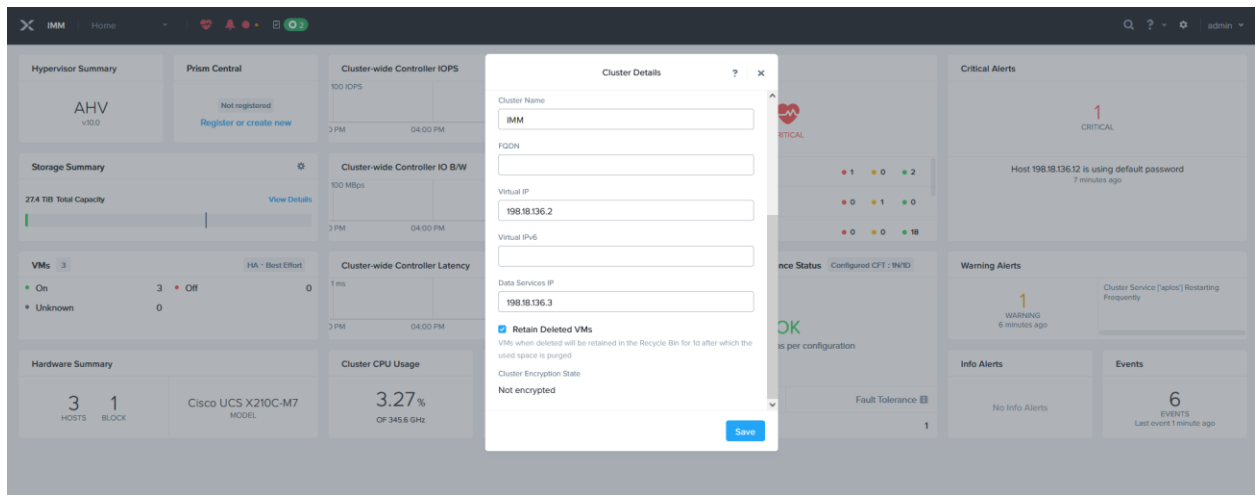
☒ **Disable Pulse. (not recommended)**
Pulse will not be enabled on this cluster.

Continue

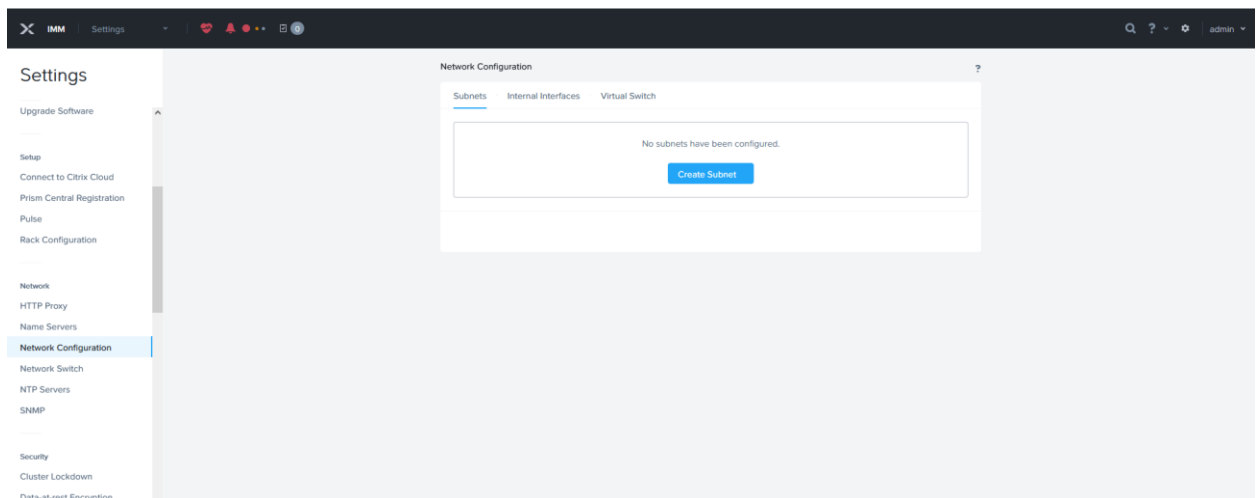
22. You should be taken to the main dashboard



23. Click the cluster name on the top left ("IMM") next to the X icon, add 198.18.136.3 in the **Data Service IP** field and click Save

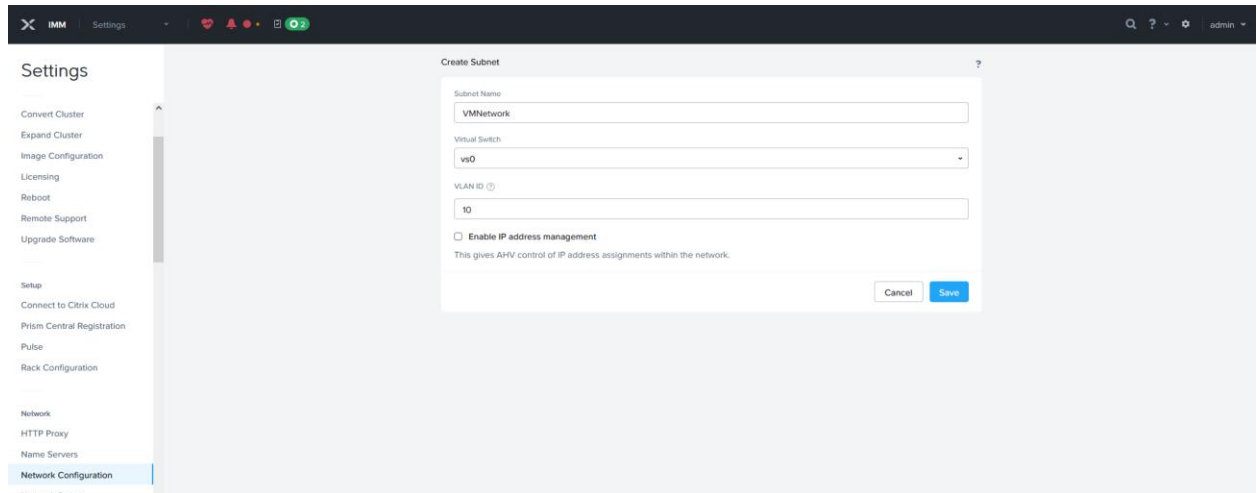


24. Let' s start with some basic configuration, click **Home** at the top and select **Settings** then **Network Configuration** and click "Create Subnet"

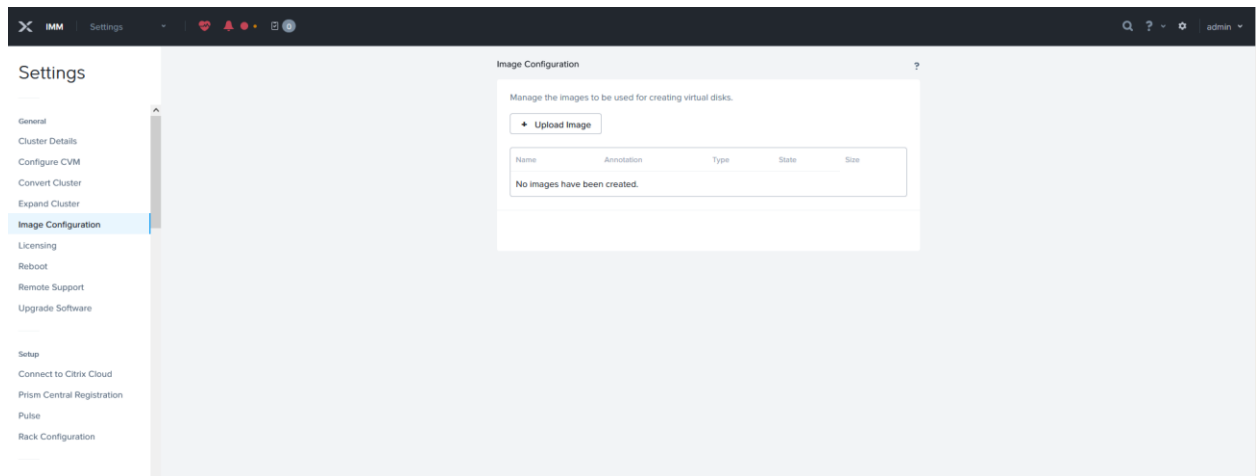


25. On "Create Subnet" use the following information

- Subnet Name = VMNetwork
- Virtual Switch= vs0(default)
- VLAN ID = 10
- Click Save

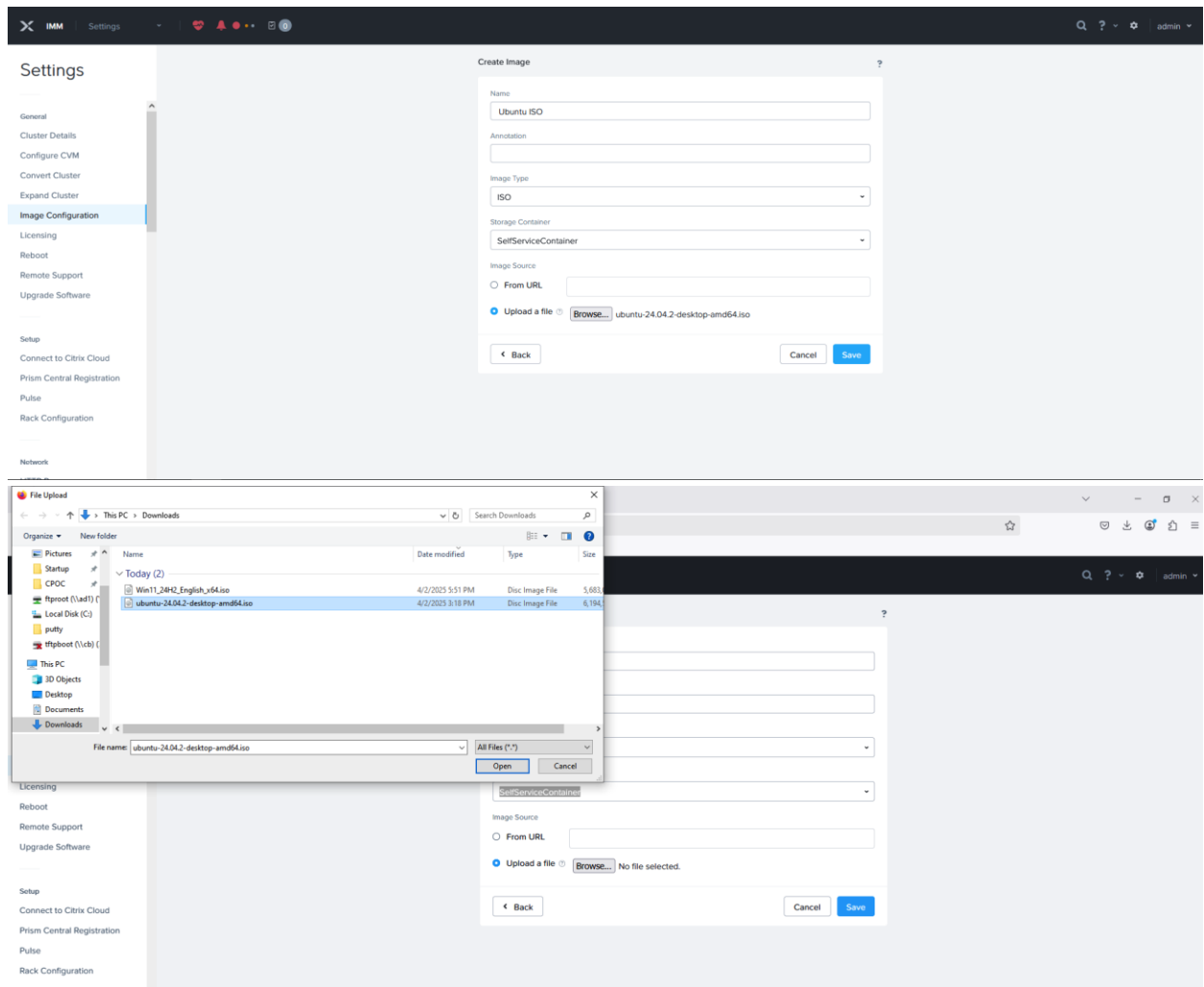


26. Then scroll up and select “Image configuration” and click “+ Upload image” .



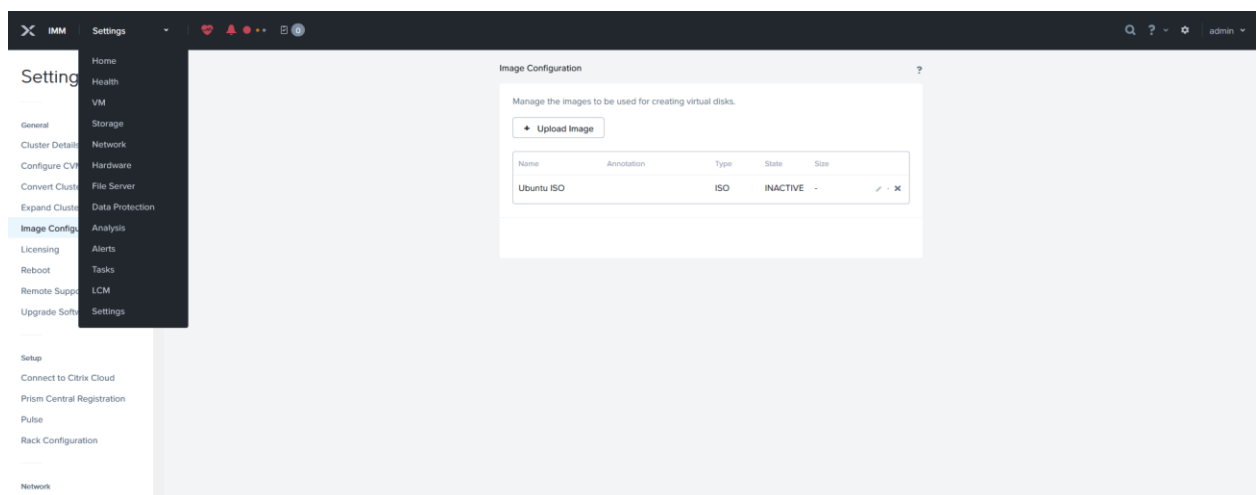
27. In this case we are going to use Ubuntu as an example

- Name = Ubuntu ISO
- Image Type = ISO
- Storage Container = SelfServiceContainer
- Image Source
 - Upload File > Click “Browse...” , from the Downloads folder select Ubuntu ISO
- Click Save

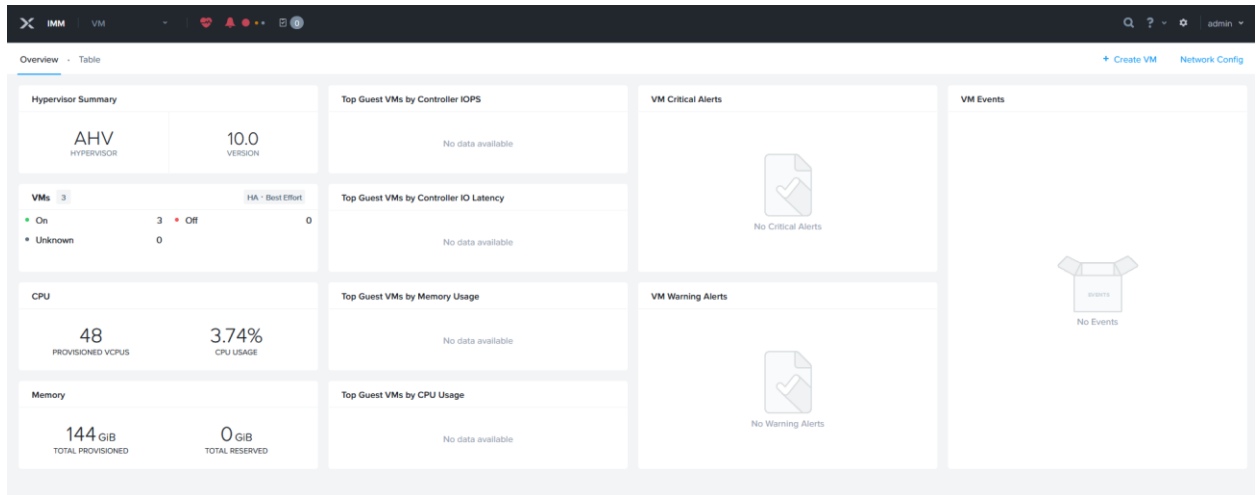


The ISO upload should take a few minutes to complete.

28. Then click the drop-down menu on the top left and select "VM"



29. In the VM dashboard click “+ Create VM” on the top right



30. Let's fill in the new VM information

- Name = Ubuntu VM
- vCPU(s)= 2
- Memory = 4
- Disk
 1. + Add a New Disk
 2. Logical size = 20
 3. Click Add
 4. Click the pencil of CD-ROM
 5. Operation = Clone from Image Service
 6. Image = select Ubuntu from the drop down
 7. Click Update
- Network Adapters (NIC)
 1. + Add New NIC
 2. Subnet Name = VMNetwork
 3. Click Add

- Click Save
- Then Click Table at the top left, next to overview

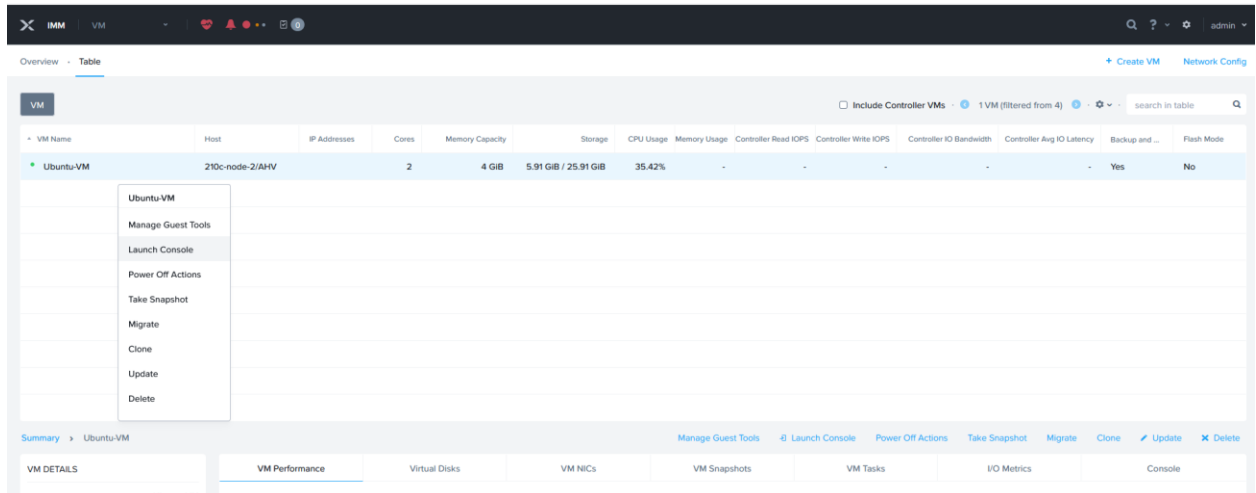
VM Name	Host	IP Addresses	Cores	Memory Capacity	Storage	CPU Usage	Memory Usage	Controller Read IOPS	Controller Write IOPS	Controller IO Bandwidth	Controller Avg IO Latency	Backup and ...	Flash Mode
Ubuntu-VM			2	4 GiB	- / 25.91 GiB	0%	0%	-	-	-	-	Yes	No

Note the new VM recently created

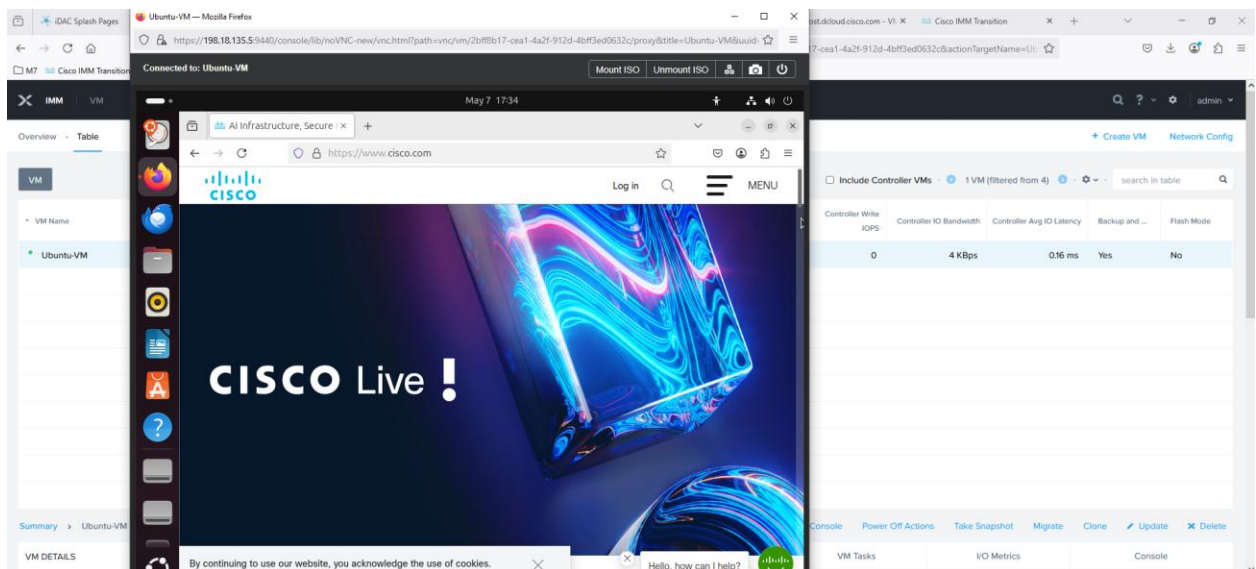
31. Right-click the Ubuntu VM and select Power On

VM Name	Host	IP Addresses	Cores	Memory Capacity	Storage	CPU Usage	Memory Usage	Controller Read IOPS	Controller Write IOPS	Controller IO Bandwidth	Controller Avg IO Latency	Backup and ...	Flash Mode
Ubuntu-VM			2	4 GiB	5.91 GiB / 25.91 GiB	0%	0%	-	-	-	-	Yes	No

32. Right-click the VM again and select Launch Console



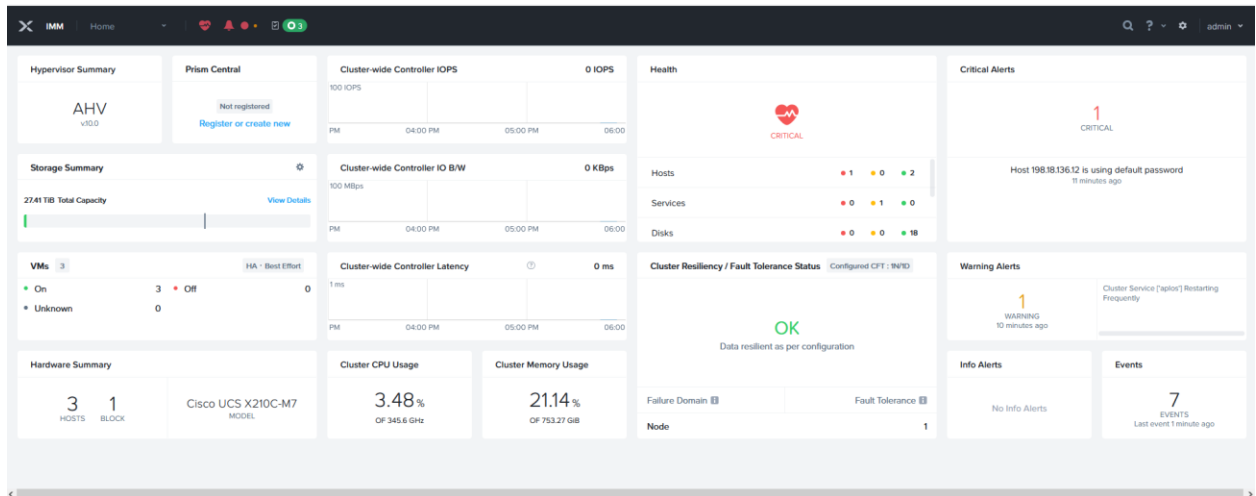
33. On the VM console, close the Installation window and open a Firefox browser and point to "cisco.com" to test VM usability and connectivity to the Internet.



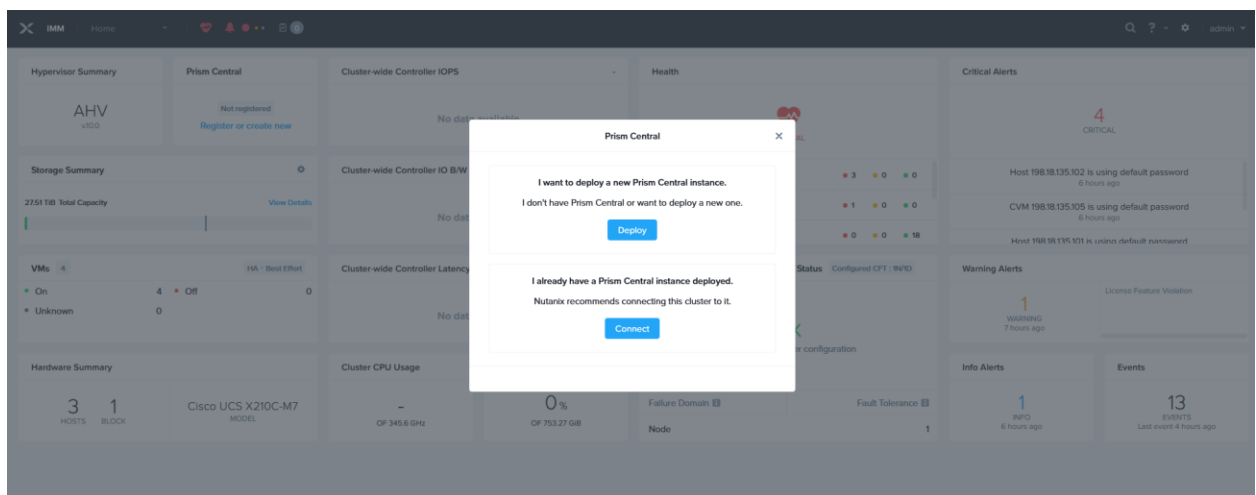
34. This concludes this scenario; you can close the Ubuntu console to continue with the next scenario

Scenario 5. Nutanix Prism Central deployment and Cluster registration

1. In the main dashboard on the Prism Central click in **Register or create new**



2. Select the first option, I want to deploy a new Prism Central Instance



3. Under Installation Image section, select "pc.2024.3.1" and click Next

Prism Central (PC) Deployment

1 PC Version 2 PC Size and Scale 3 Configuration 4 Microservices

Installation Image
Provide the Prism Central image you'd like to install. You can download the latest version from the internet or upload an installation binary from your local machine.
[Learn more about PC Versions.](#)

Available versions
☒ Only show compatible versions [Upload Installation Binary](#)

Version Number	Version Type
☐ pc_2024.3.0.1	Public version
☐ pc_2024.3.0.2	Public version
☒ pc_2024.3.1	Public version
☐ pc_2024.3.1.1	Public version

Cancel Next

- Select "Small" and click next

NOTE: Do NOT select X-Small, this option does not support cluster expansion

Prism Central (PC) Deployment

1 PC Version 2 PC Size and Scale 3 Configuration 4 Microservices

Select Prism Central (PC) Size and Scale
Each size supports given number of VMs and can be used with the minimum number of resources provided below.
[Learn more about PC sizes](#)

X-Small [New](#)

Supports up to 500 VMs | 5 Clusters | 50 hosts

4 vCPU 16 GB Memory 100 GB Disk

Start with the smallest size to explore basic Prism Central capabilities.

Small (S)

Supports up to 2500 VMs | 10 Clusters | 200 hosts

6 vCPU 28 GB Memory 500 GB Disk

☒ Scale-Out PC

Large (L)

Supports up to 12500 VMs | 25 Clusters | 500 hosts

10 vCPU 46 GB Memory 2500 GB Disk

- Add Network information:
 - Network = select existing DataNetwork
 - Subnet Mask = 255.255.192.0
 - Gateway IP = 198.18.128.1
 - DNS = 198.18.133.1
 - NTP = 198.18.128.1
 - Container = SelfServiceContainer

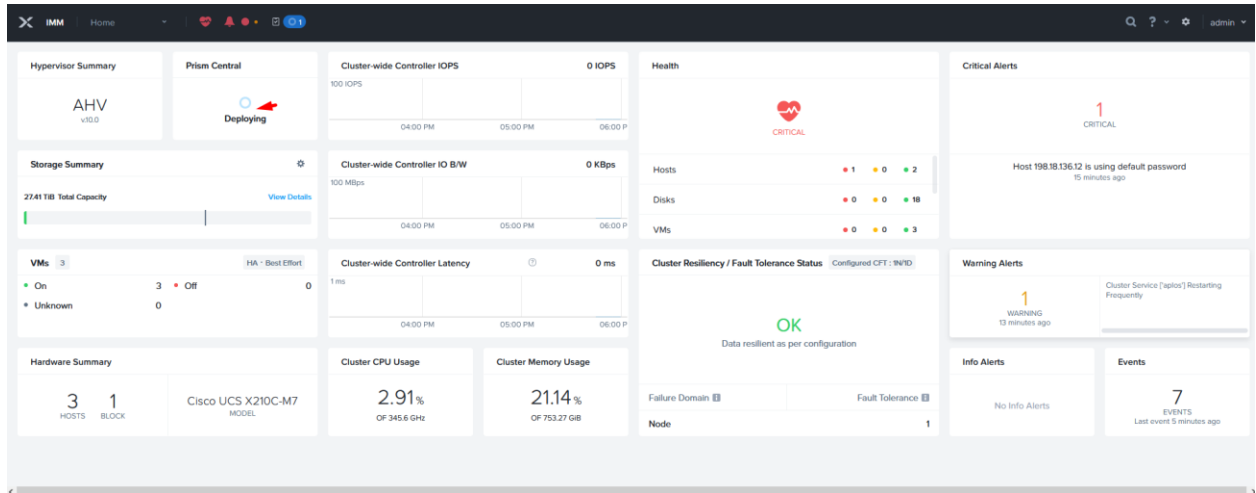
- Under General Details
- VM Name= PC-2024-3-IMM (Suggestion)
- IP Address= 198.18.136.4
- Click Next

The screenshot shows the 'Networking Details' tab of the 'Prism Central (PC) Deployment' configuration window. The 'Network' section includes fields for 'VMNetwork' (set to 'VMNetwork'), 'Subnet Mask' (255.255.192.0), 'Gateway IP' (198.18.128.1), 'DNS Address(es)' (198.18.133.1), and 'NTP Address(es)' (198.18.128.1). The 'Container' section is set to 'SelfServiceContainer'. The 'General Details' section shows 'VM Name' as 'PC-2024-3-IMM' with a resource allocation of '6 vCPUs, 28 GB RAM', and 'IP Address' as '198.18.136.4'. Navigation buttons at the bottom include '< Back', 'Cancel', and 'Next'.

- Prism Central Service Domain Name = prism-central.cluster.local
- Internal Network = Private Network[Default]
- Click Deploy

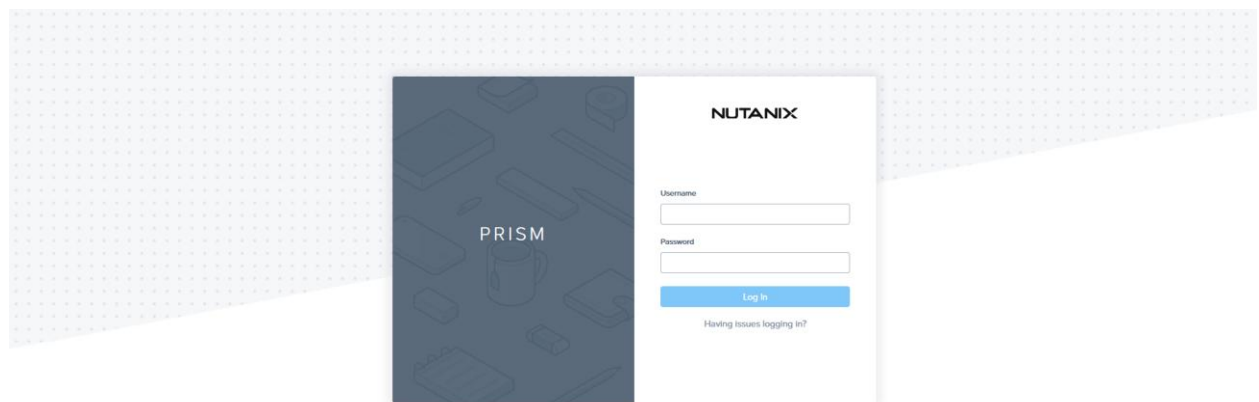
The screenshot shows the 'Microservices' tab of the 'Prism Central (PC) Deployment' configuration window. It prompts the user to 'Please enter the Prism Central domain name, select a network and provide the required inputs to enable the Microservices Infrastructure.' The 'Prism Central Service Domain Name' is set to 'prism-central.cluster.local'. The 'Internal Network' is set to 'Private Network (default)'. Under 'Use default settings (recommended)', the 'Subnet Mask' is 255.255.255.0, the 'Gateway IP Address' is 192.168.5.1, and the 'IP Address Range (5 IPs needed)' is 192.168.5.2 to 192.168.5.64. A warning icon and text state: 'Please ensure that the entered IP address range does not conflict with any reserved IPs (e.g. DHCP IP Pool) in your network.' Navigation buttons at the bottom include '< Back', 'Cancel', and 'Deploy'.

6. After the deployment begins it will take you back to the main dashboard and under the Prism Central widget you will see the deployment activity.

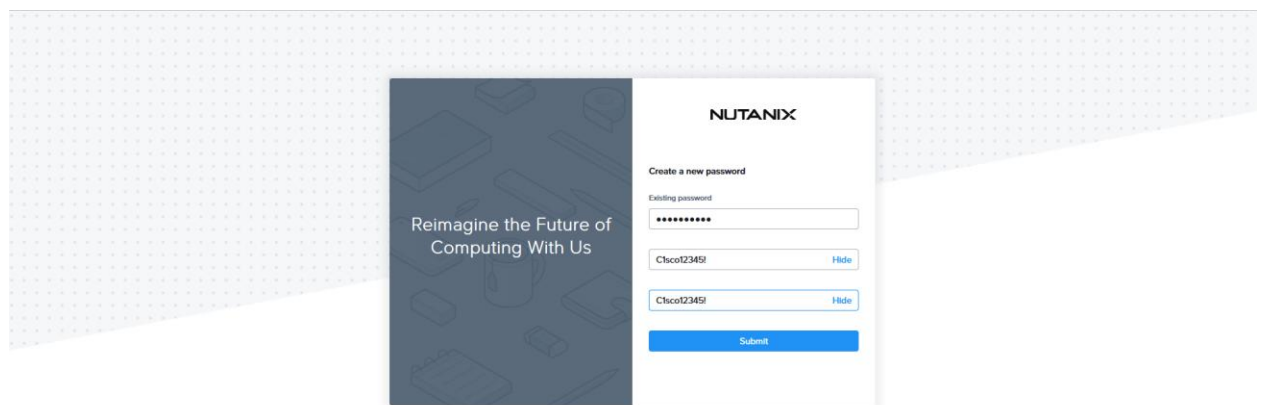
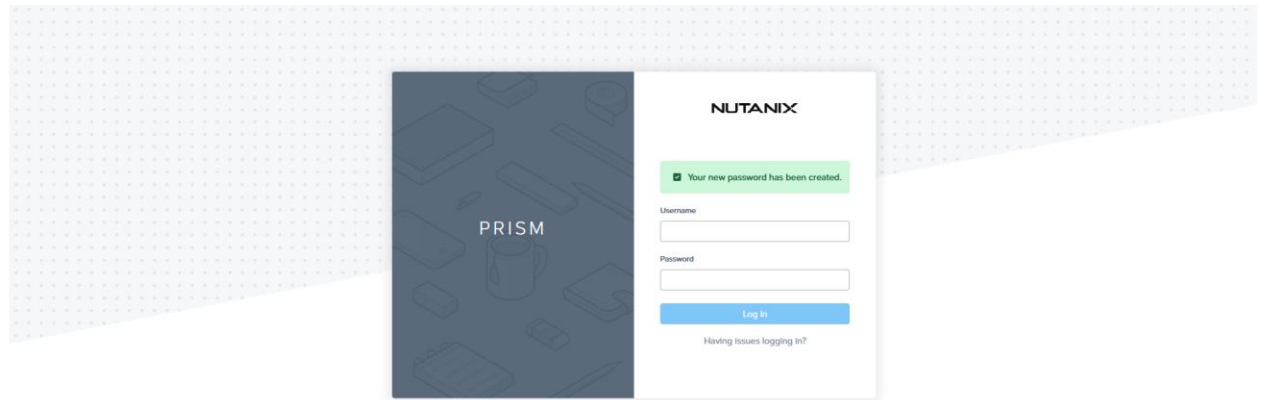


7. Once Prism Central deployment is completed, use a web browser to connect to the virtual IP assigned (198.18.136.4) and change the default

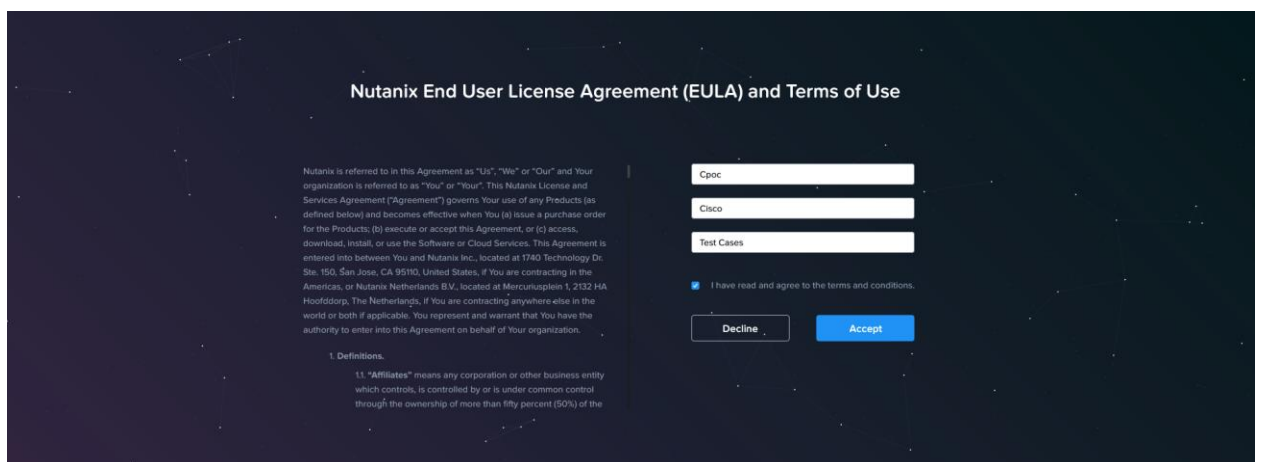
- First time logging in the credentials are:
 - Username = admin
 - Password= Nutanix/4u



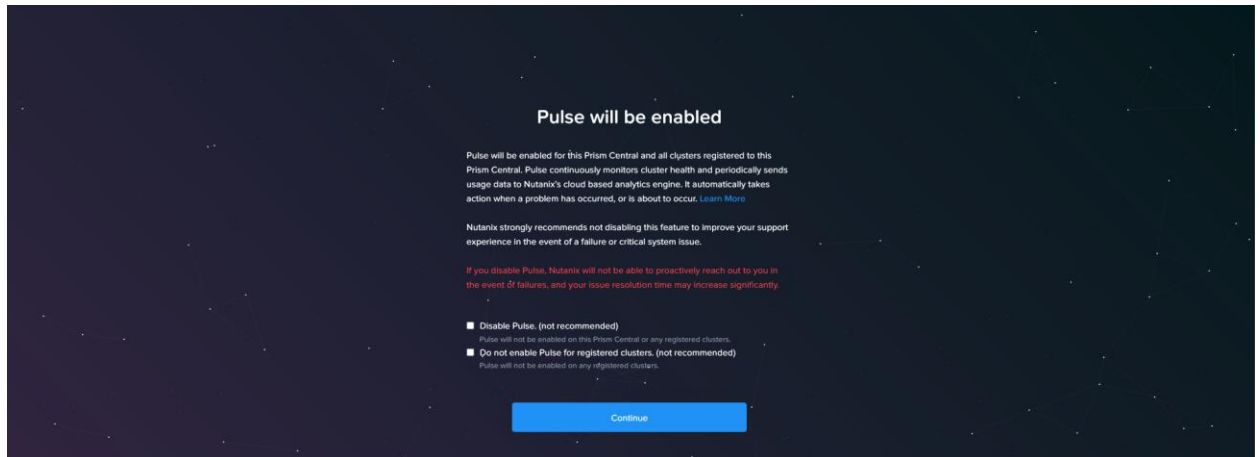
- New password **C1sco12345!**



8. After the password is changed continue with the EULA

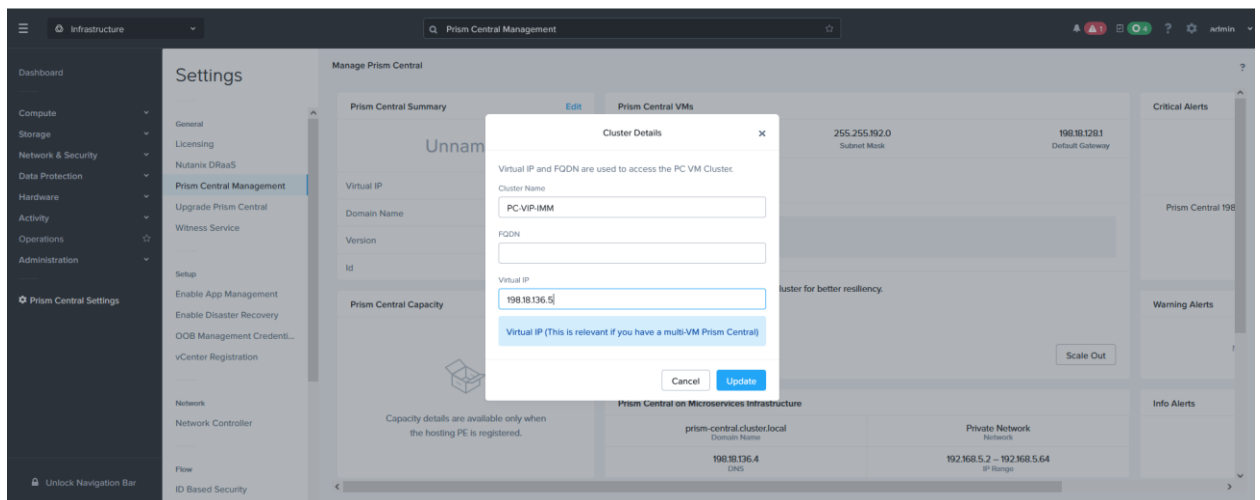


9. Keep Pulse by default and click Continue

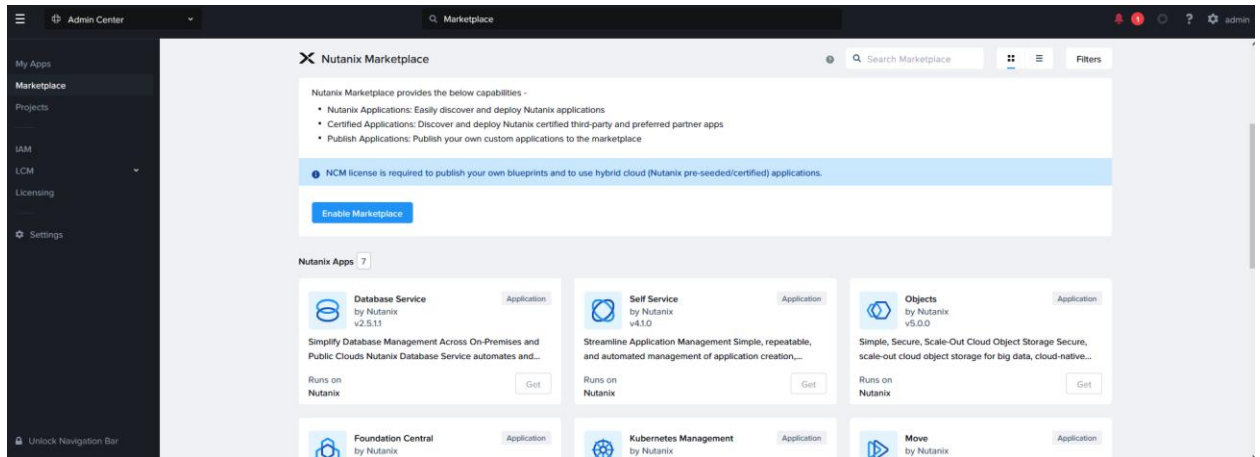


10. At the top select **Infrastructure** from the drop-down, then click on **Prism Central Settings**, select **Prism Central Management** and click edit

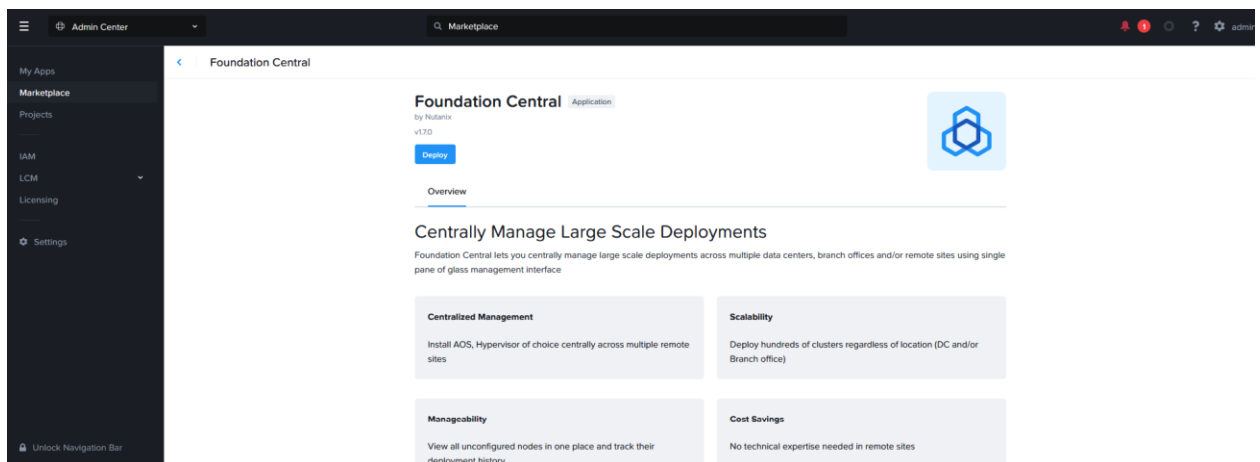
- Cluster name = PC-VIP-IMM
- Virtual IP = 198.18.136.5
- Click Update



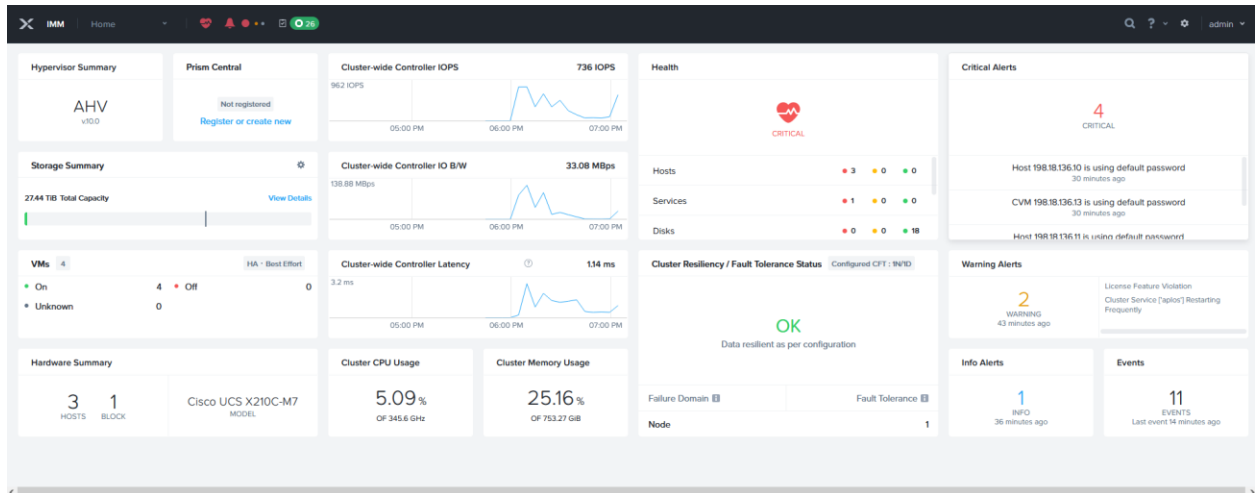
11. From the drop-down menu select **Admin Center** then click Marketplace and click **Enable Marketplace**



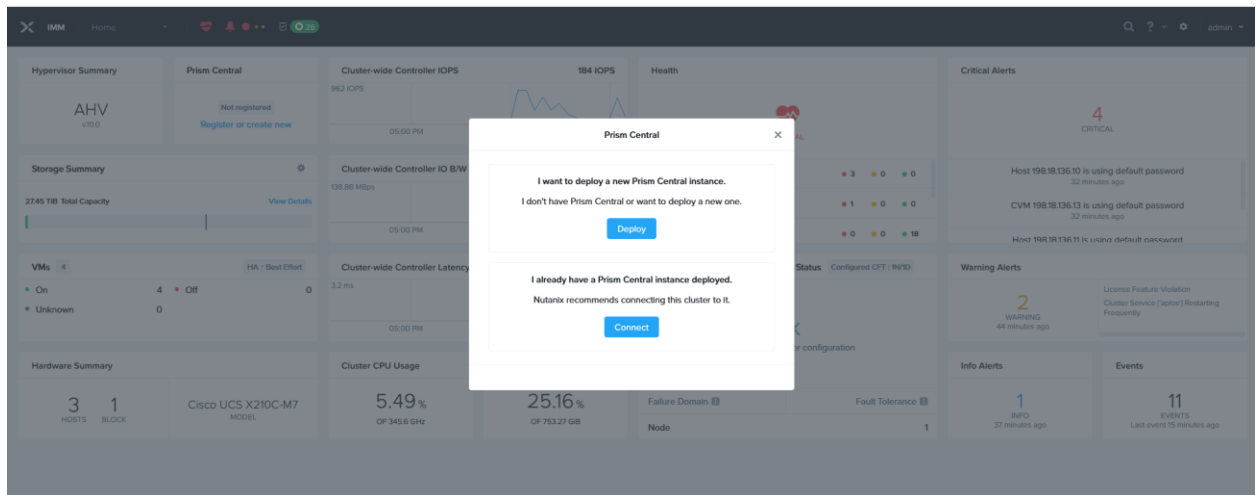
12. Once the Marketplace has been enabled, Click **“Get”** on **Foundation Central** then click **Deploy**



13. While the Foundation Central app is being deployed on Prism Central, return to the Prism Element main dashboard, and confirm then click **“Register or create a new ”**

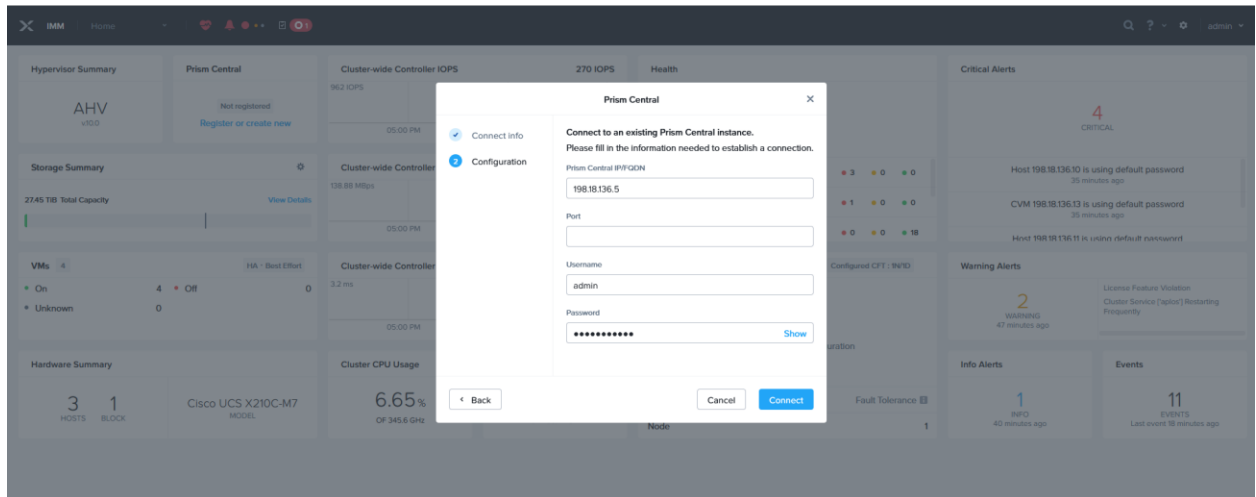


14. Now click **connect** and hit next

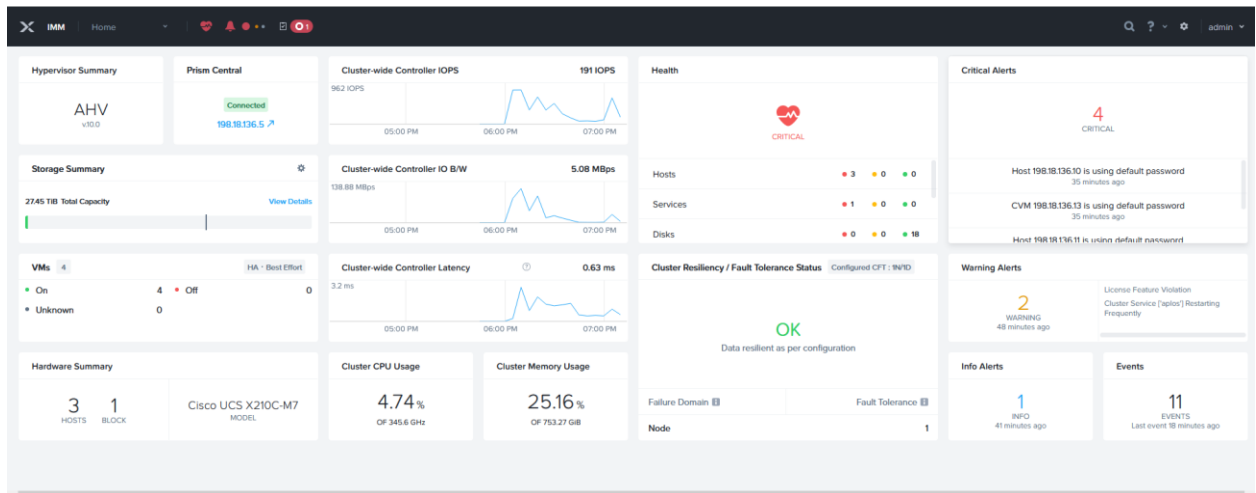


15. On the Prism Central configuration add

- Prism Central IP/FQDN = 198.18.136.5
- Username = admin
- Password = C1sco12345!
- Click Connect



16. The IMM cluster is now registered with and connected to the new Prism Central (PC 2024.3.1)

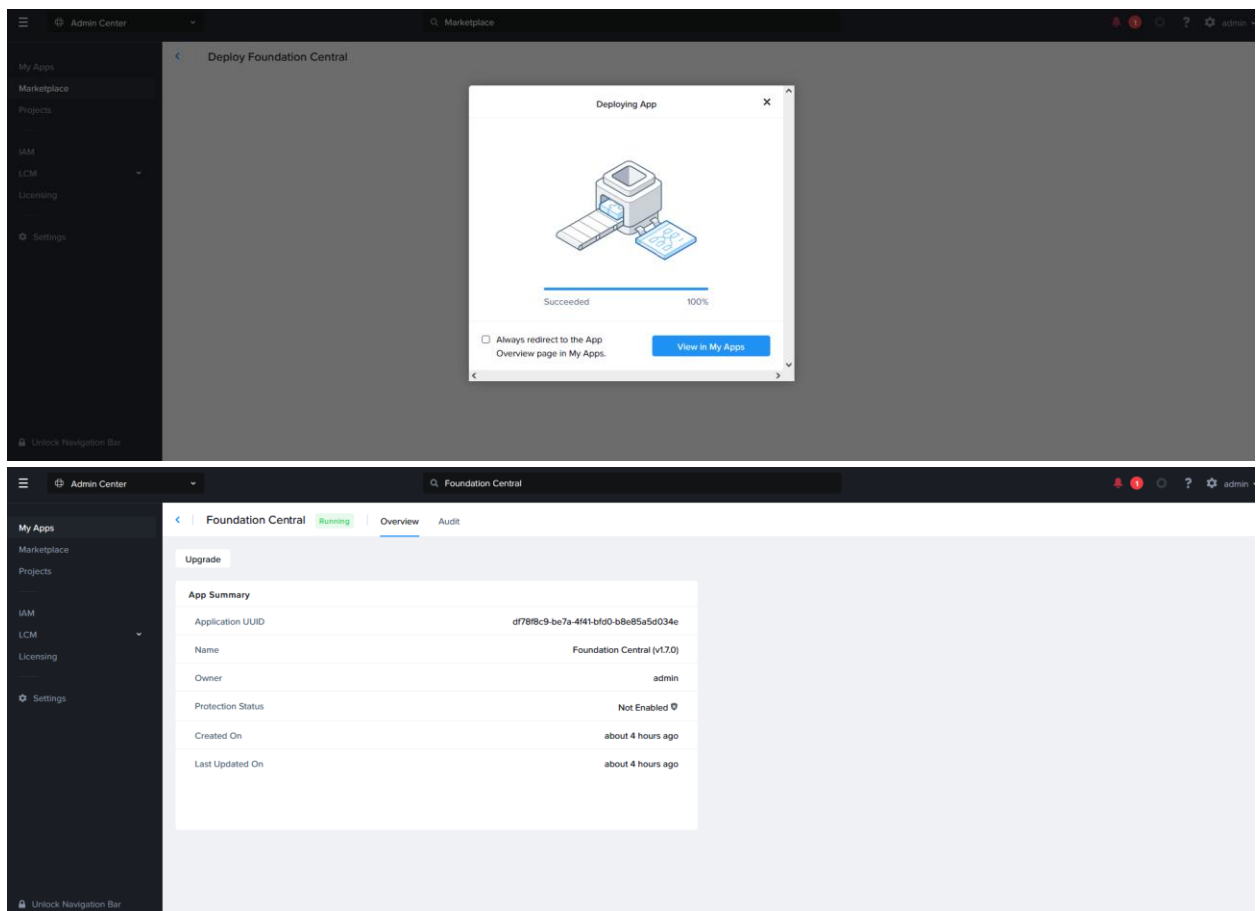


Scenario 6. Nutanix Cluster Expansion

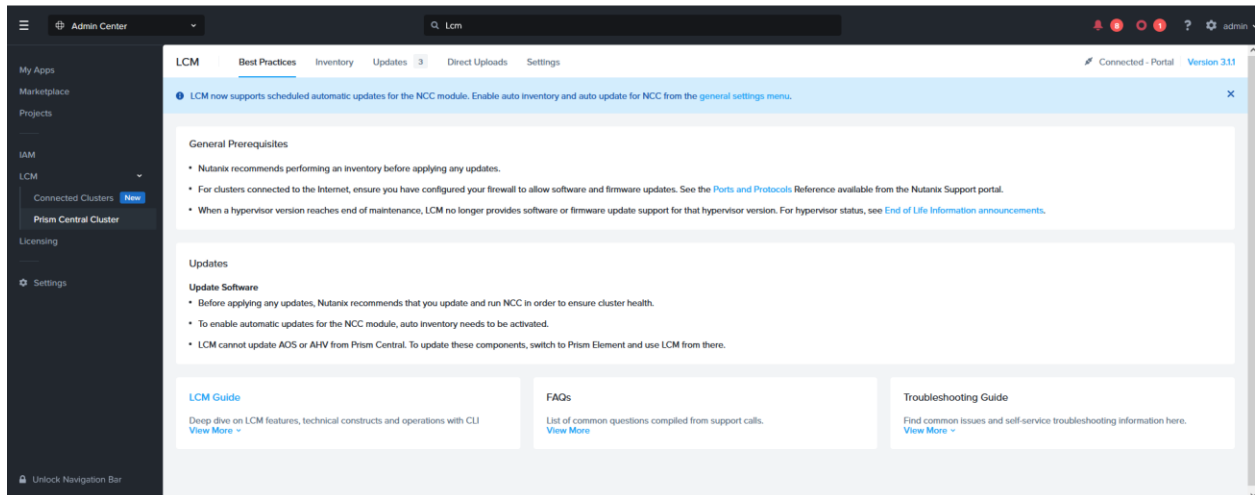
THIS SCENARIO IS JUST AN EXAMPLE OF HOW TO CREATE A CLUSTER EXPANSION, IF REQUIRED TO DO A CLUSTER EXPANSION WILL NEED TO REQUEST AN EXTRA BLADE

THE IMAGES MIGHT NOT BE ACCURATE FOR IMM CLUSTER

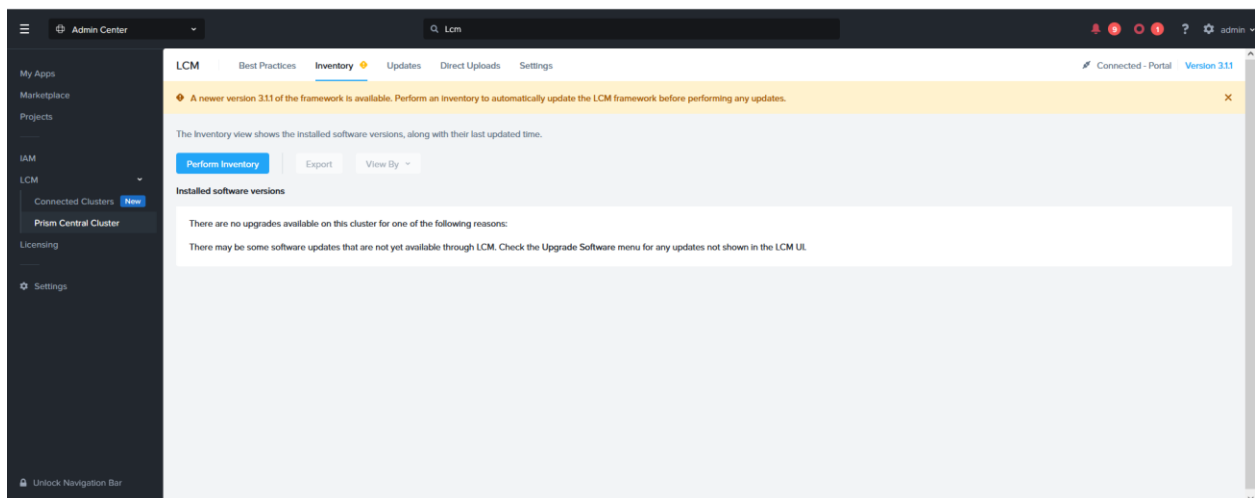
1. Return to the new Prism Central (add IP info here to make sure users will go to PC 2024 and not PC 2022) and click on View in my Apps



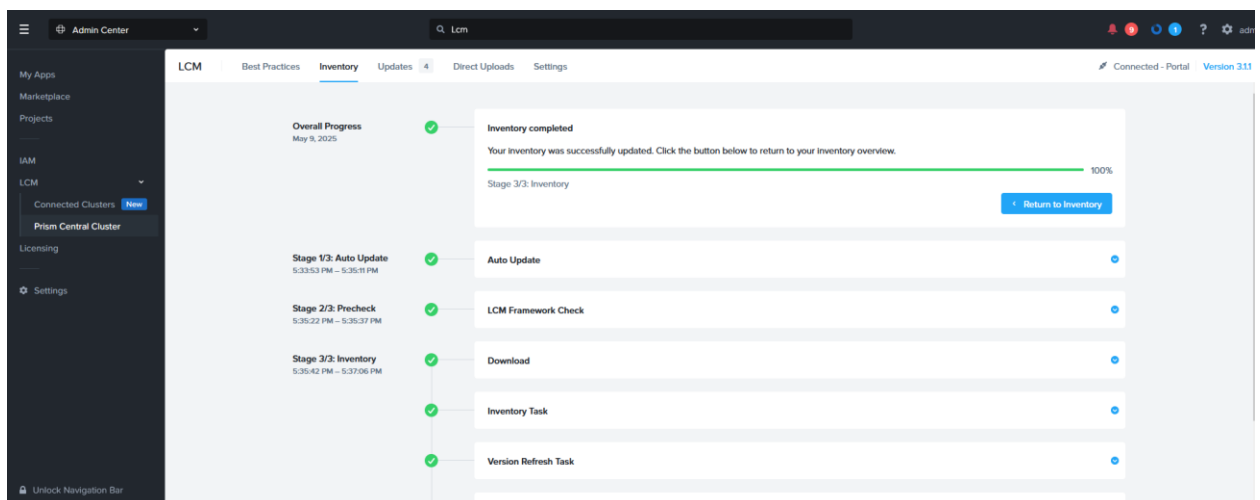
2. Once the App is on running state from **Admin Center > LCM > Prism Central Cluster**

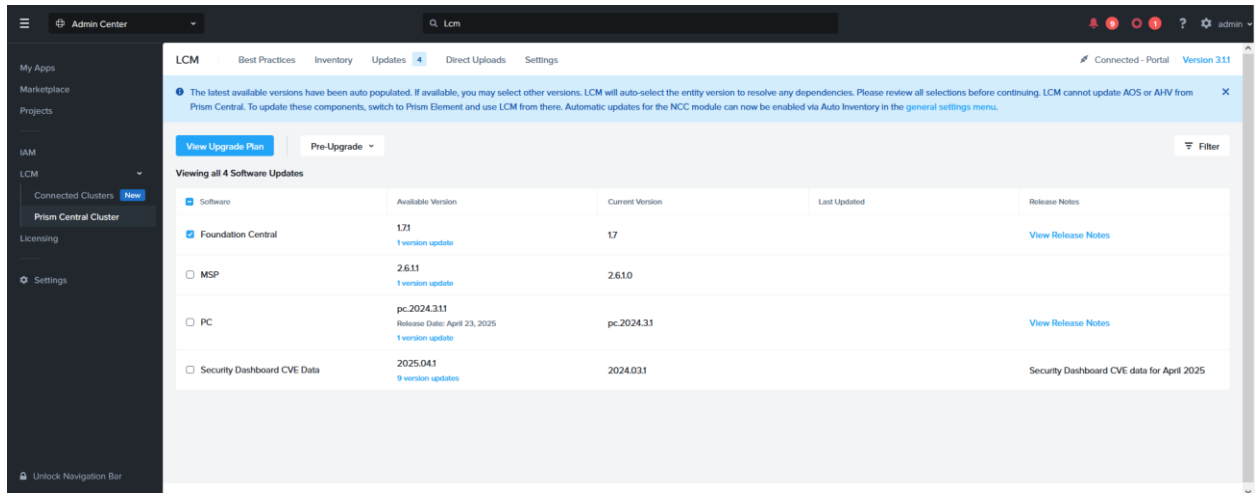


3. Click **inventory** and click **Perform Inventory** > **Proceed**



4. Wait until the Inventory is complete, then press **Return to Inventory**



5. Click Updates, check the box of Foundation Central then click **View Upgrade Plan**

LCM | Best Practices | Inventory | Updates | Direct Uploads | Settings

Connected - Portal | Version 3.11

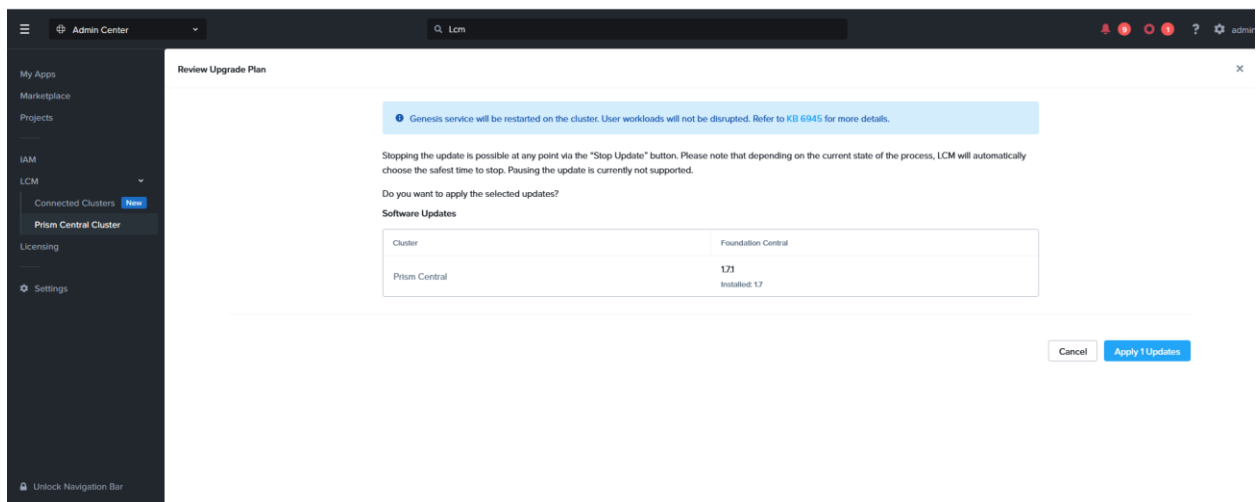
The latest available versions have been auto populated. If available, you may select other versions. LCM will auto-select the entity version to resolve any dependencies. Please review all selections before continuing. LCM cannot update AOS or AI/IV from Prism Central. To update these components, switch to Prism Element and use LCM from there. Automatic updates for the NCC module can now be enabled via Auto Inventory in the general settings menu.

[View Upgrade Plan](#) | [Pre-Upgrade](#) | Filter

Viewing all 4 Software Updates

Software	Available Version	Current Version	Last Updated	Release Notes
☒ Foundation Central	171 1 version update	17		View Release Notes
☐ MSP	2.6.11 1 version update	2.6.10		
☐ PC	pc.2024.3.11 Release Date: April 23, 2025 1 version update	pc.2024.3.1		View Release Notes
☐ Security Dashboard CVE Data	2025.04.1 9 version updates	2024.03.1		Security Dashboard CVE data for April 2025

6. Click Apply 1 Update and wait until the Foundation Central update is complete



Review Upgrade Plan

Genesis service will be restarted on the cluster. User workloads will not be disrupted. Refer to KB 6945 for more details.

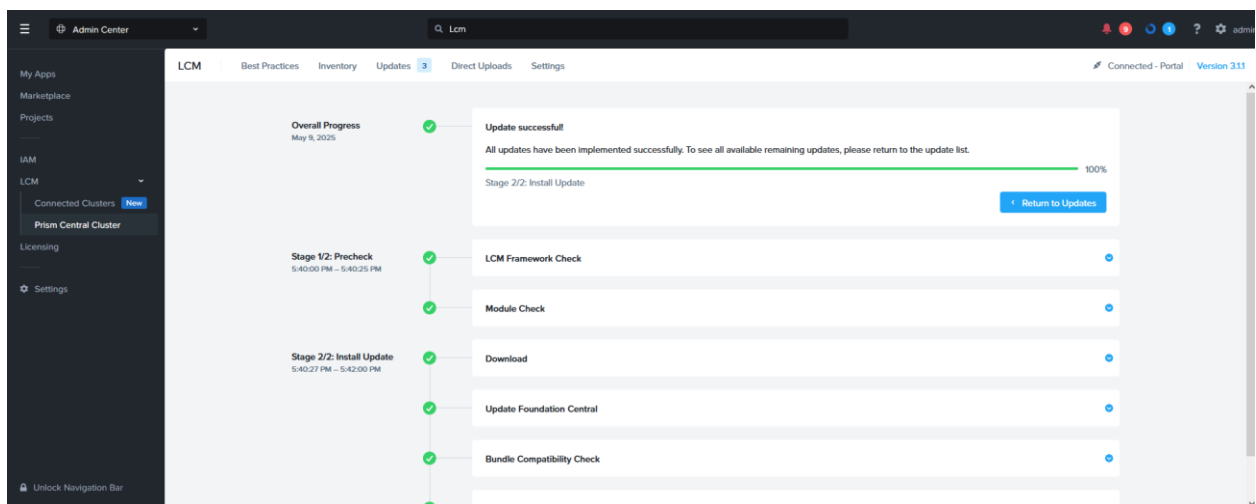
Stopping the update is possible at any point via the "Stop Update" button. Please note that depending on the current state of the process, LCM will automatically choose the safest time to stop. Pausing the update is currently not supported.

Do you want to apply the selected updates?

Software Updates

Cluster	Foundation Central
Prism Central	171 Installed: 17

[Cancel](#) [Apply 1 Updates](#)



LCM | Best Practices | Inventory | Updates | Direct Uploads | Settings

Connected - Portal | Version 3.11

Overall Progress
May 8, 2025

Update successful!
All updates have been implemented successfully. To see all available remaining updates, please return to the update list.

Stage 2/2: Install Update

[Return to Updates](#)

Stage 1/2: Precheck
5:40:00 PM - 5:40:25 PM

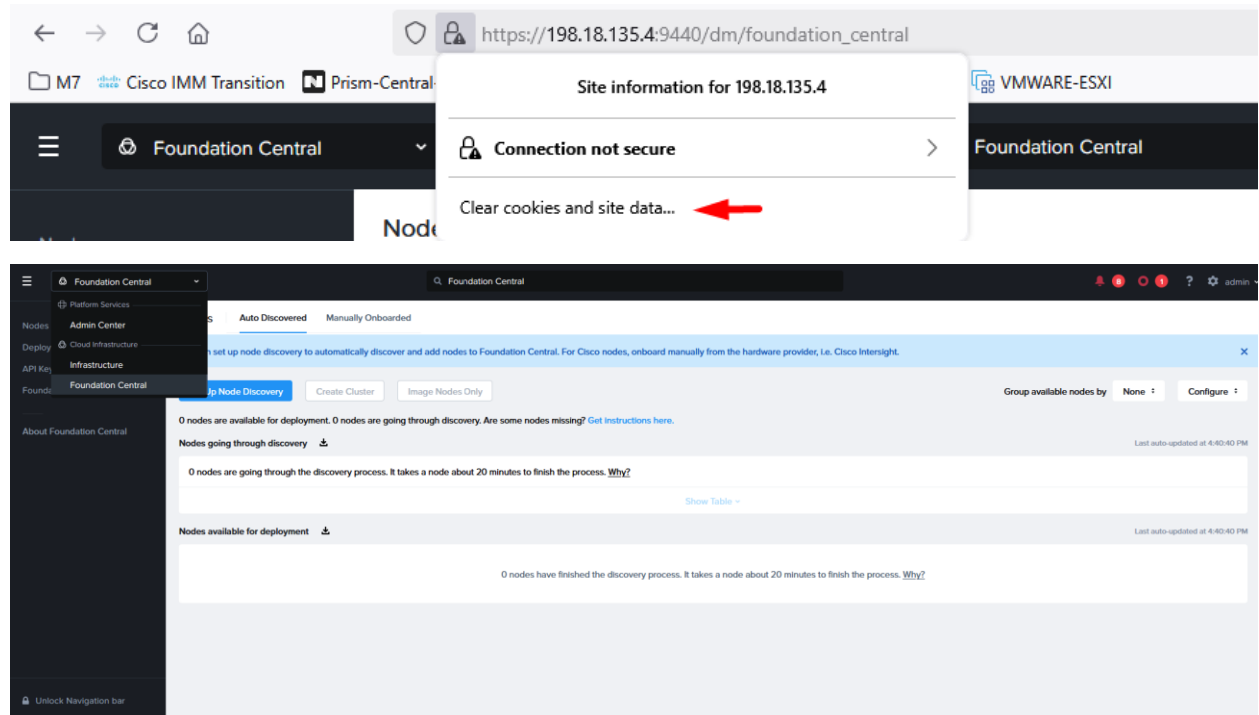
- LCM Framework Check
- Module Check

Stage 2/2: Install Update
5:40:27 PM - 5:42:00 PM

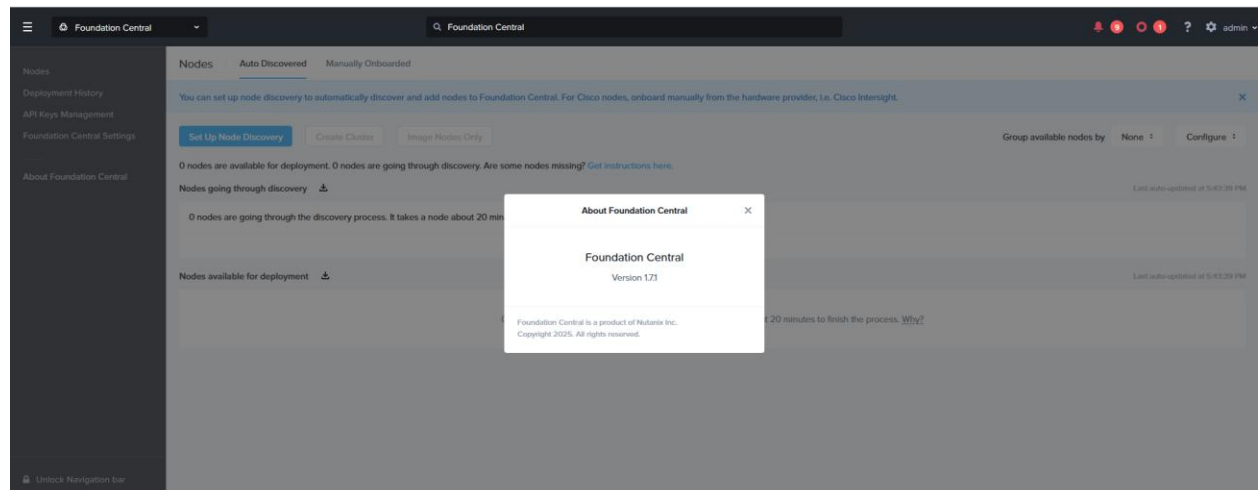
- Download
- Update Foundation Central
- Bundle Compatibility Check
- Module Compatibility Check

- Once the Foundation Central upgrade is complete, from the drop-down menu select **Foundation Central**

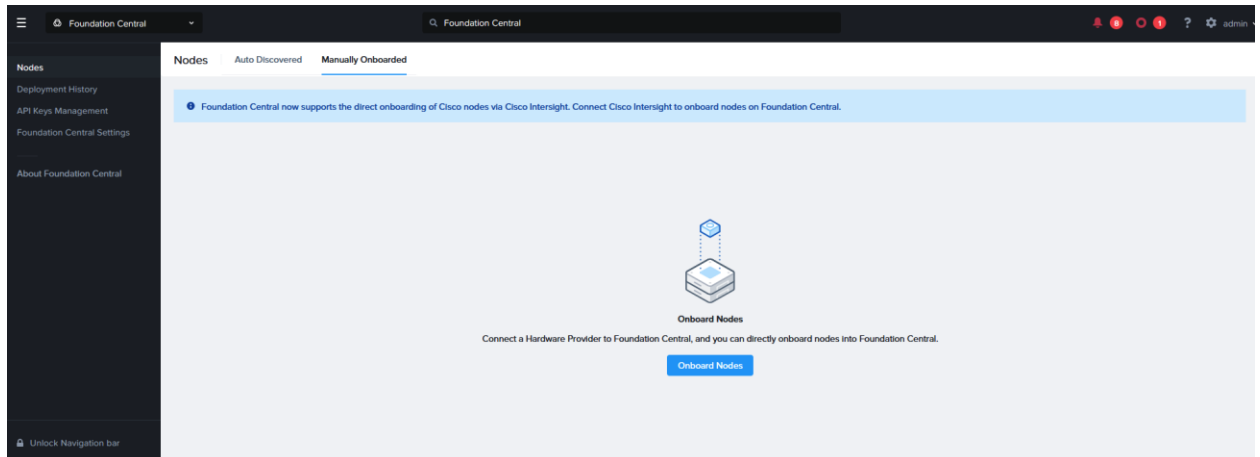
If Foundation Central does not load, just Clear the cookies and site data



- Confirm the Foundation Central by clicking on the left side About Foundation Central



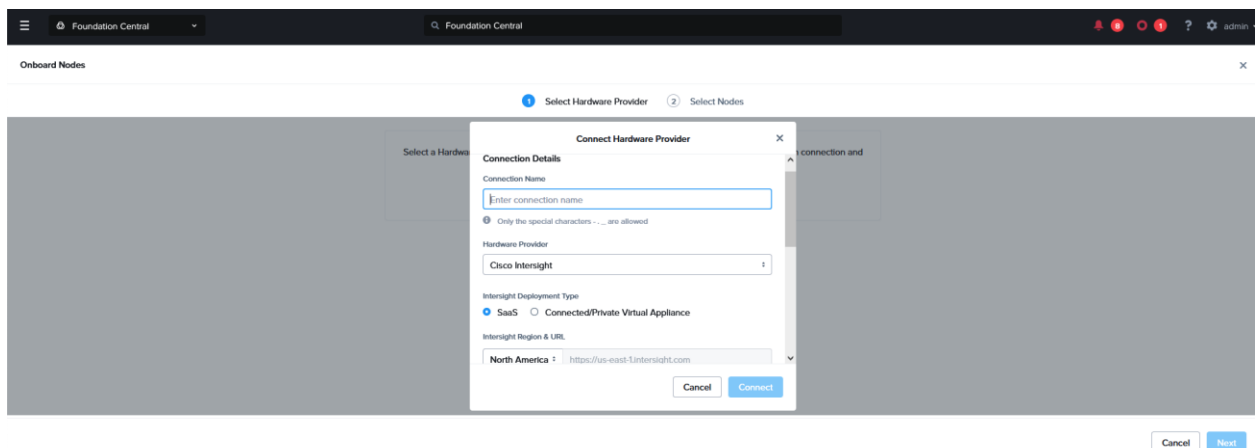
- Then under **Manually Onboarded**, click **Onboard Nodes**



10. Connect Hardware Provider

11. Fill in the required information

- Connection name= Intersight
- Hardware Provider = Cisco Intersight
- Intersight Deployment Type = SaaS
- Intersight Region & URL = North America
- Intersight API Key = <copy the one generated previously>
- Secret Key= <copy the one generated previously and saved to the desktop>



12. Select Intersight Standalone Mode and click Next

Foundation Central

Foundation Central

admin

Onboard Nodes

1 Select Hardware Provider

2 Select Nodes

Select a Hardware Provider to onboard nodes to Foundation Central. Currently, only Cisco Intersight is supported and only a single connection is allowed at a time. To add a new connection, remove the existing one.

Connection Name	Hardware Provider	URL
intersight	Cisco Intersight	https://us-east-1.intersight.com

Connection has been saved.

Node Management Mode
Select the management mode. In the next step, you will view the nodes based on the selected mode.

☐ Intersight Managed Mode

☒ Intersight Standalone Mode

Cancel

Next

13. Select the node and click Onboard

Foundation Central

Foundation Central

admin

Onboard Nodes

1 Select Hardware Provider

2 Select Nodes

Connection Name: **intersight** | Management Mode: **Intersight Standalone Mode**

You are viewing Cisco Intersight nodes without any server profile attached. To update the nodes list, click [Refresh](#).

Type in a query

Modify Filters

1 selected out of 1 Cisco Node

1-1 of 1 | 20 rows

Node Serial	Name	Model	Block Serial	Domain	Classification	Intersight Organizations	Intersight Tags	Memory (GB)	CPU (Cores)
☒ WZP28389L1	C240-WZP28389L1	UCSC-C240-M7SN	WZP28389L1	-	Nutanix	ISM	-	512	32

Back

Cancel

Onboard

14. Click "Prepare Nodes for Cluster"

Foundation Central

Foundation Central

admin

Nodes

Deployment History

API Keys Management

Foundation Central Settings

About Foundation Central

Nodes

Auto Discovered

Manually Onboarded

Onboard Nodes

Prepare Nodes for Cluster

Create Cluster

Actions

View by

Group by Management Mode

Group By: Management Mode

Modify Filters

Viewing 1 grouped out of 1 Cisco Node

Management Mode: Intersight Standalone | Empty Groups

Node Serial	Name	Node Status	Model	Domain	Intersight Organizations	Intersight Tags	Classification	Onboarded On	Last Refreshed
☐ WZP28389L1	C240-WZP28389L1	Onboarded	UCSC-C240-M7SN	-	ISM	-	Nutanix	Apr 10, 2025, 01:26 PM	Apr 10, 2025, 01:26 PM

Unlock Navigation bar

15. Select the existing ISM cluster and click Next

Foundation Central

Prepare Nodes for Cluster

1 Select Cluster 2 Select Nodes 3 AOS / Hypervisor 4 Network Settings 5 Configure Nodes 6 Security

You are viewing the clusters registered with Prism Central. Please select the cluster for which you want to prepare nodes.

Viewing 1 cluster

Cluster Name	AOS Version	Hypervisor	Node Count
ISM	70.0.5	AHV 10.0.0.1	3

Cancel Next

16. Select the node that will be added to the cluster and click Next

Foundation Central

Prepare Nodes for Cluster

1 Select Cluster 2 Select Nodes 3 AOS / Hypervisor 4 Network Settings 5 Configure Nodes 6 Security

Cluster: ISM

The following list shows the eligible nodes for the cluster. Please select nodes to prepare with the selected cluster.

Type in a query

1 selected out of 1 Cisco Node

Node Serial	Name	Model	block_serial	Insight Tags	Classification	Memory (GB)	Onboarded On
WZP283891X	C240-WZP283891X	UCSC-C240-M7SN	WZP283891X	-	Nutanix	512	May 7, 2025, 07:23 PM

Back Cancel Next

17. Return to Cisco IMM tool and copy the AOS and AHV URLs

Foundation Central

Prepare Nodes for Cluster

1 Select Cluster 2 Select Nodes 3 AOS / Hypervisor 4 Network Settings 5 Configure Nodes 6 Security

Provide the URL to download AOS and metadata from during imaging. The URLs must be accessible to Foundation Central.

AOS Download URL

https://cisco-imm.dcloud.cisco.com/repo/Nutanix-AOS-AHV-70.0.5/nutanix_installer_package-release-ganges-70.0.5

AOS Metadata URL

https://cisco-imm.dcloud.cisco.com/repo/Nutanix-AOS-AHV-70.0.5/generated-nutanix_installer_package-release-ganges-70.0.5

Hypervisor

☐ Hypervisor AHV 10.0.0.1 is included in the AOS 70.0.5 package, I have verified and want to use it.

If you won't use the included hypervisor, provide the URL below to download the hypervisor from during imaging. The URLs must be accessible to Foundation Central. The checksum will be used to avoid re-downloading an already downloaded installer.

Hypervisor Download URL

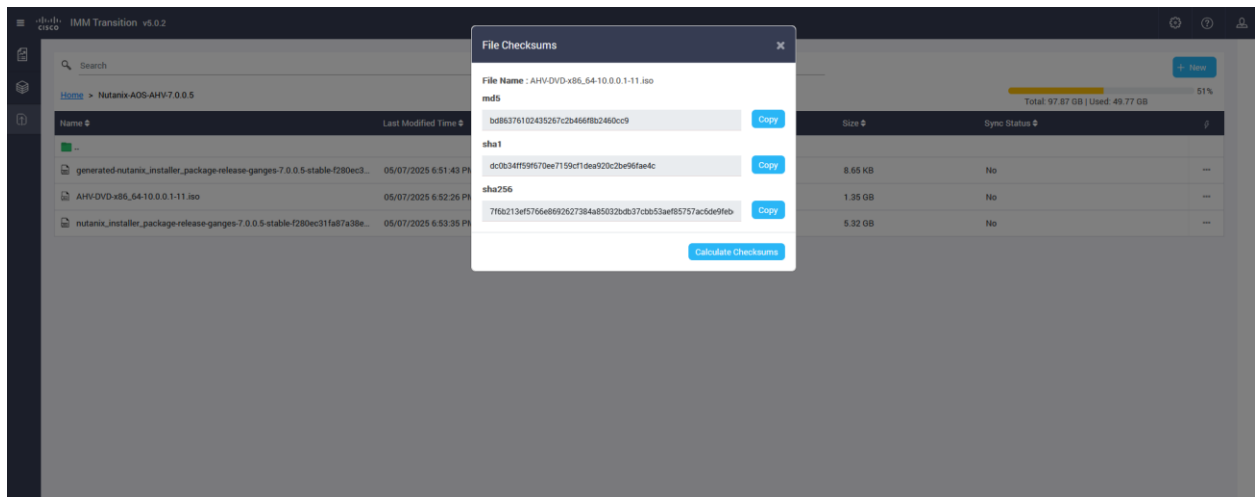
https://cisco-imm.dcloud.cisco.com/repo/Nutanix-AOS-AHV-70.0.5/AHV-DVD-x86_64-10.0.0.1-1.iso

Pre-computed SHA256 Checksum of Hypervisor Installer

7f6b23e5766e8692627384a85032bdb37cb53ae85757ac5de9f6be5e522d

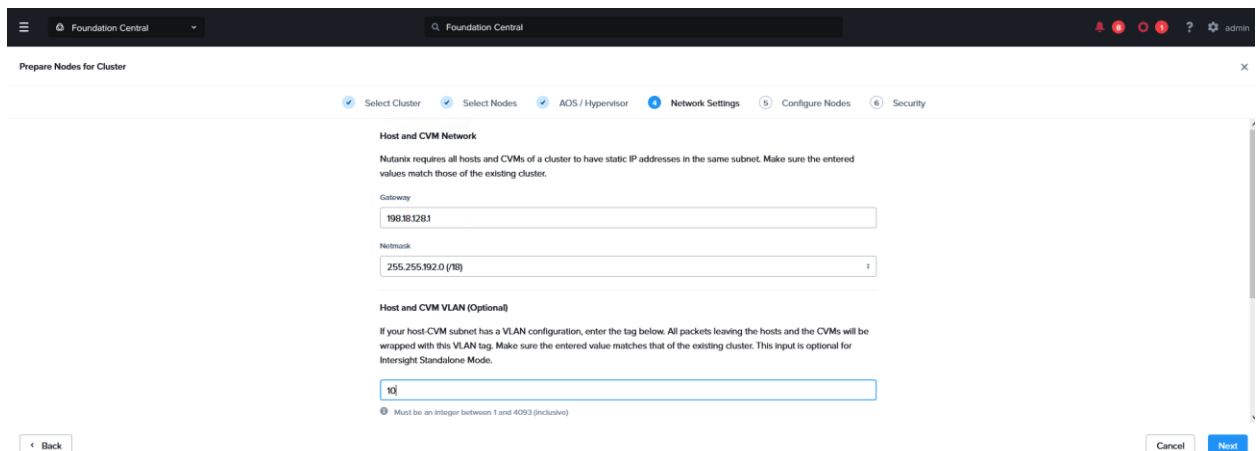
Back Cancel Next

18. For the AHV checksum, the Cisco IMM tool can be used to generate the SHA256 checksum.



19. Host and CVM Network

- Gateway= 198.18.128.1
- Netmask= 255.255.192.0 /18
- Host and CVM VLAN = 10
- Click Next



20. Fill in the IP range for the hypervisor and CVM

Foundation Central

Prepare Nodes for Cluster

Select Cluster Select Nodes AOS / Hypervisor Network Settings **Configure Nodes** Security

1 Nodes

Node Serial	Name	Hypervisor IP Set Range	CVM IP Set Range	Hypervisor Hostname Set Range
WZP283899L1	C240-WZP283899L1	198.18.135.23	198.18.135.27	c240-node-4

[Back](#) [Cancel](#) [Next](#)

21. Generate a Foundation Central API key by clicking the “+ Generate New Key” . Provide an alias and click Done. From the drop-down menu select the alias and click Submit

Foundation Central

Prepare Nodes for Cluster

Select Cluster Select Nodes AOS / Hypervisor Network Settings Configure Nodes **Security**

Foundation Central API Key

Foundation Central provides an API key to authenticate the remote nodes. It is recommended that a distinct API key be created for each remote site. You can create a new key or select from the existing ones.

Foundation Central API Key [+ Generate New Key](#)

ISM

[Back](#) [Cancel](#) [Submit](#)

API key was generated successfully.

22. The node preparation process begins

Foundation Central

Nodes
Deployment History
API Keys Management
Foundation Central Settings
About Foundation Central

Deployment History

It may take up to 15 minutes for a deployment progress to start being reported. [Why?](#)

Viewing 1 deployment Show Only Non-archived

Node Preparation for Cluster "ISM" Deployment in progress Start Date and Time: 5/9/2025, 05:50 PM [Abort](#)

Phase 1A: Node Preparation 6% 1 node in progress Phase 1B: Node Imaging 0% Not started

Cluster Details

Redundancy Factor	Host CVM Subnet	CVM NTP Servers	AOS Installer URL
2	198.18.128.1 / 255.255.192.0	198.18.128.1	https://cisco-imm.dcloud.cisco.com/repos/Nutanix-AOS-AH-V-7.0.0.5/nutanix_installer_package-release-ganges-7.0.0.5-stable-1280ec3f6d7a38eb0c07d70e4678b59d7e72ae-x86_64.tar.gz

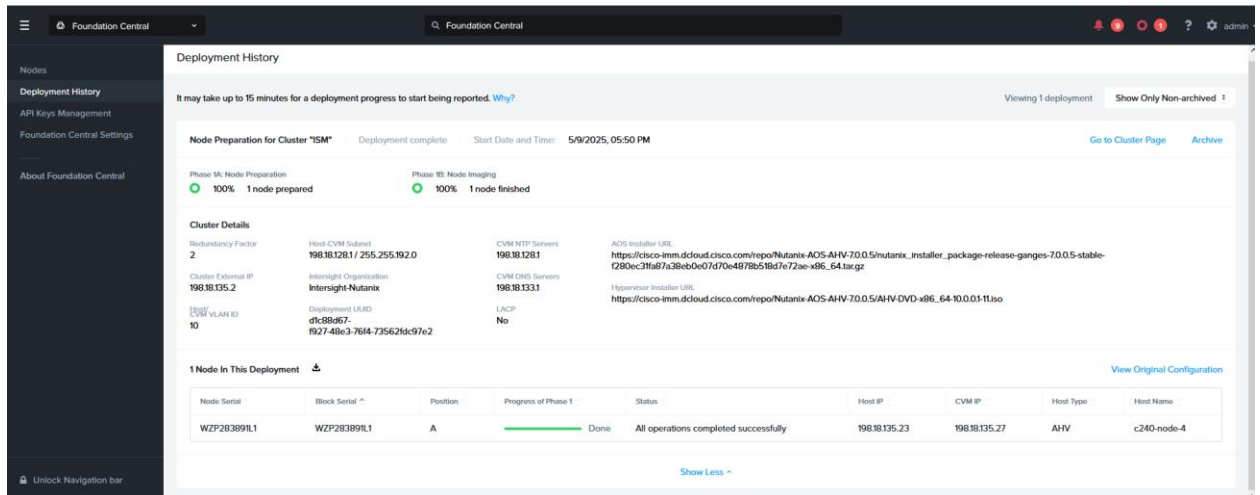
Cluster External IP	Intersight Organization	CVM DNS Servers	Hypervisor Installer URL
198.18.135.2	Intersight-Nutanix	198.18.133.1	https://cisco-imm.dcloud.cisco.com/repos/Nutanix-AOS-AH-V-7.0.0.5/AH-V-DVD-x86_64-10.0.0.1-1.iso

Deployment UUID: d1c88d67-1927-48e3-7614-735621dc97e2 LACP: No

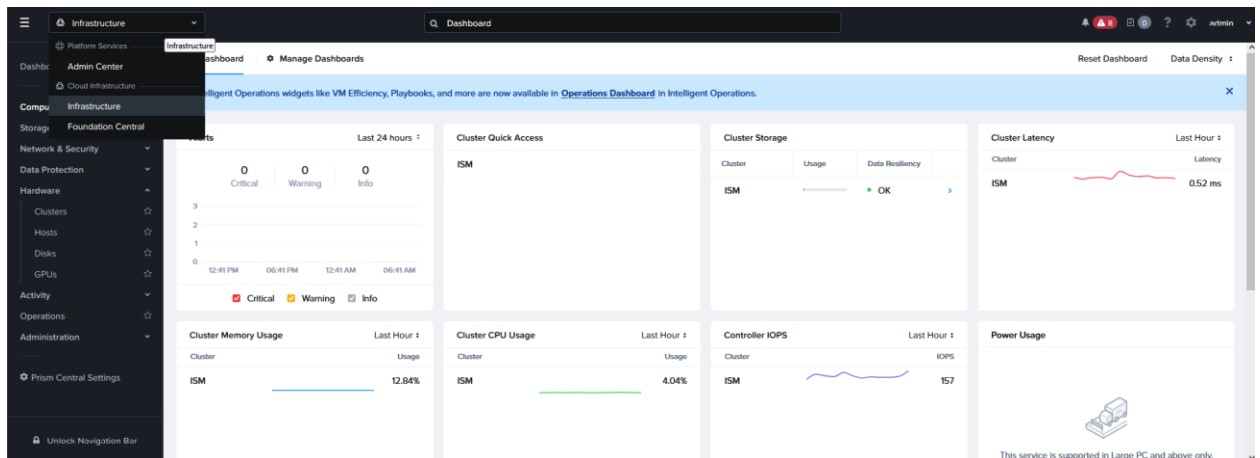
1 Node In This Deployment [View Original Configuration](#)

Node Serial	Block Serial	Position	Progress of Phase 1	Status	Host IP	CVM IP	Host Type	Host Name
WZP283899L1	WZP283899L1	A	2%	[NodeConfiguration] Performing node specific configuration in Intersight	198.18.135.23	198.18.135.27	AH-V	c240-node-4

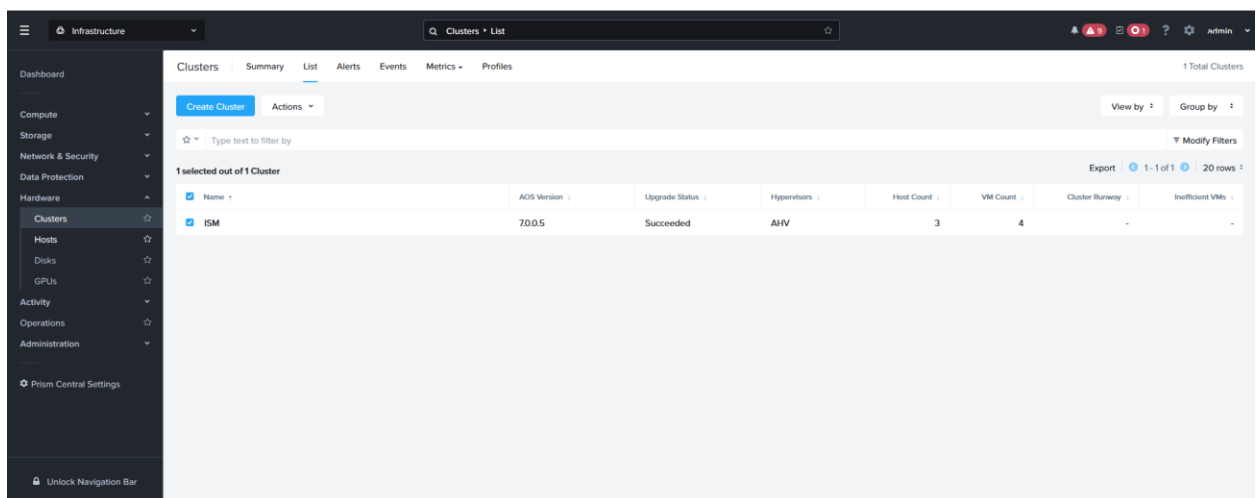
[Show Less](#)



23. Once the node preparation is completed, click at the top drop-down menu and select infrastructure



24. On the left menu, select Hardware > Clusters



25. Check the ISM box and click on actions then select Expand Cluster

The screenshot shows the Nutanix Prism Central interface. On the left, the 'Infrastructure' menu is open, and 'Clusters' is selected. A dropdown menu is visible, listing various actions such as 'Launch Prism Element', 'Upgrade Software', 'Rack Configuration', 'Manage Categories', 'Enable Efficiency Measurement', 'Disable Efficiency Measurement', 'Enable Anomaly Detection', 'Disable Anomaly Detection', 'Run Playbook', 'Manage & Backup Keys', 'Enable Data-at-Rest Encryption', 'Manage KMS Type', 'Enable Data-In-Transit Encryption', 'Disable Data-In-Transit Encryption', 'Manage Rebuild Capacity', 'Manage Recycle Bin', 'Manage Cluster Fault Tolerance', 'Maintenance Resiliency', 'Expand Cluster', and 'Destroy Cluster'. The 'Expand Cluster' option is highlighted. The main area displays the 'Clusters List' page with a table showing cluster details.

Events	Metrics	Profiles	Host Count	VM Count	Cluster Runway	Inefficient VMs
1 Total Clusters						
Export 1-1 of 1 20 rows						
AOS Version : Upgrade Status : Hypervisors : Host Count : VM Count : Cluster Runway : Inefficient VMs :						
70.0.5 Succeeded AHV 3 4 - -						

26. Check the box of the discovered node and click next

The screenshot shows the 'Expand Cluster-ISM' wizard, Step 1: Select Host. The wizard has a progress bar with steps: 1. Select Host, 2. Choose Host Type, 3. Configure Host, 4. Networking, 5. Software Check, 6. Review. The main area displays a table with discovered hosts. The 'Cisco UCS C240-M75N' host is selected.

Model	Serial Number	Available Hosts
Cisco UCS C240-M75N	WZP28389L1	1

27. On the next page select HCI from the drop-down menu and click next

The screenshot shows the 'Expand Cluster-ISM' wizard, Step 2: Choose Host Type. The wizard has a progress bar with steps: 1. Select Host, 2. Choose Host Type, 3. Configure Host, 4. Networking, 5. Software Check, 6. Review. The main area displays a table with selected hosts. The 'HCI' host type is selected.

Model/Serial	Host Position	Host Type
Cisco UCS C240-M75N WZP28389L1	A	HCI

28. Add a name c240-node-4 and then click next

Expand Cluster-ISM

✔ Select Host

✔ Choose Host Type

● Configure Host

4 Networking

5 Software Check

6 Review

Host name is required for all Hyper-V clusters, but can be skipped for AHV and ESXi clusters.
You can change a Host Name or IP by clicking on its table cell.

1 Selected Hosts

Model/Serial	Host Position	Name	CVM IPv4	CVM IPv6	Hypervisor IPv4	Hypervisor IPv6	IPMI IPv4	IPMI IPv6
Cisco UCS C240-M75N WZP2B389L1	A		198.18.135.27		198.18.135.23		198.18.143.2	

Back

CancelNext

29. On the networking step click on skip Networking

Expand Cluster-ISM

✔ Select Host

✔ Choose Host Type

✔ Configure Host

● Networking

5 Software Check

6 Review

Select the appropriate active and backup uplinks for the hosts.

1 Selected Hosts

Model/Serial	Host Position	Active Uplink	Backup Uplink
Cisco UCS C240-M75N WZP2B389L1	A	Expand row to configure switches.	Expand row to configure switches.

Back

CancelSkip NetworkingNext

30. For the software check, let' s wait for some time and then click next

Expand Cluster-ISM

✔ Select Host

✔ Choose Host Type

✔ Configure Host

✔ Networking

● Software Check

6 Review

AOS and Hypervisor version on new hosts need to be the same as the cluster.
If AOS version is lower than that of the cluster's then a reimage is not required and an upgrade will happen instead.

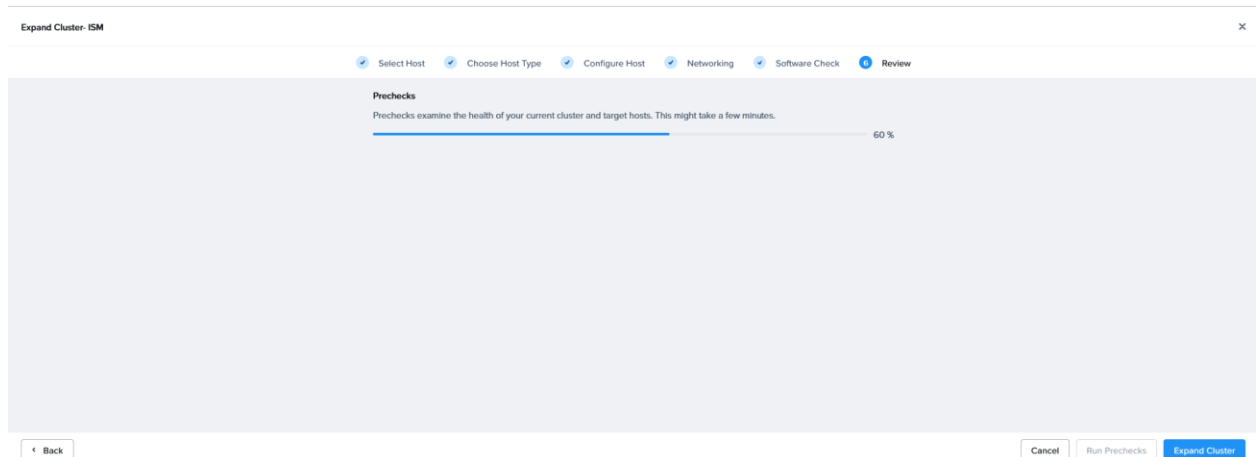
1 Selected Hosts

Model/Serial	Host Position	Host Type	Hypervisor	AOS Version	Compatibility
Cisco UCS C240-M75N WZP2B389L1	A	HCI	AHV	7.0.0.5	✔ Compatible

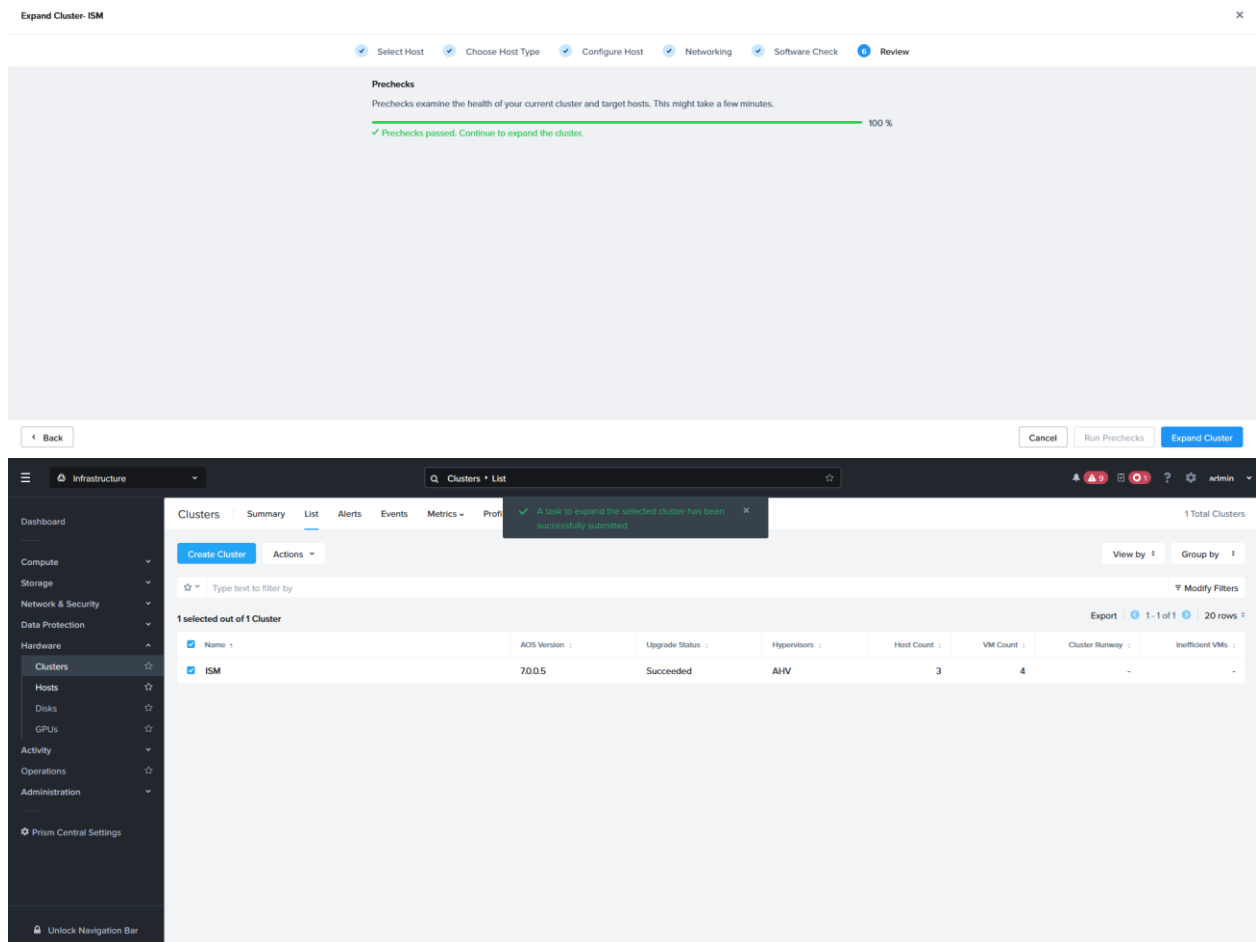
Back

CancelNext

31. Under review step, click **Run Prechecks**, wait until it completes



32. Once the precheck is completed, click on **Expand Cluster**



33. To review the progress of the cluster expansion, click task at the top right and click view all task

Infrastructure Clusters List

1 selected out of 1 Cluster

Name	AOS Version	Upgrade Status	Hypervisors
ISM	70.0.5	Succeeded	AHV

Recent Tasks

- Expand Cluster: ISM 24%
- Expand Cluster: ISM 100%
- Hypervisor Image Upload Check: ISM 100%
- Fetching cluster networks and uplinks: ISM 100%
- Discover Unconfigured Nodes: ISM 100%
- Security STIG scan 100%

Infrastructure Tasks

Viewing all 54 Tasks

Task	Entity Affected	Cluster	Progress	Status	Created On	Duration	Sub-Tasks
Node preparation for expand...	-	ISM	86%	Running	Apr 14, 2025, 10:5...	1 minute 7 seconds	5
Expand Cluster	ISM	-	86%	Running	Apr 14, 2025, 10:5...	2 minutes 9 seco...	1
Expand Cluster	ISM	-	100%	Succeeded	Apr 14, 2025, 10:5...	1 minute 15 seconds	1
Hypervisor Image Upload Che...	ISM Details	-	100%	Succeeded	Apr 14, 2025, 10:5...	24 seconds	1
Fetching cluster networks and...	ISM Details	-	100%	Succeeded	Apr 14, 2025, 10:4...	16 seconds	1
Discover Unconfigured Nodes	ISM Details	-	100%	Succeeded	Apr 14, 2025, 10:4...	23 seconds	1
Successfully finished STIG run ...	-	ISM	100%	Succeeded	Apr 13, 2025, 12:2...	Under 1 second	-
Running refresh scan	-	-	100%	Succeeded	Apr 13, 2025, 12:2...	1 minute 53 secon...	4

34. To confirm the new host has been added to the existing cluster, select Clusters on the left side

Infrastructure Clusters List

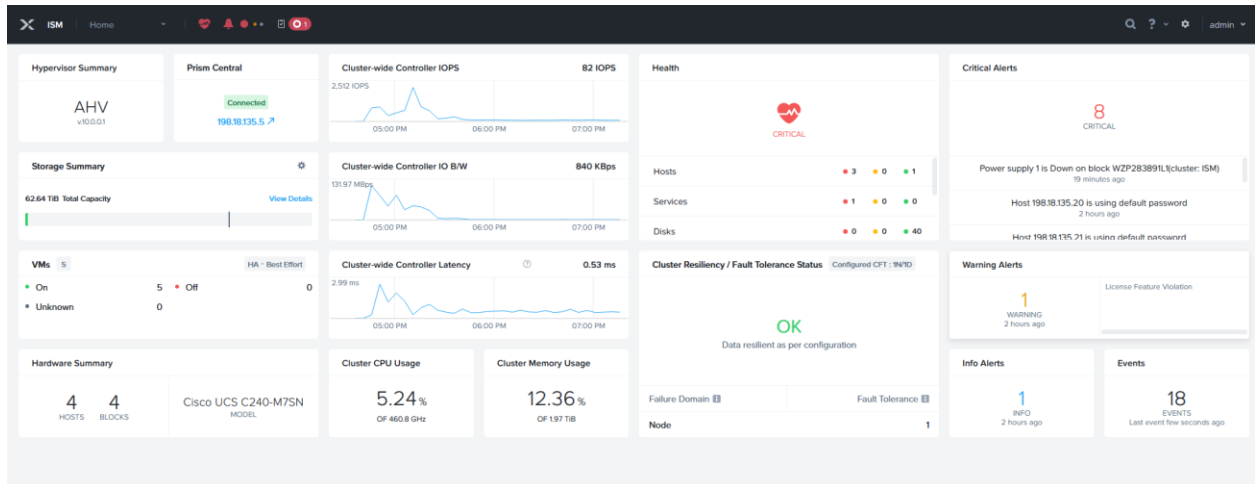
1 Total Clusters

Viewing 1 Cluster

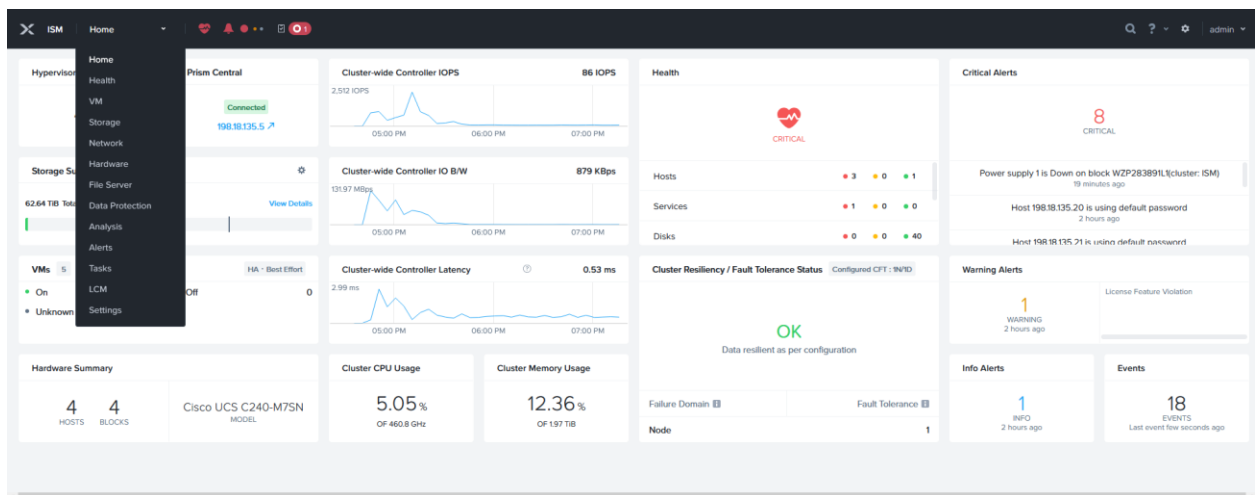
Name	AOS Version	Upgrade Status	Hypervisors	Host Count	VM Count	Cluster Runway	Inefficient VMs
ISM	70.0.5	Succeeded	AHV	4	5	-	-

35. Return to Prism Element to review on the cluster main dashboard the Node count

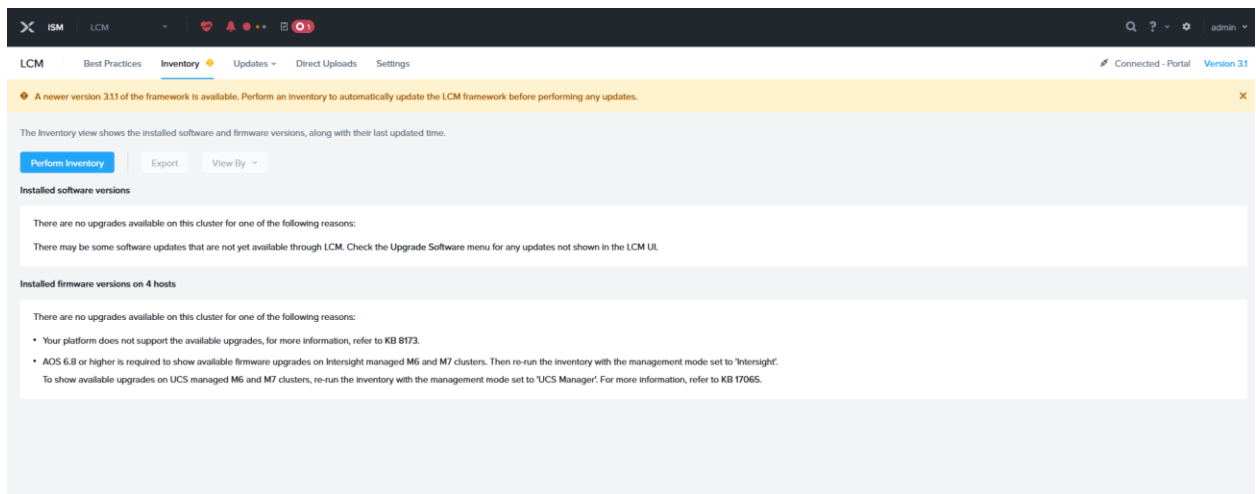
Cisco Global Demo Engineering Customer Proof of Concept



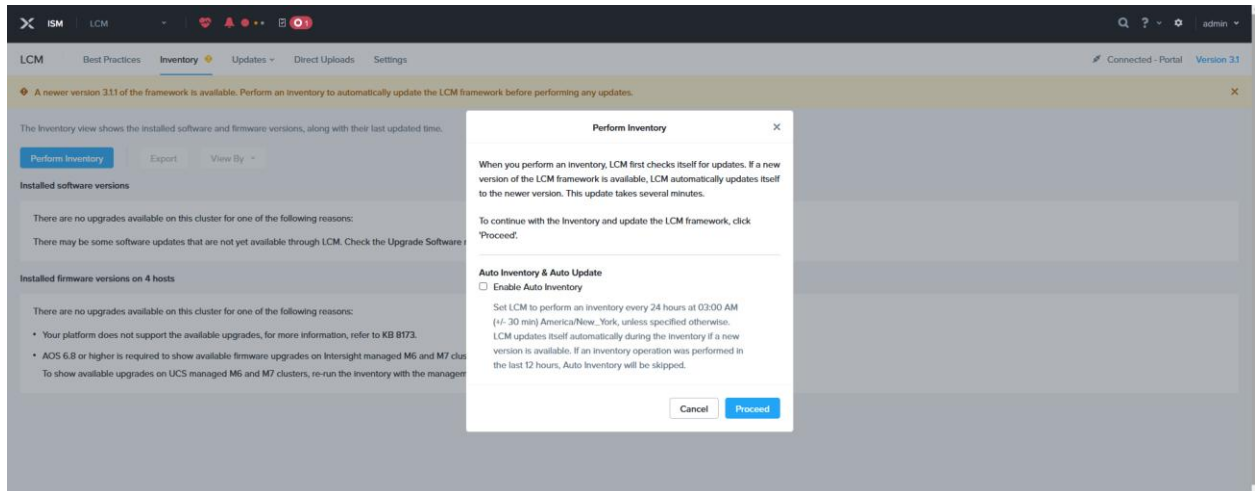
36. Click Home at the top and from the drop-down menu select LCM



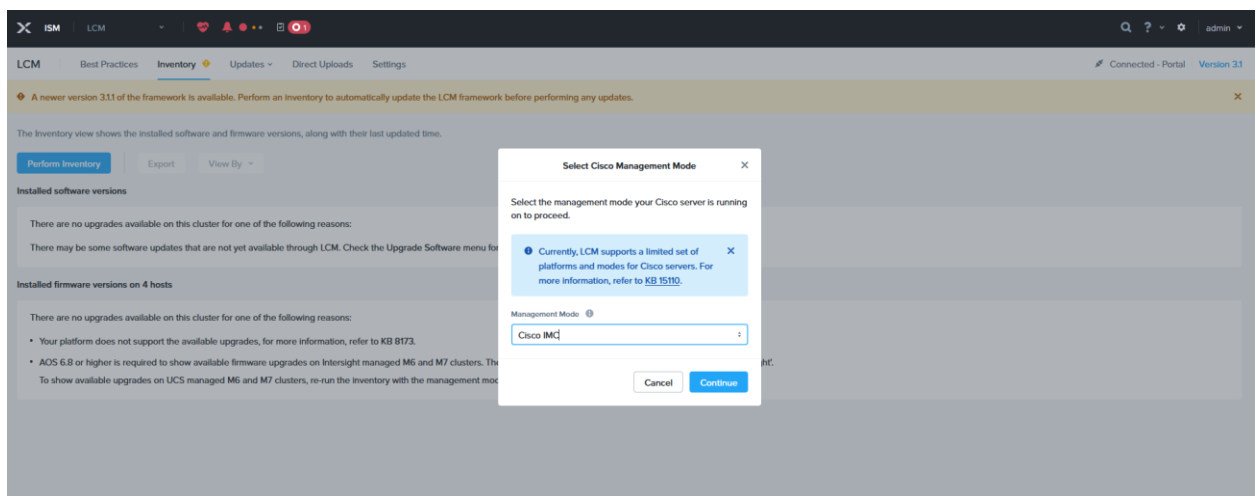
37. Select inventory and click Perform inventory



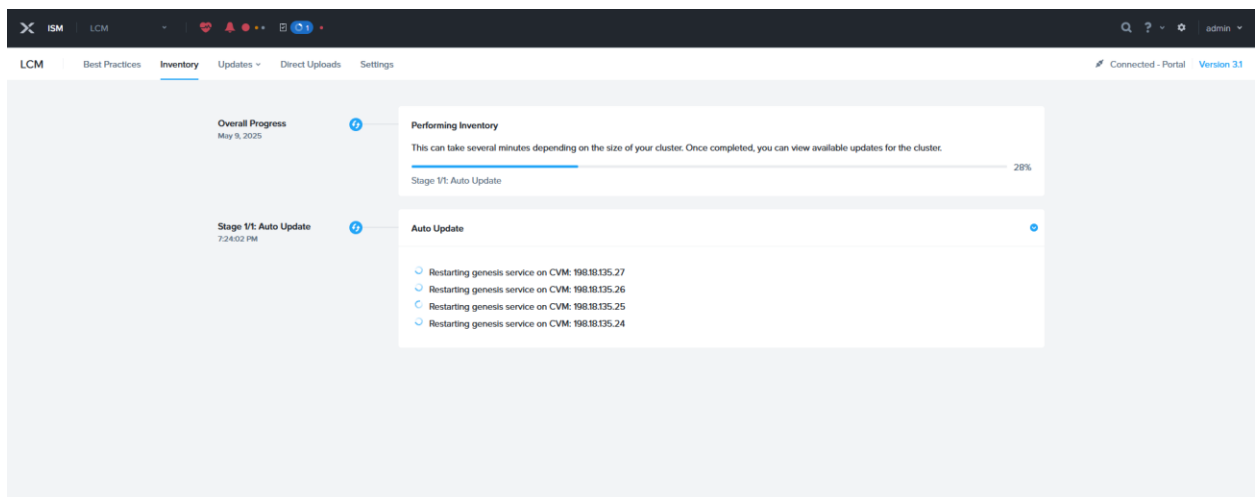
38. Click Proceed



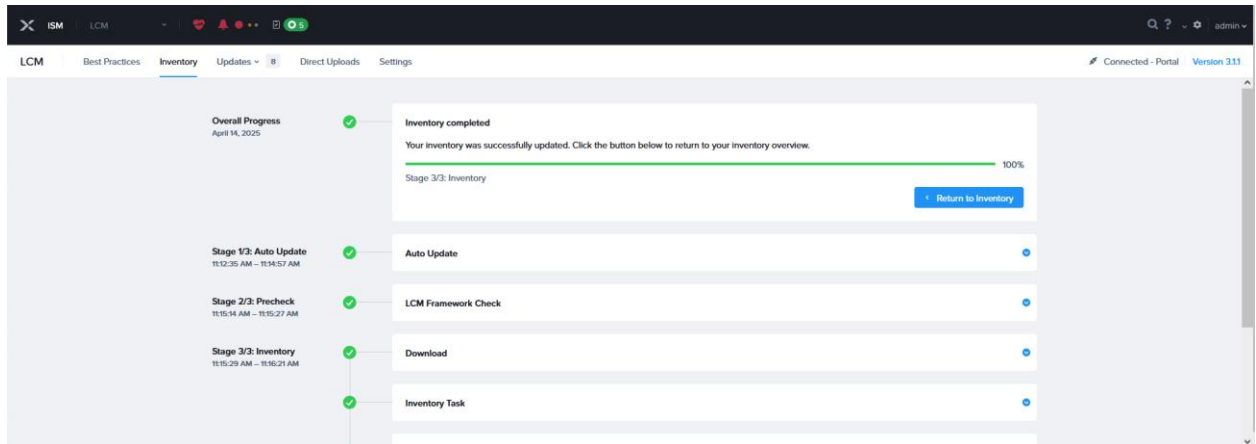
39. From the drop down select standalone Cisco IMC and click continue



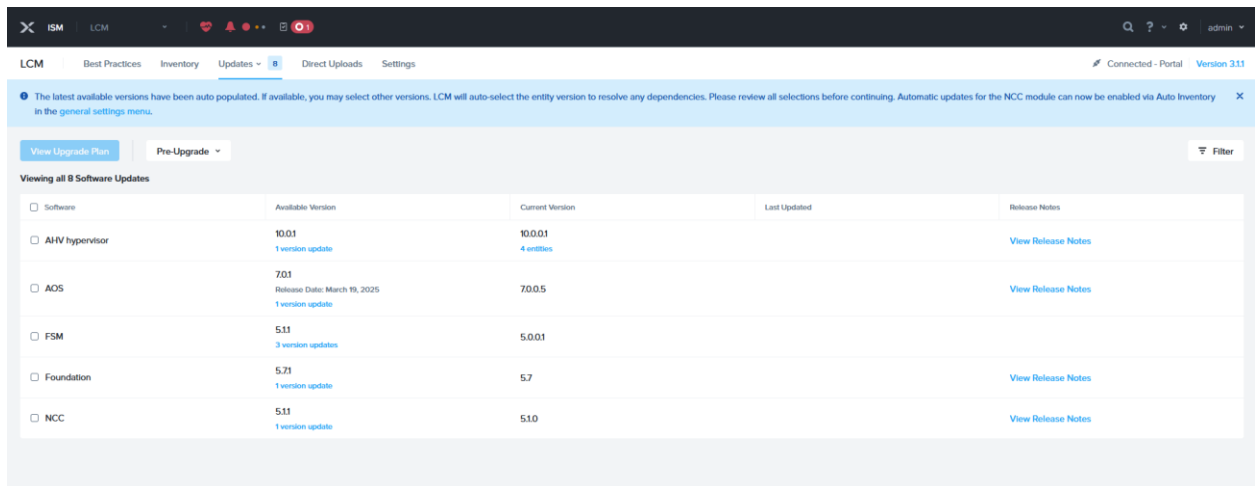
40. Wait until the Inventory completes



41. Once it's completed return to inventory



42. Click updates and select software to review all available updates



43. This is optional if you would like to apply the updates available or keep the cluster with the current version.

What's Next

Is there anything we want to point them to or provide references about?

Reference	Description